The Complex Residual Principle: How Phase, Projection, Residual, Trace, and Emergent Time Reappear across Quantum Physics, Financial Markets, and Large Language Models

How Phase, Projection, Residual, Trace, and Emergent Time Reappear across Quantum Physics, Financial Markets, and Large Language Models

A Cross-Domain Framework for Understanding Why Quantum-Like Structure May Be Rare in Substance but Common in Grammar

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Abstract

Quantum mechanics appears mysterious partly because many of its characteristic structures are normally encountered together only in microscopic physics: complex amplitudes, relative phase, interference, projection, measurement, unresolved alternatives, observer dependence, and unusual relations between phase and time. This paper asks whether some of those structures may belong to a more general organizational grammar that can also appear in macroscopic and engineered systems without making those systems literally quantum mechanical.

The investigation begins with the Capital Asset Pricing Model. CAPM already contains a projection geometry. It separates the component of an asset's return associated with the market from the component not explained by that market direction. A minimal complex extension writes an asset state as:

$Z^f = R^f + iQ^f = A^f \exp(i\theta^f)$. (0.1)

Here R^f is the component admitted by a declared financial projection, while Q^f represents structure that remains orthogonal, unresolved, unpriced, or retained as financial pressure. Q^f may include liquidity, credit, option, funding, positioning, tail, or model pressure. The imaginary coordinate does not mean that this pressure is fictional. It means that it is dynamically relevant without being fully represented in the currently admitted real-axis valuation.

When many assets are placed in a common phase representation, their collective state may be written:

$C_m = [\Sigma_i w_i \exp(i\theta_i)] / \Sigma_i w_i = \rho_m \exp(i\Phi_m)$. (0.2)

The magnitude ρ_m measures market-wide phase coherence, while Φ_m gives the dominant collective orientation. The factor $\exp(i\Phi_m)$ behaves like a clock hand. Because circular phase alone forgets completed revolutions, accumulated chronology requires an unwrapped phase $\tilde{\Phi}_m$.

A second step introduces a market admission gate G_m. Phase movement that passes the gate becomes consequential market history; phase movement that does not pass remains as unresolved selection depth:

$$d\tau_m = G_m|d\tilde{\Phi}_m| / \Omega_m. \quad (0.3)$$

$$dT_m = (1 - G_m)|d\tilde{\Phi}_m| / \Omega_m. \quad (0.4)$$

$$d\zeta_m = d\tau_m + i\, dT_m. \quad (0.5)$$

The coordinate τ_m represents ledgered market time, while T_m represents imaginary-time-like residual depth. This construction resembles the Semantic Meme Field Theory distinction between realized collapse ticks and unresolved phase rotation. In the relevant SMFT formulation, an observer projection writes a trace when collapse succeeds; when collapse is deferred, phase evolution continues without producing a new semantic tick, and the unresolved phase history accumulates along an imaginary-time axis.

The same architecture can be translated into large language model engineering. An LLM state may be represented schematically as:

$$Z^L = R^L + i\mathbf{Q}^L. \quad (0.6)$$

R^L is the emitted, accepted, cited, or actioned answer. The residual vector $\mathbf{Q}^L$ carries unresolved factual uncertainty, retrieval conflict, instruction tension, tool unreliability, ambiguity, rejected alternatives, and memory inconsistency. Under this interpretation, hallucination is not merely an incorrect token. It is a gate failure in which unresolved residual is misclassified as admitted fact.

The paper proposes the **Complex Residual Principle**:

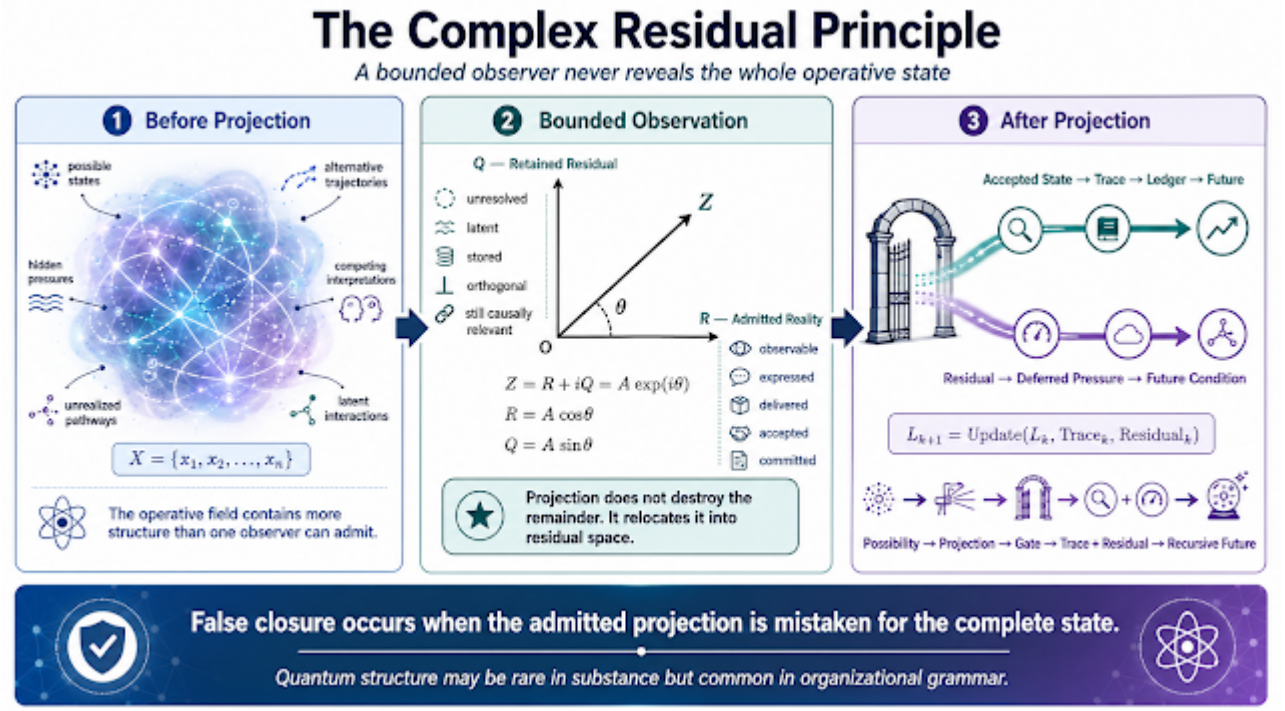
> Whenever a bounded observer projects a larger possibility field into an admitted result, dynamically active structure remains outside that projection. If the system preserves this remainder as residual, organizes alternatives by relative phase, gates consequential commitments, records accepted outcomes as trace, and allows trace to influence later projection, then quantum-like and time-like structures may emerge at the level of operational grammar.

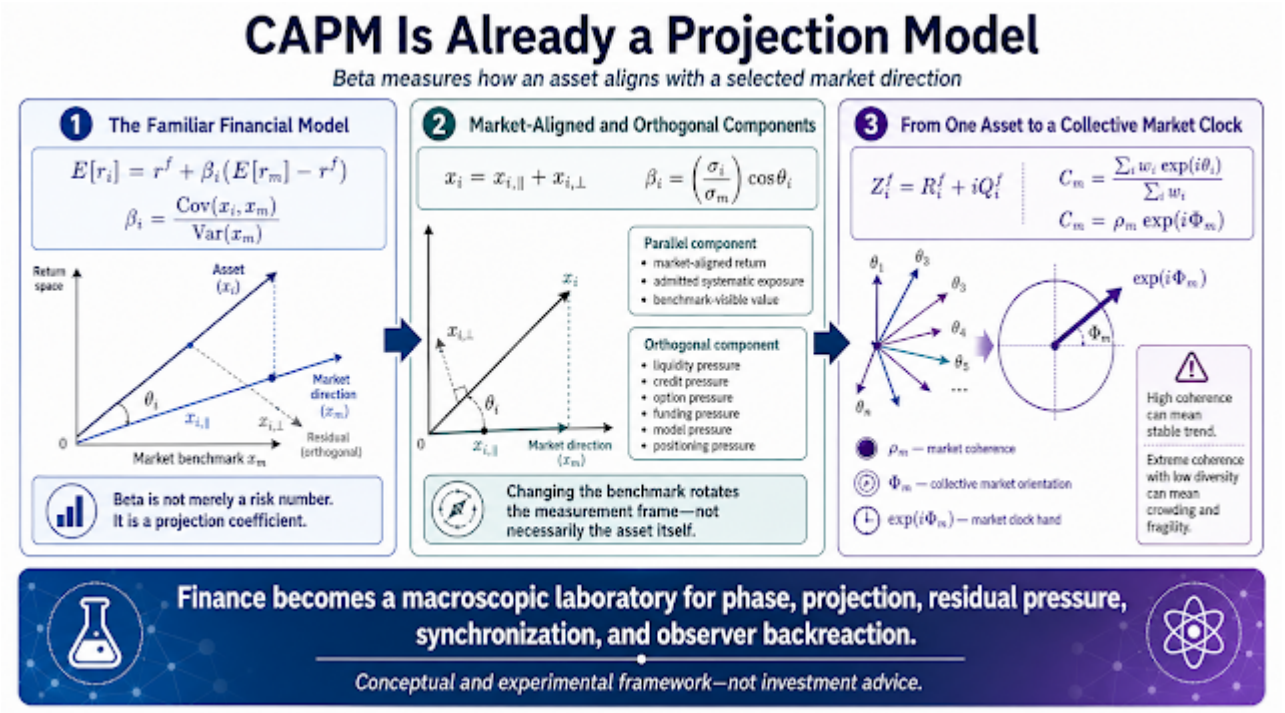
The proposal is one of functional homology, not material identity. A market is not a quantum field, an asset is not a particle, and an LLM is not a quantum computer. The Gauge Grammar of Self-Organization states the required methodological restriction clearly: quantum and gauge concepts may be transferred as disciplined functional roles under declared protocols, but not as literal claims that higher-level systems share the same physical substance.

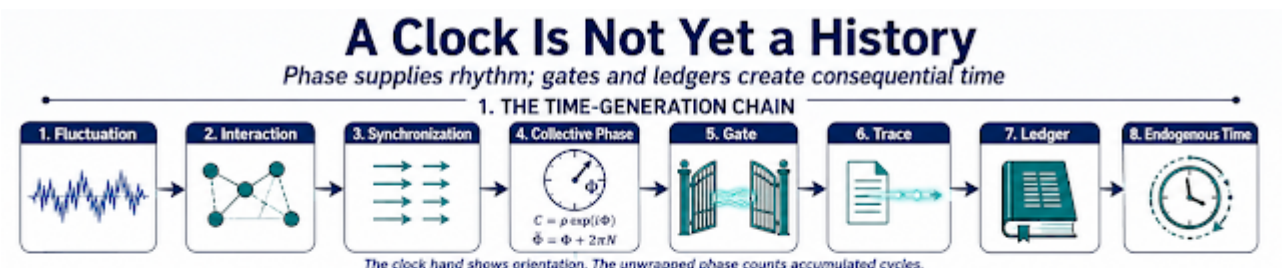
The broader hypothesis is that some characteristics regarded as mysterious in fundamental physics may become more intuitive when reconstructed inside a measurable macroscopic market and then deliberately engineered within an artificial observer. Physics may represent the deepest known physical realization of this grammar; finance may provide a visible macroscopic bridge; and LLMs may provide the most programmable experimental testbed.

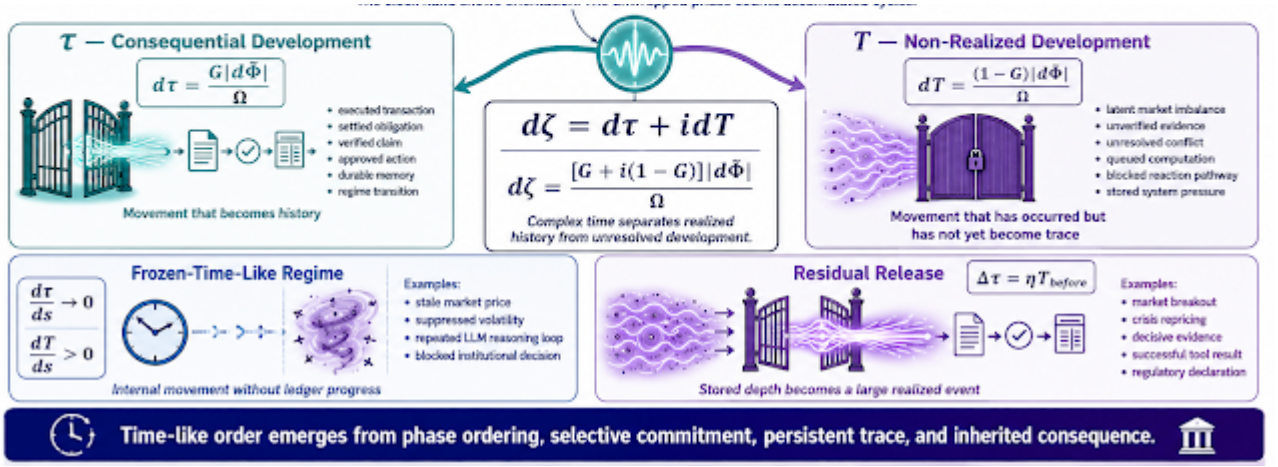
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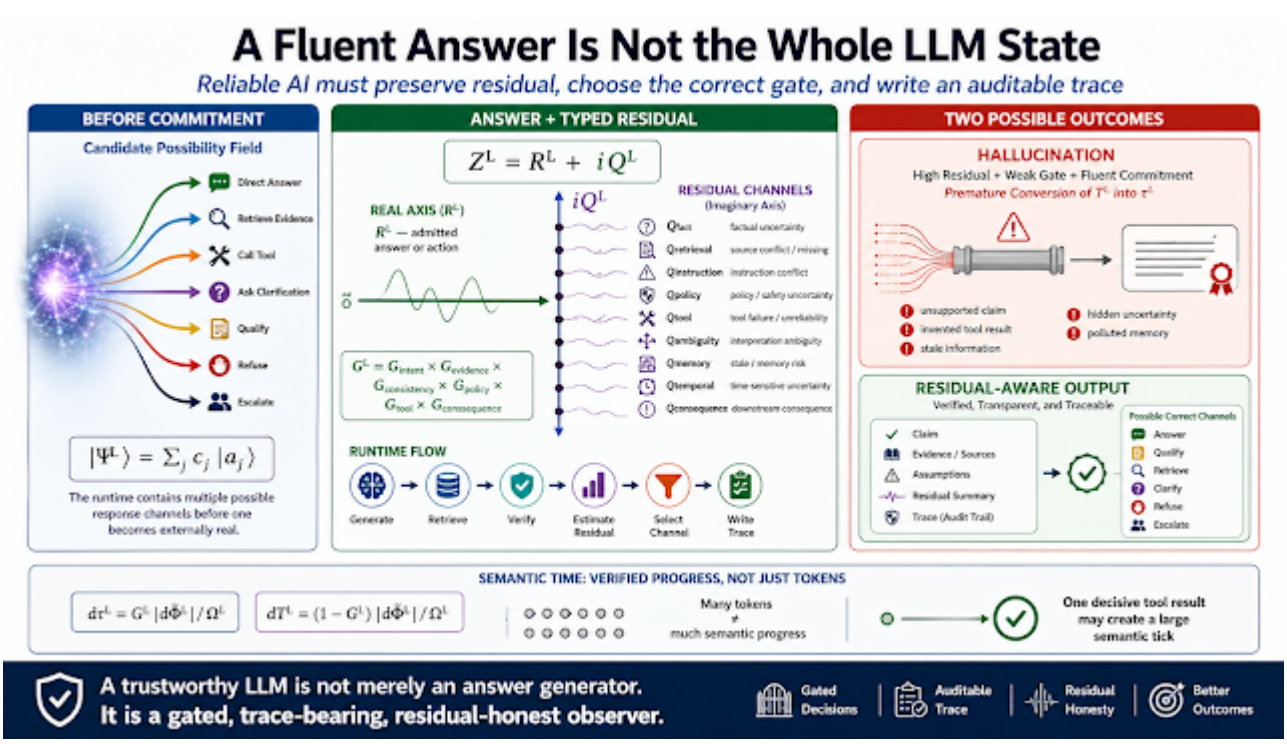
Complex residual; CAPM; quantum-like structure; phase coherence; projection; observer; financial geometry; imaginary time; emergent time; market time; large language models; residual governance; hallucination; Semantic Meme Field Theory; self-organization; trace; ledger.

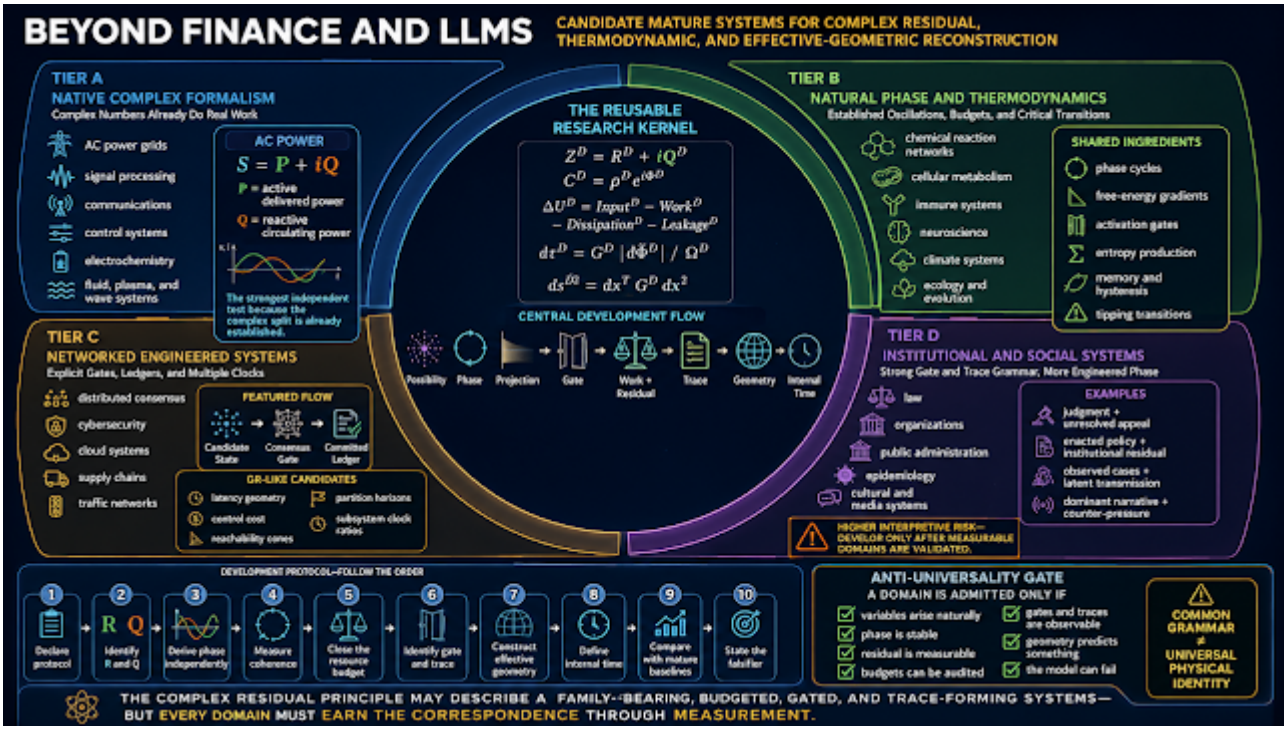







τ — Consequential Development
$d\tau = \frac{G|d\tilde{\Phi}|}{\Omega}$
executed transaction
settled obligation
verified claim
approved action
durable memory
regime transition
Movement that becomes history
$d\zeta = d\tau + idT$
$d\zeta = \frac{[G + i(1-G)]|d\tilde{\Phi}|}{\Omega}$
Complex time separates realized history from unresolved development.
T — Non-Realized Development
$dT = \frac{(1-G)|d\Phi|}{\Omega}$
latent market imbalance
unverified evidence
unresolved conflict
queued computation
blocked reaction pathway
stored system pressure
Movement that has occurred but has not yet become trace
Frozen-Time-Like Regime
$\frac{d\tau}{ds} \to 0$
$\frac{dT}{ds} > 0$
Examples:
stale market price
suppressed volatility
repeated LLM reasoning loop
blocked institutional decision
Internal movement without ledger progress
Residual Release
$\Delta\tau = \eta T_{before}$
Examples:
market breakout
crisis repricing
decisive evidence
successful tool result
regulatory declaration
Stored depth becomes a large realized event
Time-like order emerges from phase ordering, selective commitment, persistent trace, and inherited consequence.


A Fluent Answer Is Not the Whole LLM State
Reliable AI must preserve residual, choose the correct gate, and write an auditable trace
BEFORE COMMITMENT
Candidate Possibility Field
Direct Answer
Retrieve Evidence
Call Tool
Ask Clarification
Qualify
Refuse
Escalate
$|\Psi^L\rangle = \sum_j c_j |a_j\rangle$
The runtime contains multiple possible response channels before one becomes externally real.
ANSWER + TYPED RESIDUAL
$Z^L = R^L + iQ^L$
REAL AXIS (R^L)
R^L — admitted answer or action
iQ^L
RESIDUAL CHANNELS
(Imaginary Axis)
Q_{fact} factual uncertainty
$Q_{retrieval}$ source conflict / missing
$Q_{instruction}$ instruction conflict
Q_{policy} policy / safety uncertainty
Q_{tool} tool failure / unreliability
$Q_{ambiguity}$ interpretation ambiguity
Q_{memory} stale / memory risk
$Q_{temporal}$ time-sensitive uncertainty
$Q_{consequence}$ downstream consequence
$G^L = G_{intent} \times G_{evidence} \times G_{consistency} \times G_{policy} \times G_{tool} \times G_{consequence}$
RUNTIME FLOW
Generate
Retrieve
Verify
Estimate Residual
Select Channel
Write Trace
TWO POSSIBLE OUTCOMES
HALLUCINATION
High Residual + Weak Gate + Fluent Commitment
Premature Conversion of T^L into τ^L
unsupported claim
invented tool result
stale information
hidden uncertainty
polluted memory
RESIDUAL-AWARE OUTPUT
Verified, Transparent, and Traceable
Claim
Evidence / Sources
Assumptions
Residual Summary
Trace (Audit Trail)
Possible Correct Channels
Answer
Qualify
Retrieve
Clarify
Refuse
Escalate
SEMANTIC TIME: VERIFIED PROGRESS, NOT JUST TOKENS
$d\tau^L = G^L |d\tilde{\Phi}^L| / \Omega^L$
$dT^L = (1-G^L) |d\tilde{\Phi}^L| / \Omega^L$
Many tokens ≠ much semantic progress
One decisive tool result may create a large semantic tick
A trustworthy LLM is not merely an answer generator.
It is a gated, trace-bearing, residual-honest observer.
Gated Decisions
Auditable Trace
Residual Honesty
Better Outcomes


BEYOND FINANCE AND LLMS
CANDIDATE MATURE SYSTEMS FOR COMPLEX RESIDUAL, THERMODYNAMIC, AND EFFECTIVE-GEOMETRIC RECONSTRUCTION
TIER A
NATIVE COMPLEX FORMALISM
Complex Numbers Already Do Real Work
AC power grids
signal processing
communications
control systems
electrochemistry
fluid, plasma, and wave systems
AC POWER
$S = P + iQ$
P = active delivered power
Q = reactive circulating power
The strongest independent test because the complex split is already established.
THE REUSABLE RESEARCH KERNEL
$Z^D = R^D + iQ^D$
$C^D = \rho^D e^{i\Phi^D}$
$\Delta U^D = Input^D - Work^D - Dissipation^D - Leakage^D$
$d\tau^D = G^D |d\tilde{\Phi}^D| / \Omega^D$
$ds^{2D} = dx^T G^D dx^2$
CENTRAL DEVELOPMENT FLOW
Possibility
Phase
Projection
Gate
Work + Residual
Trace
Geometry
Internal Time
TIER B
NATURAL PHASE AND THERMODYNAMICS
Established Oscillations, Budgets, and Critical Transitions
chemical reaction networks
cellular metabolism
immune systems
neuroscience
climate systems
ecology and evolution
SHARED INGREDIENTS
phase cycles
free-energy gradients
activation gates
entropy production
memory and hysteresis
tipping transitions
TIER C
NETWORKED ENGINEERED SYSTEMS
Explicit Gates, Ledgers, and Multiple Clocks
distributed consensus
cybersecurity
cloud systems
supply chains
traffic networks
FEATURED FLOW
Candidate State
Consensus Gate
Committed Ledger
GR-LIKE CANDIDATES
latency geometry
partition horizons
control cost
subsystem clock cycles
reachability cones
TIER D
INSTITUTIONAL AND SOCIAL SYSTEMS
Strong Gate and Trace Grammar, More Engineered Phase
law
organizations
public administration
epidemiology
cultural and media systems
EXAMPLES
judgment + unresolved appeal
enacted policy + institutional residual
observed cases + latent transmission
dominant narrative + counter-pressure
HIGHER INTERPRETIVE RISK — DEVELOP ONLY AFTER MEASURABLE DOMAINS ARE VALIDATED.
DEVELOPMENT PROTOCOL—FOLLOW THE ORDER
1 Declare protocol
2 Identify R and Q
3 Derive phase independently
4 Measure coherence
5 Close the resource budget
6 Identify gate and trace
7 Construct effective geometry
8 Define internal time
9 Compare with mature baselines
10 State the falsifier
ANTI-UNIVERSALITY GATE
A DOMAIN IS ADMITTED ONLY IF
variables arise naturally
phase is stable
residual is measurable
budgets can be audited
gates and traces are observable
geometry predicts something
the model can fail
COMMON GRAMMAR ≠ UNIVERSAL PHYSICAL IDENTITY
THE COMPLEX RESIDUAL PRINCIPLE MAY DESCRIBE A FAMILY-BEARING, BUDGETED, GATED, AND TRACE-FORMING SYSTEMS—
BUT EVERY DOMAIN MUST EARN THE CORRESPONDENCE THROUGH MEASUREMENT.

0. Reader Contract and Scope

0.1 What this paper is trying to establish

This paper develops a cross-domain structural model connecting three very different systems:

quantum and field-theoretic physics;

financial markets and CAPM;

large language models and agentic AI runtimes.

The proposed common grammar is:

Possibility Field → Interaction → Phase Organization → Projection → Gate → Trace + Residual → Updated Field. (0.7)

The article asks whether this sequence can explain why several structures repeatedly appear across domains:

complex representation;

relative phase;

constructive and destructive interference;

projection-dependent observation;

collapse-like commitment;

unresolved alternatives;

observer backreaction;

synchronization;

regime transitions;

trace-based irreversibility;

variable internal tick rates;

imaginary-time-like residual depth.

The claim is not that every item has the same physical meaning in every domain. The claim is that each may perform a comparable **organizational role**.

The governing distinction is:

Functional Homology ≠ Material Identity. (0.8)

A useful cross-domain correspondence states:

Physics Role → General Function → Protocol-Bound Domain Role. (0.9)

It does not state:

Physics Object = Financial Object = LLM Object. (0.10)

The Gauge Grammar follows precisely this discipline. It treats fields, mediators, gates, traces, and invariants as recurring functional roles, while explicitly rejecting literal equations such as “market = Yang–Mills field” or “AI verifier = physical boson.”

0.2 Three levels of claim

The paper should be read at three levels.

Level 1 — Structural thesis

Physics, finance, and LLM runtimes can be described using comparable roles:

field;

orientation;

projection;

gate;

trace;

residual;

feedback;

invariance.

This is the safest thesis.

Level 2 — Operational thesis

The shared grammar can produce measurable improvements.

In finance, it may support:

market coherence measurement;

residual-risk separation;

regime detection;

intrinsic market-time construction;

volatility-release prediction.

In LLMs, it may support:

multidimensional uncertainty representation;

hallucination diagnosis;

evidence-gate design;

semantic-progress measurement;

residual-aware memory and tool use.

This is the main engineering thesis.

Level 3 — Foundational hypothesis

Quantum structure and physical time may be the most fundamental known realization of a wider possibility–projection–residual grammar required by observer-compatible self-organization.

This is a speculative research direction, not an established conclusion.

The structural and operational results must remain useful even if the foundational hypothesis is rejected.

0.3 Why bounded observers are the starting point

No real observer accesses total reality.

A physical instrument has finite resolution.

A market participant sees only selected prices, reports, counterparties, models, and time horizons.

An LLM runtime sees only its prompt, context window, retrieved artifacts, tool outputs, system instructions, and retained memory.

Observation therefore begins with inequality:

TotalField > ObserverCapacity. (0.11)

A bounded observer extracts some structure and leaves a remainder:

$\text{ObservedState}_P = \text{ExtractedStructure}_P + \text{Residual}_P$. (0.12)

The subscript P matters. P denotes a declared protocol:

$P = (B, \Delta, h, u)$. (0.13)

Where:

B = system boundary;

Δ = observation or aggregation rule;

h = time or state horizon;

u = admissible intervention family.

The same world may produce different visible structures under different protocols.

A financial regulator, high-frequency trader, pension fund, options dealer, and central bank do not observe the same “market,” even when they refer to the same collection of transactions.

Likewise, an LLM evaluated for factual accuracy, stylistic quality, legal safety, or tool reliability is being projected through different frames.

The bounded-observer framework therefore gives the first universal split:

$X \rightarrow R_p + \text{Residual}_p$. (0.14)

The Complex Residual Principle proposes that the second term should not merely be discarded. It may require its own coordinate.

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1. Quantum Mystery and Macroscopic Familiarity

1.1 What appears mysterious

Quantum mechanics is often introduced through a collection of concepts that appear alien to ordinary experience:

a state is represented by a complex amplitude;

relative phase changes observable probabilities;

alternatives interfere;

measurement depends on the chosen observable;

one outcome becomes recorded while others do not;

the observer cannot be treated as completely irrelevant;

measurement changes the state assigned to the system;

unobserved structure remains essential to prediction;

imaginary time appears in important physical calculations;

microscopic reversible evolution somehow gives rise to irreversible records.

Taken together, these features make quantum theory seem unlike the macroscopic world.

Yet several parts are already common outside quantum mechanics.

Classical waves interfere.

Oscillators synchronize.

Nonlinear systems undergo phase transitions.

Order parameters emerge from many interacting components.

Measurement changes biological, economic, and social systems.

Coarse-graining converts reversible microdynamics into effective irreversibility.

Feedback allows previous outputs to change subsequent dynamics.

The mystery may therefore be partly composite. Some components are genuinely quantum. Others are general properties of waves, bounded observation, self-organization, statistical aggregation, and irreversible recording.

The task is not to erase the difference.

The task is to separate:

Quantum-Specific Structure from General Organizational Grammar. (1.1)

1.2 Substance and grammar

Consider language.

English, Chinese, mathematics, computer code, and music are not made from the same material and do not perform the same functions. Yet they may share grammatical roles:

symbols;

composition;

boundary;

sequence;

reference;

transformation;

closure.

The recurrence of grammar does not imply identity of substance.

The same distinction can be applied here.

A physical wavefunction, a financial state vector, and an LLM candidate distribution are not the same object.

But they may all contain:

multiple admissible configurations;

relative orientations;

interaction among alternatives;

observer-dependent projections;

commitment thresholds;

residual states;

historical traces.

The central methodological proposition is:

Shared Grammar Does Not Require Shared Substrate. (1.2)

This permits a disciplined comparison without reducing physics to metaphor or elevating finance and AI into unearned physical theories.

The Gauge Grammar calls this a functional role translation. It begins from bounded observers, requires a protocol, identifies recurring roles such as field, identity, mediation, binding, gate, trace, and invariance, and tests whether the translation improves diagnosis or intervention.

1.3 Why complex numbers deserve reinterpretation

A complex number is normally written:

$Z = R + iQ$. (1.3)

In elementary mathematics:

R is the real component;

Q is the imaginary component;

$i^2 = -1$.

In physics, the complex state supports amplitude and phase:

$Z = A \exp(i\theta)$. (1.4)

Using Euler's relation:

$\exp(i\theta) = \cos\theta + i \sin\theta$. (1.5)

Therefore:

$R = A \cos\theta$. (1.6)

$Q = A \sin\theta$. (1.7)

$A^2 = R^2 + Q^2$. (1.8)

The usual language can mislead intuition. "Imaginary" sounds unreal, invented, or dispensable. But Q is not mathematically less real than R. It is a coordinate orthogonal to the chosen real axis.

This suggests an observer-relative interpretation:

R = structure admitted by the current projection. (1.9)

Q = structure retained outside the current projection. (1.10)

Under a different projection angle α:

$R' = R\cos\alpha + Q\sin\alpha$. (1.11)

$Q' = -R\sin\alpha + Q\cos\alpha$. (1.12)

What was residual in one frame may become visible in another.

This is crucial.

The Complex Residual Principle does not define Q as error.

Q may contain:

hidden structure;

unresolved tension;

option value;

incompatible evidence;

unselected alternatives;

deferred commitment;

uncertainty;

pressure awaiting another measurement frame.

Thus:

Residual ≠ Nothing. (1.13)

Residual ≠ Necessarily Error. (1.14)

Residual = Unadmitted but Potentially Consequential Structure. (1.15)

1.4 The real axis is a declared axis

Suppose a two-dimensional state is represented by the vector:

$\mathbf{v} = (R,Q)$. (1.16)

Calling R “real” does not mean that nature supplied a label saying this axis is ontologically privileged.

The axis is privileged by a measurement, accounting rule, decision protocol, or observer.

In finance, the privileged axis may be:

market-aligned return;

current discount model;

recognized accounting value;

executable price;

regulatory capital treatment.

In an LLM, it may be:

the emitted answer;

the top-ranked token;

the verified claim;

the permitted tool action;

the memory item accepted for future reuse.

A projection operator $\hat{O}_p$ selects visible structure:

$R_p = \hat{O}_p(X)$. (1.17)

The remainder is:

$Q_p = X - \text{Reconstruct}_p(R_p)$. (1.18)

This expression is schematic because X and R_p need not occupy the same mathematical space. Its conceptual point is exact:

Projection creates visibility by selecting a direction, and selection necessarily creates a residual.

The SMFT framework similarly treats observation as active projection rather than passive copying. An observer selects a frame, commits an interpretation, and writes a trace; pre-commitment buildup is retained through a complex semantic-time structure.

1.5 The overlooked macroscopic possibility

Once complex representation is interpreted as admitted structure plus retained residual, several quantum-like characteristics become less alien.

Interference

If two complex states combine:

$Z = Z_1 + Z_2$. (1.19)

Then:

$|Z|^2 = |Z_1|^2 + |Z_2|^2 + 2\text{Re}(Z_1 Z_2^*)$. (1.20)

The cross term depends on relative phase.

In a market, aligned flows can amplify a trend while opposed positions cancel.

In an LLM, compatible evidence and instructions reinforce a candidate answer, while conflicting sources and constraints weaken it.

These are not necessarily quantum interference. But they instantiate the general role:

Outcome Depends on Relative Orientation, Not Magnitude Alone. (1.21)

Projection dependence

The observed component changes when the measurement frame changes.

A market beta depends on the selected market portfolio and time horizon.

An LLM answer depends on prompt framing, retrieval set, system policy, and evaluation rule.

Again, this does not reproduce quantum measurement in full. But it illustrates:

Observed Structure Is Frame-Dependent. (1.22)

Backreaction

A market price changes subsequent orders, collateral, narratives, and risk limits.

An LLM answer changes the user's next prompt, the conversation history, memory, and future retrieval.

Thus:

$\text{Observation}_k \rightarrow \text{UpdatedField}_{k+1}$. (1.23)

Trace

An executed trade is not merely a number. It enters balance sheets, charts, risk systems, and institutional memory.

An LLM tool call or memory write changes future admissible actions.

Thus:

Record ≠ Passive Archive. (1.24)

Trace = Record that Changes Future Routing. (1.25)

These recurrences suggest that at least part of quantum mystery may become more intuitive when viewed through a broader observer–projection–trace architecture.

1.6 The first major hypothesis

The article's first major hypothesis can now be stated.

General Projection–Residual Hypothesis

Any system containing:

a larger possibility field X;

a bounded observer $\hat{O}_p$;

a selected visible component R_p;

a retained residual Q_p;

a gate controlling commitment;

a trace that modifies later states;

may exhibit structures resembling phase dependence, interference, collapse, backreaction, and emergent temporal order.

In compact form:

$X \to \hat{O}_p \to R_p + iQ_p \to \text{Gate}_p \to \text{Trace}_p \to X'$. (1.26)

The hypothesis is deliberately broad.

It does not say that every such system is quantum.

It says that quantum-like grammar may recur wherever bounded projection and consequential commitment recur.

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2. The Complex Residual Principle

2.1 Definition

Definition 2.1 — Complex Residual State

Under a declared protocol P, a system state may be represented by:

$Z_p = R_p + iQ_p$. (2.1)

Where:

R_p is the structure admitted by the observer and protocol;

Q_p is the dynamically relevant structure not admitted into R_p;

i marks orthogonality to the current admission axis.

In polar form:

$Z_p = A_p \exp(i\theta_p)$. (2.2)

With:

$R_p = A_p \cos\theta_p$. (2.3)

$Q_p = A_p \sin\theta_p$. (2.4)

$A_p^2 = R_p^2 + Q_p^2$. (2.5)

The angle θ_p measures the orientation of the system relative to the current projection frame.

2.2 The Complex Residual Principle

Principle 2.2

Whenever a bounded observer projects a larger possibility field into an admitted result, unadmitted but dynamically active structure remains. If this remainder is preserved as a residual coordinate rather than erased, the combined state can be represented by a complex or multidimensional residual geometry.

In compact form:

Bounded Projection ⇒ Admitted Structure + Residual. (2.6)

Or:

$\hat{O}_p(X) \Rightarrow R_p + iQ_p$. (2.7)

The principle has five consequences.

Consequence 1 — Completeness is protocol-relative

R_p may be complete for one purpose and incomplete for another.

A price may be sufficient for immediate execution but insufficient for understanding liquidity risk.

An LLM answer may satisfy formatting rules but fail factual verification.

Therefore:

$\text{Complete}_{p1}(R) \not\Rightarrow \text{Complete}_{p2}(R)$. (2.8)

Consequence 2 — Residual can later become observable

A frame rotation or new instrument may project Q into R.

$Q_p \rightarrow R_p'$ under FrameChange($P \rightarrow P'$). (2.9)

An unpriced liquidity risk may later dominate price.

An unresolved LLM source conflict may become decisive after retrieval.

Consequence 3 — Residual affects future dynamics

Even while unobserved, Q can alter stability, gate pressure, and transition probability.

FutureState = F(R,Q,Trace,Environment). (2.10)

Consequence 4 — Flattening residual creates false closure

If the system reports only R while silently discarding Q:

ReportedState = R. (2.11)

But the operative state remains:

OperativeState = R + iQ. (2.12)

The difference creates hidden risk:

ClosureError = OperativeState − ReportedState. (2.13)

Consequence 5 — Mature observation requires residual governance

A mature observer must decide whether residual should be:

disclosed;

bounded;

monitored;

investigated;

deferred;

transferred;

converted into another observable;

retained for future revision.

This is consistent with the Gauge Grammar premise that intelligence is not merely structure extraction but structure extraction plus residual governance and trace-guided intervention.

2.3 Scalar residual and residual vector

The simplest model uses one residual coordinate:

Z = R + iQ. (2.14)

This is useful for visualization and first-order derivation.

But many real systems contain multiple unresolved dimensions.

Define:

$\mathbf{Q} = (Q_1, Q_2, \ldots, Q_n)^T$. (2.15)

Then a generalized residual state is:

$Z = R + i\mathbf{Q}$. (2.16)

This is notation rather than an ordinary scalar complex number. It represents one admitted coordinate coupled to a residual space.

A generalized magnitude may be defined:

$A^2 = R^2 + \mathbf{Q}^T G \mathbf{Q}$. (2.17)

Where G is a positive semidefinite residual metric.

If G = I:

$A^2 = R^2 + \Sigma_j Q_j^2$. (2.18)

If G contains off-diagonal terms, residual pressures overlap or interact:

$\mathbf{Q}^T G \mathbf{Q} = \Sigma_j G_{jj} Q_j^2 + 2\Sigma_{j<k} G_{jk} Q_j Q_k$. (2.19)

This distinction matters in both finance and LLMs.

In finance:

liquidity pressure may amplify credit pressure;

volatility pressure may interact with option convexity;

funding stress may correlate with forced selling.

In LLMs:

retrieval conflict may amplify factual uncertainty;

tool failure may increase instruction ambiguity;

long-context drift may weaken evidence consistency.

A one-dimensional Q hides these interactions.

The scalar complex number is therefore the entry point.

The residual metric is the mature model.

2.4 Residual is observer-relative but not arbitrary

Because Q depends on protocol, one might conclude that it is merely subjective.

That conclusion is too strong.

Observer-relative does not mean unconstrained.

A valid residual model should satisfy:

declared boundary;

declared feature map;

reproducible projection rule;

measurable trace;

frame-robust relation;

falsifiable prediction.

Let:

$P = (B, \Delta, h, u)$. (2.20)

Let φ be the declared feature map.

Then:

$R_p = \text{Projection}(X \mid B,\Delta,h,\varphi)$. (2.21)

Residual must be defined relative to the same declaration:

$Q_p = \text{Residual}(X \mid R_p,B,\Delta,h,\varphi)$. (2.22)

Two observers may obtain different values because they use different protocols. But equivalent protocols should produce compatible relations.

A cross-frame requirement may be written:

$\text{Inv}(R_p,Q_p) = \text{Inv}(R_p',Q_p')$. (2.23)

The invariant need not be the individual coordinates. It may be:

total magnitude;

ordering;

risk bound;

transition probability;

conserved relation;

reproducible gate outcome.

Thus the framework avoids both extremes:

naive objectivism, in which one projection is mistaken for total reality;

uncontrolled relativism, in which every projection is equally valid.

The correct standard is:

2.5 Projection, gate, and trace are different operations

These terms must not be collapsed into one.

Projection

Projection makes a candidate structure visible:

$V_k = \hat{O}_p(X_k)$. (2.25)

Gate

The gate decides whether the visible candidate becomes committed:

$C_k = Gate_p(V_k, R_k, Q_k)$. (2.26)

Where:

$C_k \in \{reject, defer, accept, escalate\}$. (2.27)

Trace

If accepted, the result enters a ledger:

$L_{k+1} = Update(L_k, C_k, Metadata_k, Residual_k)$. (2.28)

Field update

The new trace changes the next system state:

$X_{k+1} = F(X_k, L_{k+1}, Environment_k)$. (2.29)

The full cycle is:

$X_k \rightarrow \hat{O}_p \rightarrow V_k \rightarrow Gate_p \rightarrow L_{k+1} + Residual_k \rightarrow X_{k+1}$. (2.30)

This distinction will become essential later.

In finance:

a quote is visible;

a transaction passes the gate;

settlement and records create trace.

In an LLM:

a candidate answer is generated;

a verifier or policy gate evaluates it;

an emitted answer, tool action, or memory write creates trace.

In SMFT, projection selects an interpretation, gate conditions determine whether collapse succeeds, and a successful event becomes a tick that writes history. The framework's complex semantic time $\zeta = \tau + iT$ separates trace-producing commitment from pre-commitment buildup.

2.6 Residual and imaginary time

The residual coordinate Q and imaginary-time-like coordinate T are related but not identical.

Q describes unresolved state structure:

$$Z = R + iQ. \quad (2.31)$$

T describes accumulated unresolved evolution:

$$\zeta = \tau + iT. \quad (2.32)$$

A simple relation may be:

$$dT/ds = f(Q,G,\theta). \quad (2.33)$$

Where:

s is an external process parameter;

Q is unresolved pressure;

G is the commitment gate;

θ is phase velocity.

One possible form is:

$$dT = (1 - G)|d\tilde{\theta}| / \Omega^*. \quad (2.34)$$

This means unresolved time grows when:

the system continues changing in phase;

the gate does not admit a consequential event.

Realized time grows when:

$$d\tau = G|d\tilde{\theta}| / \Omega^*. \quad (2.35)$$

Therefore:

$$d\zeta = [G + i(1 - G)]|d\tilde{\theta}| / \Omega^*. \quad (2.36)$$

This formula will later be specialized for markets and LLMs.

It provides the article's central temporal distinction:

> Phase motion supplies an internal clock signal, but only gated trace formation supplies history.

Or:

Phase Is Not Yet Time. (2.37)

Phase + Gate + Trace = Time-Like Order. (2.38)

The SMFT source makes a closely related distinction. It treats τ as observer-synchronized collapse rhythm and T as pre-commitment buildup; a projection that commits an interpretation writes a trace, while unresolved evolution remains on the imaginary-time axis.

2.7 The minimal cross-domain kernel

The general model can now be compressed into four equations.

Complex residual state

$$Z^D = R^D + iQ^D. \quad (2.39)$$

Collective phase state

$$C^D = \rho^D \exp(i\Phi^D). \quad (2.40)$$

Gate-partitioned complex time

$$d\zeta^D = [G^D + i(1 - G^D)]|d\tilde{\Phi}^D| \,/\, \Omega^D. \quad (2.41)$$

Trace-conditioned update

$$L^D{}_{,k+1} = \text{Update}(L^D{}_{,k}, \text{Trace}^D{}_{,k}, \text{Residual}^D{}_{,k}). \quad (2.42)$$

Where:

$$D \in \{\text{Physics, Finance, LLM}\}. \quad (2.43)$$

The symbols do not have identical meanings across domains.

They identify corresponding roles:

Symbol	General role
Z^D	domain state with admitted and residual structure
R^D	currently admitted component
Q^D	dynamically active unresolved component
ρ^D	collective coherence
Φ^D	dominant phase orientation

Symbol	General role
G^D	commitment gate
τ^D	ledgered consequential time
T^D	unresolved phase-development depth
L^D	accumulated trace ledger

The rest of the article will derive these roles first in finance and then translate them into LLM engineering.

2.8 First major conclusion

The first two sections establish a limited but powerful result.

Complex numbers need not be interpreted only as mysterious mathematical machinery imported from microscopic physics.

They can also be interpreted as a disciplined representation of a recurring observer problem:

> A bounded observer must distinguish between what has been admitted into visible reality and what remains dynamically consequential outside that admission.

The fundamental split is:

State = Admitted Projection + Retained Residual. (2.44)

When phase, gate, trace, and feedback are added:

Complex State → Relative Phase → Projection → Gate → Trace + Residual → Recursive Time. (2.45)

This does not prove that markets or LLMs are quantum systems.

It establishes the conceptual and mathematical kernel required to investigate why quantum-like grammar may reappear in them.

The next stage begins with the most unexpectedly suitable bridge: CAPM, a familiar financial model that already contains projection, relative orientation, observer dependence, and an unexploited route toward complex residual geometry.

3. CAPM Is Already a Projection Model

3.1 Why CAPM is the natural bridge

The Capital Asset Pricing Model is usually introduced as an expected-return equation:

$E[r_i] = r^f + \beta_i(E[r_m] - r^f)$. (3.1)

Where:

$E[r_i]$ = expected return of asset i;

r^f = risk-free rate;

$E[r_m]$ = expected market return;

β_i = sensitivity of asset i to the market.

In this familiar form, CAPM appears to be a scalar pricing rule. It states that the expected excess return of an asset is proportional to the expected excess return of the market.

Define excess returns:

$x_i = r_i - r^f$. (3.2)

$x_m = r_m - r^f$. (3.3)

Then:

$E[x_i] = \beta_i E[x_m]$. (3.4)

The deeper geometric content becomes visible when beta is written as:

$\beta_i = Cov(x_i,x_m) / Var(x_m)$. (3.5)

If x_i and x_m are treated as centered vectors of observed returns over a declared time window, then:

$Cov(x_i,x_m) \propto \langle x_i,x_m \rangle$. (3.6)

$Var(x_m) \propto \langle x_m,x_m \rangle$. (3.7)

Therefore:

$\beta_i = \langle x_i,x_m \rangle / \langle x_m,x_m \rangle$. (3.8)

This is the coefficient obtained when the asset-return vector is projected onto the market-return vector.

CAPM is therefore already an observer-dependent projection model.

It does not describe the entire asset.

It describes the part of the asset that is visible along one declared market direction.

3.2 The market direction as a measurement axis

Let:

$\hat{e}_m = x_m / \|x_m\|$. (3.9)

This is the normalized market direction.

The component of asset i along the market direction is:

$x_{i,\parallel} = \langle x_i, \hat{e}_m \rangle \hat{e}_m$. (3.10)

The orthogonal component is:

$x_{i,\perp} = x_i - x_{i,\parallel}$. (3.11)

Therefore:

$x_i = x_{i,\parallel} + x_{i,\perp}$. (3.12)

The parallel component is the part explained by the market factor.

The orthogonal component is the part not explained by that factor.

Using the angle θ_i between x_i and x_m:

$\cos\theta_i = \langle x_i, x_m \rangle / (\|x_i\| \|x_m\|)$. (3.13)

Since:

$\sigma_i \propto \|x_i\|$. (3.14)

$\sigma_m \propto \|x_m\|$. (3.15)

We obtain:

$\beta_i = (\sigma_i / \sigma_m) \cos\theta_i$. (3.16)

This equation separates two distinct properties:

relative scale, σ_i / σ_m;

relative orientation, $\cos\theta_i$.

Beta is therefore not merely “how risky the asset is.”

It combines:

Asset Magnitude × Market Alignment. (3.17)

An asset can have high total variability but low market alignment.

Another asset can have moderate variability but strong alignment with the market.

The scalar beta compresses both effects into one coefficient.

3.3 CAPM as a bounded observation protocol

CAPM does not operate without a declared frame.

The observer must choose:

which market portfolio counts as the market;

which currency is used;

which risk-free rate is used;

which data frequency is used;

which time window is used;

whether returns are arithmetic or logarithmic;

whether beta is static or conditional;

whether the system includes one factor or several.

Let the financial observation protocol be:

$P^f = (B^f, \Delta^f, h^f, u^f)$. (3.18)

Where:

B^f = declared asset and market boundary;

Δ^f = sampling and aggregation rule;

h^f = estimation horizon;

u^f = admissible intervention or trading family.

Then beta should be written more precisely as:

$\beta_i,P = Cov_p(x_i,x_m) / Var_p(x_m)$. (3.19)

The angle also becomes protocol-relative:

$\theta_i,P = \arccos[Cov_p(x_i,x_m) / (\sigma_i,P\sigma_m,P)]$. (3.20)

This protocol dependence is not a defect.

It is a declaration of the observer frame.

A monthly beta relative to a global equity index is not the same observable as an intraday beta relative to a domestic sector index.

Different protocols disclose different structures.

The Gauge Grammar makes the same methodological point in broader form: no claim about a market is stable until the boundary, observation rule, horizon, and admissible intervention family have been declared. It also warns that quantum-style language should remain functional rather than literal.

3.4 The orthogonal component is not nothing

Traditional CAPM often writes:

$x_i = \alpha_i + \beta_i x_m + \varepsilon_i$. (3.21)

Where:

α_i = intercept or abnormal component;

ε_i = residual disturbance.

The conventional interpretation frequently treats ε_i as noise.

But this interpretation may be too aggressive.

The residual may contain:

sector-specific information;

liquidity effects;

credit risk;

event risk;

option convexity;

regulatory exposure;

funding constraints;

hidden leverage;

model misspecification;

structural change;

unobserved factors.

Therefore:

Residual ≠ Pure Noise. (3.22)

A more careful decomposition is:

$x_i = x_{i,market} + x_{i,residual}$. (3.23)

Where:

$x_{i,market} = \beta_i x_m$. (3.24)

And:

$x_{i,residual} = \alpha_i + \varepsilon_i$. (3.25)

The residual becomes especially important when the market regime changes.

A component that was previously orthogonal may become dominant.

For example:

liquidity may be secondary during normal conditions but decisive during crisis;

credit risk may appear negligible until refinancing fails;

option convexity may remain hidden until volatility changes;

geopolitical exposure may remain dormant until a shock occurs.

This gives the first financial expression of the Complex Residual Principle:

CAPM Projection ⇒ Market-Admitted Component + Retained Residual. (3.26)

3.5 Rotating the asset or rotating the measurement basis

There are two mathematically related but conceptually different ways to describe a change in θ.

Active interpretation

The asset state rotates relative to a fixed market basis.

$Z_i' = \exp(i\alpha)Z_i$. (3.27)

Passive interpretation

The measurement basis rotates while the asset state remains fixed.

$R_i' = R_i \cos\alpha + Q_i \sin\alpha$. (3.28)

$Q_i' = -R_i \sin\alpha + Q_i \cos\alpha$. (3.29)

The passive interpretation is often more appropriate in finance.

An asset does not need to change physically for its measured risk to change.

The observer may change:

market benchmark;

valuation horizon;

discount rate;

regulatory treatment;

factor model;

reporting basis.

A technology company may appear highly market-aligned under a growth-equity frame but highly rate-sensitive under a duration frame.

A bank may appear stable under a price-volatility frame but fragile under a funding-liquidity frame.

Thus:

Measurement Rotation Can Reclassify Risk without Changing the Underlying Asset. (3.30)

This closely resembles the general principle that observation depends on the chosen basis.

It does not reproduce quantum measurement in full.

But it demonstrates that basis dependence is already natural in ordinary financial modelling.

3.6 From scalar beta to operator notation

Let the market projection operator be:

$\hat{P}_m = |\hat{e}_m\rangle\langle\hat{e}_m|$. (3.31)

Then the market-aligned component of asset i is:

$|x_i,\parallel\rangle = \hat{P}_m|x_i\rangle$. (3.32)

The residual projector is:

$\hat{Q}_m = I - \hat{P}_m$. (3.33)

Then:

$|x_i,\perp\rangle = \hat{Q}_m|x_i\rangle$. (3.34)

And:

$\hat{P}_m + \hat{Q}_m = I$. (3.35)

This gives:

$|x_i\rangle = \hat{P}_m|x_i\rangle + \hat{Q}_m|x_i\rangle$. (3.36)

The notation is elementary linear algebra.

Its importance is conceptual.

The asset is not reduced to the market projection.

The market projection and residual projection together reconstruct the declared state.

The loss occurs only when the observer keeps:

$\hat{P}_m|x_i\rangle$. (3.37)

And discards:

$\hat{Q}_m|x_i\rangle$. (3.38)

This is the financial form of false closure.

3.7 Alpha as structured residual

Alpha is often interpreted as excess performance beyond CAPM.

But in the present framework, alpha may have several meanings:

genuine skill;

omitted factor exposure;

temporary mispricing;

regime-specific residual;

measurement error;

compensation for hidden constraints.

Therefore:

$\alpha_i = \text{StructuredResidual}_i + \text{EstimationError}_i$. (3.39)

This suggests that alpha should not automatically be celebrated as unexplained value.

It should be decomposed.

A residual-aware asset model asks:

What part of alpha is stable across frames?

What part disappears under another factor basis?

What part is caused by illiquidity?

What part is a temporary selection effect?

What part survives out-of-sample testing?

What part becomes trace-confirmed through realized cash flow?

The mature question is not:

Does the asset have alpha?

It is:

What residual structure remains after the current projection, and which parts survive admissible frame changes? (3.40)

3.8 First financial conclusion

CAPM already contains four ingredients required by the broader framework:

a declared observation direction;

a projection coefficient;

an orthogonal residual;

observer dependence through benchmark and horizon.

The minimal CAPM decomposition is:

Asset State = Market Projection + Orthogonal Residual. (3.41)

This is not yet a complex theory.

But it supplies the exact geometry required to construct one.

---

4. From CAPM Projection to a Complex Asset State

4.1 The complex representation

Define the effective financial state of asset i as:

$Z_i^f = R_i^f + iQ_i^f$. (4.1)

In polar form:

$Z_i^f = A_i^f \exp(i\theta_i^f)$. (4.2)

Therefore:

$R_i^f = A_i^f \cos\theta_i^f$. (4.3)

$Q_i^f = A_i^f \sin\theta_i^f$. (4.4)

$A_i^{f2} = R_i^{f2} + Q_i^{f2}$. (4.5)

The superscript f indicates the financial domain.

The interpretation depends on the declared protocol.

R_i^f may represent:

admitted value;

market-aligned expected return;

recognized accounting value;

executable price;

present value under a selected discount rule;

currently observable risk contribution.

Q_i^f represents structure not captured by that admitted coordinate.

It is not necessarily one specific form of risk.

It is the total residual under the chosen frame.

4.2 The financial meaning of i

The imaginary unit i should not be interpreted as “imaginary money.”

Its role is geometric.

It marks a direction that:

is orthogonal to the admitted axis;

affects total state magnitude;

may rotate into visibility under another frame;

can influence future dynamics;

cannot be recovered from R alone.

Therefore:

iQ^f = Retained Financial Structure outside the Current Admission Axis. (4.6)

This interpretation avoids two errors.

Error 1 — Treating Q as fictional

Liquidity pressure, hidden leverage, and option convexity are not fictional.

They may be difficult to observe, but they can dominate realized outcomes.

Error 2 — Treating Q as automatically negative

Residual does not always mean danger.

Q may include:

upside option value;

strategic flexibility;

unused borrowing capacity;

latent demand;

positive convexity;

unrecognized intellectual property;

diversification value.

The residual coordinate may hold both opportunity and threat.

4.3 One Q is not enough

A realistic financial residual is multidimensional.

Define:

$\mathbf{Q}_i^f = [Qmarket, Qliquidity, Qcredit, Qoption, Qrate, Qfunding, Qtail, Qtax, Qmodel]^T$. (4.7)

Then:

$Z_i^f = R_i^f + i\mathbf{Q}_i^f$. (4.8)

This is a generalized complex-residual notation rather than an ordinary scalar complex number.

Define the total financial magnitude:

$A_i^{f2} = R_i^{f2} + \mathbf{Q}_i^{fT} G_i^f \mathbf{Q}_i^f$. (4.9)

Where G_i^f is the residual metric.

If residual dimensions are treated as orthogonal:

$G_i^f = I$. (4.10)

Then:

$A_i^{f2} = R_i^{f2} + \Sigma_j Q_{ij}^2$. (4.11)

But financial risks are rarely independent.

A more realistic metric contains interactions:

$$G_i^f = \begin{bmatrix} G_{11} & G_{12} & \dots & G_{1n} \\ G_{21} & G_{22} & \dots & G_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ G_{n1} & G_{n2} & \dots & G_{nn} \end{bmatrix}. \quad (4.12)$$

Then:

$\mathbf{Q}^T G \mathbf{Q} = \Sigma_j G_{jj} Q_j^2 + 2\Sigma_{j<k} G_{jk} Q_j Q_k$. (4.13)

Examples:

liquidity stress can amplify credit loss;

rate changes can amplify duration and funding pressure;

volatility can amplify option hedging flows;

leverage can amplify tail risk;

regulatory constraints can convert market risk into forced selling.

Thus the geometry itself changes under stress.

4.4 Dynamic phase

The asset angle should not be treated as static.

Let:

$\theta_i^f = \theta_i^f(t;P^f)$. (4.14)

The angle can change because:

the asset state changes;

the market state changes;

the observer frame changes;

the protocol changes;

the horizon changes.

A dynamic equation may be written schematically as:

$d\theta_i^f/dt = \omega_i + K_{im} \sin(\Phi_m - \theta_i^f) + \xi_i(t)$. (4.15)

Where:

ω_i = intrinsic asset phase velocity;

K_{im} = coupling to the market phase;

Φ_m = collective market phase;

$\xi_i(t)$ = idiosyncratic perturbation.

This has the form of a synchronization model.

It says that the asset has its own tendency, but market coupling pulls it toward the collective phase.

When K_{im} is weak:

θ_i evolves largely independently. (4.16)

When K_{im} is strong:

$\theta_i \to \Phi_m$. (4.17)

This is a natural model of regime formation.

It does not require quantum mechanics.

It requires interacting oscillatory or directional states.

4.5 Market alignment and residual pressure

As θ_i changes:

$R_i^f = A_i^f \cos\theta_i^f$. (4.18)

$Q_i^f = A_i^f \sin\theta_i^f$. (4.19)

If $\theta_i \approx 0$:

$R_i^f \approx A_i^f$. (4.20)

$Q_i^f \approx 0$. (4.21)

The asset is highly aligned with the selected market projection.

If $\theta_i \approx \pi/2$:

$R_i^f \approx 0$. (4.22)

$Q_i^f \approx A_i^f$. (4.23)

The asset is almost entirely orthogonal to the current market axis.

If $\theta_i \approx \pi$:

$R_i^f \approx -A_i^f$. (4.24)

The asset is anti-aligned.

These cases can represent:

high positive beta;

low beta;

market neutrality;

defensive or negative-beta behaviour;

factor rotation.

The angle therefore gives a richer interpretation than beta alone.

4.6 Discounting as selection depth

Suppose the base discount factor over horizon h is:

$Dbase(h) = [1 / (1 + rbase)]^h$. (4.25)

Suppose the CAPM-adjusted discount factor is:

$DCAPM(h) = [1 / (1 + rCAPM)]^h$. (4.26)

The ratio is:

$$DCAPM(h) / Dbase(h) = [(1 + rbase) / (1 + rCAPM)]^h. \quad (4.27)$$

Define:

$$\kappa = \ln[(1 + rCAPM) / (1 + rbase)]. \quad (4.28)$$

Then:

$$DCAPM(h) / Dbase(h) = \exp(-\kappa h). \quad (4.29)$$

Define the accumulated financial selection depth:

$$\sigma^f = \kappa h. \quad (4.30)$$

Therefore:

$$DCAPM / Dbase = \exp(-\sigma^f). \quad (4.31)$$

If we associate the admitted ratio with a cosine:

$$\cos\theta^f = \exp(-\sigma^f). \quad (4.32)$$

Then:

$$\theta^f = \arccos[\exp(-\sigma^f)]. \quad (4.33)$$

And:

$$\sin\theta^f = \sqrt{[1 - \exp(-2\sigma^f)]}. \quad (4.34)$$

Therefore:

$$R^f / A^f = \exp(-\sigma^f). \quad (4.35)$$

$$Q^f / A^f = \sqrt{[1 - \exp(-2\sigma^f)]}. \quad (4.36)$$

This creates a direct bridge between:

discounting;

exponential suppression;

angular geometry;

residual accumulation.

The coordinate σ^f is additive under constant κ:

$$\sigma^f(h_1 + h_2) = \sigma^f(h_1) + \sigma^f(h_2). \quad (4.37)$$

But θ^f is bounded:

$$0 \leq \theta^f < \pi/2. \quad (4.38)$$

Therefore σ^f is a better candidate for an imaginary-time-like selection depth than θ^f itself.

The angle visualizes the rotation.

The depth records the accumulated suppression.

4.7 Realized value and retained pressure

Under this construction:

$R^f = A^f \exp(-\sigma^f)$. (4.39)

$Q^f = A^f\sqrt{[1 - \exp(-2\sigma^f)]}$. (4.40)

At $\sigma^f = 0$:

$R^f = A^f$. (4.41)

$Q^f = 0$. (4.42)

As σ^f increases:

R^f decreases. (4.43)

Q^f increases. (4.44)

As $\sigma^f \to \infty$:

$R^f \to 0$. (4.45)

$Q^f \to A^f$. (4.46)

This should not be interpreted as literal conservation of money between R and Q.

It is a normalized geometric representation.

It says that increasing filtering depth moves more of the total declared state outside the admitted real-axis component.

The unresolved part remains represented rather than erased.

4.8 The financial state as a measurement result

A market price is often treated as if it were the asset itself.

The Complex Residual Principle rejects that identification.

Let:

$\text{Price}_i(t) = \hat{O}\text{price}[Z_i^f(t)]$. (4.47)

The price is one projection.

Other projections include:

$\text{CreditSpread}_i(t) = \hat{O}\text{credit}[Z_i^f(t)]$. (4.48)

$OptionSkew_i(t) = \hat{O}option[Z_i^f(t)]$. (4.49)

$Liquidity_i(t) = \hat{O}liquidity[Z_i^f(t)]$. (4.50)

$AccountingValue_i(t) = \hat{O}accounting[Z_i^f(t)]$. (4.51)

The same underlying state can yield different observables.

None should automatically be treated as total reality.

The asset is the latent financial state.

The observable is a projection under a declared measurement protocol.

4.9 A hierarchy of financial closure

The framework distinguishes four closure levels.

Level 1 — Quote closure

A price is displayed.

Level 2 — Transaction closure

Capital commits at the price.

Level 3 — Settlement closure

Legal and accounting transfer completes.

Level 4 — Economic closure

The value is validated through future cash flow, repayment, use, or liquidation.

These levels need not agree.

A displayed quote may not transact.

A transaction may fail to settle.

A settled asset may later default.

Therefore:

Visible Price ≠ Final Economic Closure. (4.52)

Each level produces a stronger trace.

This will become important when market time is defined through consequential events rather than through every displayed fluctuation.

---

5. The Market as a Collective Phase Field

5.1 Why the market cannot be represented by price alone

A market index compresses many asset states into one reported number.

But the same index level can occur under very different internal conditions:

broad participation or narrow leadership;

high liquidity or fragile liquidity;

low leverage or extreme leverage;

dispersed expectations or crowded consensus;

stable credit or deteriorating credit;

low volatility or suppressed volatility.

Therefore:

Market Index ≠ Full Market State. (5.1)

A richer market representation must preserve:

magnitude;

phase;

coherence;

dispersion;

residual pressure;

coupling structure.

Let each asset have a normalized phase state:

$u_i = \exp(i\theta_i)$. (5.2)

Define weights $w_i \geq 0$.

Let:

$W = \Sigma_i w_i$. (5.3)

Define the collective market order parameter:

$C_m = (1/W)\Sigma_i w_i \exp(i\theta_i)$. (5.4)

Write:

$C_m = \rho_m \exp(i\Phi_m)$. (5.5)

Where:

$\rho_m = |C_m|$. (5.6)

$\Phi_m = \arg(C_m)$. (5.7)

This is the central market-phase construction.

5.2 Interpretation of market coherence

The coherence parameter satisfies:

$0 \le \rho_m \le 1$. (5.8)

If asset phases are widely dispersed:

$\rho_m \approx 0$. (5.9)

If asset phases are strongly aligned:

$\rho_m \approx 1$. (5.10)

The market can therefore be classified.

Coherence	Interpretation
Low ρ_m	fragmented regime, weak common direction
Rising ρ_m	synchronization and regime formation
High ρ_m	broad collective phase-lock
Very high ρ_m with low diversity	crowding, fragility, saturation risk

The important point is that high coherence is not automatically healthy.

It may indicate:

strong coordination;

efficient consensus;

broad trend confirmation.

Or:

one-way positioning;

herding;

suppressed disagreement;

disappearance of liquidity on the other side.

The meaning depends on the accompanying residual structure.

5.3 Weighted phase and economic importance

Weights may represent:

market capitalization;

traded volume;

risk contribution;

liquidity;

economic output;

collateral importance;

network centrality.

Different weights define different market observers.

A capitalization-weighted market phase is:

$C_m,cap = [\Sigma_i Cap_i \exp(i\theta_i)] / \Sigma_i Cap_i$. (5.11)

A liquidity-weighted phase is:

$C_m,liq = [\Sigma_i Liq_i \exp(i\theta_i)] / \Sigma_i Liq_i$. (5.12)

A risk-weighted phase is:

$C_m,risk = [\Sigma_i RC_i \exp(i\theta_i)] / \Sigma_i RC_i$. (5.13)

These phases need not agree.

Their disagreement is itself informative.

Define cross-frame phase dispersion:

$Dframe = Var\{\Phi_m,cap,\Phi_m,liq,\Phi_m,risk,...\}$. (5.14)

High Dframe means that the market appears coherent under one observer but fragmented under another.

This is a financial form of frame dependence.

5.4 Constructive and destructive market interference

Consider two assets:

$Z_1 = A_1 \exp(i\theta_1)$. (5.15)

$Z_2 = A_2 \exp(i\theta_2)$. (5.16)

Their combined state is:

$Z_{12} = Z_1 + Z_2$. (5.17)

Then:

$|Z_{12}|^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\theta_1 - \theta_2)$. (5.18)

If:

$\theta_1 - \theta_2 \approx 0$. (5.19)

Then:

$\cos(\theta_1 - \theta_2) \approx 1$. (5.20)

The two states reinforce.

If:

$\theta_1 - \theta_2 \approx \pi$. (5.21)

Then:

$\cos(\theta_1 - \theta_2) \approx -1$. (5.22)

They cancel.

In finance, reinforcement may appear as:

synchronized sector movement;

common factor exposure;

shared funding pressure;

index rebalancing;

coordinated risk reduction.

Cancellation may appear as:

long-short hedging;

sector rotation;

defensive positioning;

offsetting currency exposure;

duration hedging.

This is classical phase interference, not proof of quantum interference.

But it demonstrates that relative phase can materially alter aggregate outcomes at the macroscopic level.

5.5 Trend as collective phase organization

A trend is often defined only by price direction.

A richer definition is:

Trend = Direction + Coherence + Persistence + Ledger Reinforcement. (5.23)

Let the collective phase velocity be:

$\Omega_m = d\tilde{\Phi}_m/dt$. (5.24)

Where $\tilde{\Phi}_m$ is the unwrapped phase.

A persistent trend requires:

$\text{sign}(\Omega_m)$ stable over a declared window. (5.25)

It also requires non-trivial coherence:

$\rho_m \geq \rho^*$. (5.26)

And persistence:

$\Pr[\text{sign}(\Omega_m, t+\Delta t) = \text{sign}(\Omega_m, t)] \geq p^*$. (5.27)

A stronger trend condition is:

$\text{Trend}_m = [\rho_m \geq \rho^*] \wedge [|\Omega_m| \geq \Omega^*] \wedge [\text{Persistence} \geq p^*]$. (5.28)

This separates:

random price drift;

narrow asset leadership;

broad phase-organized regime movement.

A long-term trend can therefore be interpreted as a stable collective phase trajectory.

5.6 The market clock hand

The factor:

$\exp(i\Phi_m)$. (5.29)

acts like a clock hand.

It indicates the current position in a collective market cycle.

But:

$\exp[i(\Phi_m + 2\pi)] = \exp(i\Phi_m)$. (5.30)

Therefore visible phase alone cannot distinguish one completed cycle from another.

Define:

$\tilde{\Phi}_m = \Phi_m + 2\pi N_m$. (5.31)

Where N_m counts completed phase revolutions.

Then:

Φ_m = visible circular phase. (5.32)

$\tilde{\Phi}_m$ = accumulated phase chronology. (5.33)

A primitive phase-time coordinate is:

$\tau_{phase,m} = [\tilde{\Phi}_m(t) - \tilde{\Phi}_m(to)] / \Omega_{ref}$. (5.34)

Where Ω_{ref} is a declared reference phase velocity.

This resembles how physical clocks infer elapsed duration from accumulated oscillatory phase.

But this is not yet full time.

It is only phase chronology.

5.7 Why phase chronology is not enough

Suppose the market phase rotates continuously but leaves no durable consequence.

Prices fluctuate.

Positions change briefly.

The system returns to its earlier configuration.

No important trace survives.

In that case, accumulated phase exists, but little historical time has been generated in the stronger sense.

This reveals a distinction:

Phase Order ≠ Consequential History. (5.35)

To generate time-like structure, the system must distinguish between:

movement that merely occurs;

movement that changes future admissibility.

The missing operation is the gate.

A market event becomes a time-producing event only when it enters a ledger and changes future conditions.

Therefore:

Market Time Requires Phase + Gate + Trace. (5.36)

5.8 Phase diversity and saturation

Define normalized phase bins with probabilities p_j.

The market phase entropy is:

$H_{phase} = -\Sigma_j p_j \ln p_j$. (5.37)

High Hphase indicates dispersed orientations.

Low Hphase indicates concentrated orientations.

The pair (ρ_m,Hphase) distinguishes several regimes.

ρ_m	Hphase	Interpretation
Low	High	dispersed, noisy market
Moderate	Moderate	forming regime
High	Moderate	coherent but differentiated trend
High	Very low	over-locked, crowded regime
Falling	Rising	decoherence or regime breakdown

This resolves an apparent contradiction.

Phase alignment is needed for collective organization.

But complete phase uniformity can eliminate useful difference.

A healthy collective regime may require:

Coherence without Total Homogeneity. (5.38)

This principle also clarifies the SMFT phase-lock discussion. The uploaded imaginary-time paper argues that unresolved phase evolution can continue while ordinary collapse ticks slow or stop in an over-saturated phase-lock regime.

For finance, the analogous claim is restrained:

Extreme synchronization may reduce price discovery and increase latent residual pressure. (5.39)

5.9 Collective market residual

The market itself also has an admitted and residual state.

Define:

$Z_m^f = R_m^f + iQ_m^f$. (5.40)

Or:

$Z_m^f = R_m^f + i\mathbf{Q}_m^f$. (5.41)

Where R_m^f may include:

index level;

admitted risk premium;

observed volatility;

recognized liquidity;

recorded credit spread.

The residual vector may include:

$\mathbf{Q}_m^f = [Qcrowding, Qleverage, Qliquidity, Qcredit, Qvolatility, Qpolicy, Qmodel, Qtail]^T$. (5.42)

The market can therefore appear stable on the real axis while residual pressure grows.

For example:

R_m stable, $\|\mathbf{Q}_m\|$ rising. (5.43)

This condition may describe:

volatility suppression;

hidden leverage growth;

narrowing breadth;

weakening liquidity;

increasing refinancing pressure;

growing policy dependence.

The Complex Residual Principle treats this as a distinct state rather than as “nothing happening.”

5.10 Market state update

Let the market ledger at event k be $L_{m,k}$.

The market projection produces a visible candidate:

$V_{m,k} = \hat{O}_{m,P}(Z_{m,k})$. (5.44)

A gate determines whether the candidate becomes consequential:

$C_{m,k} = Gate_{m,P}(V_{m,k}, \mathbf{Q}_{m,k})$. (5.45)

The ledger updates:

$L_{m,k+1} = \text{Update}(L_{m,k}, C_{m,k}, \text{Residual}_{m,k})$. (5.46)

The market state then changes:

$Z_{m,k+1} = F_m(Z_{m,k}, L_{m,k+1}, \text{Environment}_k)$. (5.47)

This creates the self-referential cycle:

Market State → Price/Signal → Ledger → Participant Response → New Market State. (5.48)

The market is therefore not merely measured.

It partly measures itself through its own prices and traces.

5.11 Second major conclusion

The collective market model now contains:

many interacting states;

relative phase;

interference;

synchronization;

an order parameter;

phase entropy;

observer-relative weighting;

residual pressure;

recursive backreaction.

The minimal market field is:

$C_m = \rho_m \exp(i\Phi_m)$. (5.49)

The complex market state is:

$Z_m^f = R_m^f + i\mathbf{Q}_m^f$. (5.50)

The unwrapped phase supplies a primitive chronology:

$\tilde{\Phi}_m = \Phi_m + 2\pi N_m$. (5.51)

But a true internal market time still requires one further step:

Some phase movements must become accepted events, while others remain unresolved.

That distinction will generate the market analogue of real time and imaginary time.

Financial scope note: The framework developed here is theoretical and experimental. It is not investment advice, a valuation recommendation, or a trading system.

6. The Clock Hand and the Timeline

6.1 Why the market phase resembles a clock

The collective market phase was defined as:

$C_m = \rho_m \exp(i\Phi_m)$. (6.1)

The factor $\exp(i\Phi_m)$ gives the current angular orientation of the market's collective state.

This resembles a clock hand because the same visible position can recur after one complete revolution:

$\exp[i(\Phi_m + 2\pi)] = \exp(i\Phi_m)$. (6.2)

The circular phase therefore records **where the market currently is within a cycle**, but not how many cycles have already occurred.

A clock face has the same limitation.

A hand pointing at twelve does not reveal whether one minute, one hour, or one day has passed. Duration appears only when the system also counts completed rotations.

Define the unwrapped market phase:

$\tilde{\Phi}_m = \Phi_m + 2\pi N_m$. (6.3)

Where:

Φ_m = visible circular phase;

N_m = number of completed collective phase cycles;

$\tilde{\Phi}_m$ = accumulated phase history.

The primitive market-phase time may then be written:

$\tau_{phase,m} = [\tilde{\Phi}_m(t) - \tilde{\Phi}_m(to)] / \Omega ref$. (6.4)

Where Ωref is a declared reference angular velocity.

This creates the first time-like characteristic:

Elapsed Internal Duration $\propto$ Accumulated Phase Change. (6.5)

The same principle appears in many physical clocks.

An atomic clock, oscillator, pendulum, or electromagnetic clock does not observe "time itself." It observes a repeatable phase process and counts accumulated cycles.

The market model therefore does not need to begin with a mysterious time coordinate. It can begin with an interacting phase field.

6.2 External time and internal market time

Calendar time t exists independently of the proposed market clock.

The market may be observed at:

seconds;

minutes;

trading days;

quarters;

years.

But equal intervals of calendar time need not contain equal amounts of meaningful market change.

A quiet month may contain little structural transformation.

A single crisis day may contain:

multiple liquidity failures;

rapid repricing;

margin calls;

defaults;

regulatory intervention;

forced portfolio restructuring.

Therefore:

$\Delta t_1 = \Delta t_2$ does not imply $\Delta \tau_{m,1} = \Delta \tau_{m,2}$. (6.6)

This distinction is familiar in other fields.

A biological organism can undergo rapid internal transformation during a short external interval.

A computer may perform many state transitions in one second and almost none in another.

A legal dispute may remain dormant for months and then change irreversibly in one hearing.

The market model therefore distinguishes:

t = external calendar or execution time. (6.7)

τ_m = internal consequential market time. (6.8)

The central problem is then:

What converts ordinary phase movement into consequential market time?

Phase change alone cannot answer this.

A phase may rotate without producing any durable change.

The missing ingredient is selective admission.

6.3 Why phase alone is insufficient

Suppose asset phases fluctuate continuously:

$$d\tilde{\Phi}_m \neq 0. \quad (6.9)$$

But suppose:

no major transaction occurs;

no risk limit changes;

no price level becomes institutionally recognized;

no balance-sheet effect is created;

no market participant revises future behaviour;

no trace survives beyond the immediate fluctuation.

Then the system has changed momentarily, but little durable market history has been created.

This gives:

$$\text{Phase Movement} \neq \text{Historical Commitment}. \quad (6.10)$$

A rotating system can possess rhythm without possessing an arrow of time.

A clock hand may return to the same position.

A pure cycle can repeat.

Historical time requires that some event becomes irreversible relative to the system's subsequent state.

Therefore:

$$\text{Time-Like Order Requires Phase} + \text{Selection} + \text{Trace}. \quad (6.11)$$

The role of selection is played by a gate.

The role of irreversibility is played by a ledger.

6.4 The market gate

Define a market gate:

$G_m \in [0,1]$. (6.12)

The gate measures the degree to which current market-phase movement becomes a consequential event.

A binary version is:

$G_m =$
{
1, event admitted,
0, event not admitted.
}. (6.13)

A continuous version allows partial commitment.

The gate may depend on several components:

G_m = Galign Gdiff Gpressure Gcommit Gtrace. (6.14)

Where:

Galign = collective phase coherence;

Gdiff = remaining distinguishability or usable gradient;

Gpressure = accumulated economic tension;

Gcommit = capital, liquidity, or institutional commitment;

Gtrace = durability of the resulting record.

Each component performs a separate role.

Alignment

The market must possess enough coherence for the event to represent more than isolated noise.

Difference

The market must still contain enough distinction for a new event to carry information.

Perfect homogeneity can eliminate price discovery.

Pressure

There must be sufficient economic force to cross a threshold.

Commitment

Participants must commit capital, collateral, inventory, or legal obligation.

Trace

The event must alter future market conditions.

The gate should therefore not be reduced to “price moved.”

A price movement may be visible but historically weak.

6.5 Examples of market gates

Different protocols may define different gate events.

Transaction gate

A quote becomes an executed trade.

Closing gate

An intraday movement survives to the official close.

Settlement gate

A trade becomes legally and financially completed.

Breadth gate

A move spreads across a sufficiently large part of the market.

Liquidity gate

The move occurs with enough traded depth to represent commitment.

Credit gate

A risk perception becomes visible in borrowing terms or default pricing.

Margin gate

A price change becomes large enough to trigger collateral transfer or forced liquidation.

Policy gate

A central-bank or regulatory action changes admissible market behaviour.

Accounting gate

An economic condition becomes recognized in financial statements.

Regime gate

A transient pattern becomes a statistically persistent state.

Each gate defines a different kind of market tick.

The framework therefore permits multiple clocks:

$\tau_{trade}, \tau_{settlement}, \tau_{credit}, \tau_{policy}, \tau_{regime}$. (6.15)

These clocks need not advance at the same rate.

6.6 Realized market time

Define the realized market-time increment:

$d\tau_m = G_m |d\tilde{\Phi}_m| / \Omega^*$. (6.16)

Where:

$|d\tilde{\Phi}_m|$ = magnitude of collective phase movement;

G_m = degree of admission;

Ω^* = normalization scale.

When:

$G_m \approx 1$. (6.17)

Then:

$d\tau_m \approx |d\tilde{\Phi}_m| / \Omega^*$. (6.18)

Most phase movement becomes realized market history.

When:

$G_m \approx 0$. (6.19)

Then:

$d\tau_m \approx 0$. (6.20)

The system may continue moving internally, but little consequential market time is generated.

This means that market time is not merely accumulated volatility.

High volatility can occur without durable regime formation.

Likewise, low volatility can coexist with growing hidden pressure.

The gate determines whether movement becomes history.

6.7 Imaginary-time-like market depth

Phase movement that does not pass the gate should not automatically be treated as nonexistent.

Define:

$$dT_m = (1 - G_m)|d\tilde{\Phi}_m| / \Omega^*. \quad (6.21)$$

Where T_m is unresolved market depth.

Then:

$$d\tau_m + dT_m = |d\tilde{\Phi}_m| / \Omega^*. \quad (6.22)$$

The two coordinates partition total phase development.

When the gate is open:

$d\tau_m$ dominates. (6.23)

When the gate is closed:

dT_m dominates. (6.24)

The complex market-time coordinate is:

$$d\zeta_m = d\tau_m + i\, dT_m. \quad (6.25)$$

Therefore:

$$d\zeta_m = [G_m + i(1 - G_m)]|d\tilde{\Phi}_m| / \Omega^*. \quad (6.26)$$

This equation is the central construction of the financial time model.

The real-time-like component records phase change that becomes consequential and ledgered.

The imaginary-time-like component records phase change that remains unresolved, uncommitted, or unable to generate a new market tick.

6.8 What T_m may contain

T_m should not be interpreted as one hidden physical substance.

It is an operational depth variable.

Its measurable proxies may include:

suppressed volatility;

narrowing market breadth;

rising leverage;

worsening liquidity;

unresolved valuation disagreement;

unexecuted order imbalance;

increasing option skew;

growing refinancing pressure;

unstable correlations;

repeated failed breakouts;

policy dependence;

crowded one-way positioning.

A possible composite measure is:

$T_m = a_1 T_{vol} + a_2 T_{liq} + a_3 T_{lev} + a_4 T_{breadth} + a_5 T_{credit} + a_6 T_{position}$. (6.27)

Where the coefficients a_j are protocol-specific.

The important structural claim is:

> The market may continue accumulating consequential pressure even while little appears on the admitted price axis.

This is the financial analogue of unresolved phase evolution.

The attached SMFT imaginary-time model similarly distinguishes realized collapse ticks from continuous unresolved phase rotation. In that framework, semantic time does not advance when collapse fails, but unresolved phase memory continues to accumulate.

6.9 A simple example

Suppose the market has a steadily changing collective phase:

$|d\tilde{\Phi}_m| / \Omega^* = 1$. (6.28)

Regime A — Open gate

Let:

$G_m = 0.9$. (6.29)

Then:

$d\tau_m = 0.9$. (6.30)

$dT_m = 0.1$. (6.31)

Most movement becomes realized market history.

This may represent:

broad trend confirmation;

strong transaction volume;

persistent repricing;

institutional acceptance.

Regime B — Closed gate

Let:

$G_m = 0.1$. (6.32)

Then:

$d\tau_m = 0.1$. (6.33)

$dT_m = 0.9$. (6.34)

Most movement remains unresolved.

This may represent:

unstable internal rotation;

unresolved disagreement;

compressed volatility;

weak commitment;

crowded but fragile positioning.

The same quantity of phase movement can therefore produce very different temporal outcomes.

6.10 The first time-like asymmetry

The gate introduces an asymmetry:

Admitted Phase → Trace. (6.35)

Unadmitted Phase → Residual Depth. (6.36)

A pure oscillator treats all rotations symmetrically.

The gate breaks this symmetry.

Some phase movements become historically privileged.

This is the first step toward an arrow of time.

But the arrow remains incomplete until the admitted event is preserved and allowed to alter future

dynamics.

That requires a ledger.

---

7. Gate-Partitioned Complex Market Time

7.1 Complex time as a bookkeeping geometry

The proposed complex market time is:

$\zeta_m = \tau_m + iT_m$. (7.1)

This is not a claim that finance possesses literal physical imaginary time.

It is a bookkeeping geometry that separates two modes of evolution:

realized, trace-producing evolution;

unresolved, non-trace-producing evolution.

The distinction is useful because ordinary calendar time mixes them.

Two markets may exist for the same number of days but possess very different values of τ_m and T_m.

A mature market may have:

high τ_m from frequent consequential settlement;

low T_m because unresolved pressure is quickly disclosed.

A fragile market may have:

low τ_m because apparent stability suppresses new information;

high T_m because leverage and disagreement accumulate out of view.

Therefore:

Calendar Duration ≠ Market Development Depth. (7.2)

7.2 Dynamic equations

Let t be calendar time.

The realized-time rate is:

$d\tau_m/dt = G_m|d\tilde{\Phi}_m/dt| / \Omega^*$. (7.3)

The unresolved-depth rate is:

$dT_m/dt = (1 - G_m)|d\tilde{\Phi}_m/dt| / \Omega^*$. (7.4)

The complex-time rate is:

$d\zeta_m/dt = [G_m + i(1 - G_m)]|d\tilde{\Phi}_m/dt| / \Omega^*$. (7.5)

Integrating from to to t:

$\tau_m(t) = \int_t 0^t G_m(s)|\dot{\tilde{\Phi}}_m(s)| / \Omega^* ds$. (7.6)

$T_m(t) = \int_t 0^t [1 - G_m(s)]|\dot{\tilde{\Phi}}_m(s)| / \Omega^* ds$. (7.7)

$\zeta_m(t) = \tau_m(t) + iT_m(t)$. (7.8)

These equations separate:

how rapidly the field changes;

how much of the change becomes consequential.

7.3 Gate dependence on state

The gate itself should be dynamic.

A possible logistic gate is:

$G_m = 1 / \{1 + \exp[-S_m]\}$. (7.9)

Where:

$S_m = a\rho_m + bD_m + cP_m + dC_m - eF_m$. (7.10)

Where:

ρ_m = coherence;

D_m = distinguishability;

P_m = pressure above threshold;

C_m = capital or commitment;

F_m = fragility penalty.

The gate rises when coherence, usable difference, pressure, and commitment become sufficient.

It falls when fragility or saturation dominates.

A threshold form is:

$G_m = H(S_m - S^*)$. (7.11)

Where H is the Heaviside step function.

The continuous gate is usually more realistic because financial commitment is rarely perfectly binary.

7.4 Why extreme phase-lock can reduce the gate

At first glance, high coherence should increase G_m.

But extreme coherence may reduce distinguishability.

Let phase diversity be $D\theta$.

A simple difference term is:

$D_m = D\theta \exp(-\lambda \rho_m)$. (7.12)

When coherence is low, the market lacks organization.

When coherence is moderate, collective structure appears.

When coherence becomes extreme, diversity may collapse.

Therefore G_m may be highest at an intermediate region:

$G_m = f(\rho_m)$ with $f'(\rho_m) > 0$ at low coherence and $f'(\rho_m) < 0$ near saturation. (7.13)

A simple example is:

$G_m = 4\rho_m(1 - \rho_m)$. (7.14)

This function satisfies:

$G_m = 0$ at $\rho_m = 0$. (7.15)

$G_m = 1$ at $\rho_m = 0.5$. (7.16)

$G_m = 0$ at $\rho_m = 1$. (7.17)

The exact functional form is not asserted as universal.

Its purpose is conceptual:

no coherence produces no collective event;

moderate coherence produces meaningful selection;

total lock can eliminate novelty and price discovery.

This resolves the apparent contradiction between:

phase alignment enabling collapse;

over-alignment suppressing new collapse.

The attached SMFT paper uses a related idea: ordinary collapse ticks require phase alignment, but extreme phase saturation can eliminate interpretive gradients and halt further meaningful ticks.

7.5 Stored unresolved depth and sudden release

Suppose unresolved market depth accumulates:

$T_m(t) \uparrow$. (7.18)

The accumulated depth may remain latent until a trigger opens the gate.

Let the release event at t^* satisfy:

$G_m(t^*) \to 1$. (7.19)

A release function may be:

$\Delta\tau_m(t^*) = g[T_m(t^*-), \text{Shock}(t^*), \text{Liquidity}(t^*)]$. (7.20)

Where $T_m(t^*-)$ is the unresolved depth immediately before release.

A simple proportional model is:

$\Delta\tau_m = \eta T_m(t^*-)$. (7.21)

Where $0 \le \eta \le 1$.

Then the residual depth updates:

$T_m(t^*+) = [1 - \eta]T_m(t^*-)$. (7.22)

This resembles stored pressure being converted into a realized market event.

Possible financial manifestations include:

breakout after volatility compression;

crash after leverage accumulation;

yield jump after policy uncertainty;

spread widening after prolonged credit denial;

liquidity gap after one-sided positioning;

currency devaluation after defended imbalance.

The interpretation is not that “imaginary time causes the crash.”

It is that the T_m coordinate records unresolved development that may later be released through a gate.

7.6 Market-time dilation

Define the market-time rate relative to calendar time:

$\gamma_m = d\tau_m/dt$. (7.23)

If:

$\gamma_m \gg 1$. (7.24)

The market is generating many consequential transitions per unit calendar time.

This may occur during:

crisis;

policy shock;

default cascade;

rapid regime transition.

If:

$\gamma_m \ll 1$. (7.25)

The market generates few consequential transitions.

This may occur during:

stagnation;

artificial price stabilization;

market closure;

extreme crowding;

low-information repetition.

The phrase “time dilation” should be used carefully.

The model does not reproduce relativistic time dilation.

It illustrates the more general characteristic:

Internal Tick Rate Varies with State Geometry. (7.26)

The market’s effective time can accelerate or slow relative to the external calendar.

7.7 Frozen market time

A frozen-time-like regime occurs when:

$d\tau_m/dt \to 0$. (7.27)

While:

$dT_m/dt > 0$. (7.28)

The market continues evolving internally, but the ledgered state barely advances.

Possible examples include:

prolonged trading suspension;

price limits preventing clearing;

no-bid conditions;

fixed exchange-rate defence;

suppressed volatility under heavy intervention;

illiquid assets with stale prices;

one-way consensus with no counterparties.

The system is not static.

Its unresolved structure may continue changing.

Therefore:

Displayed Stability ≠ Internal Stillness. (7.29)

This is one of the most important practical implications of the model.

7.8 Multiple market clocks

Different subsystems may possess different gate structures.

For asset i:

$d\tau_i/dt = G_i|d\tilde{\theta}_i/dt| / \Omega_i$. (7.30)

For the market:

$d\tau_m/dt = G_m|d\tilde{\Phi}_m/dt| / \Omega_m$. (7.31)

Define the asset-to-market clock ratio:

$\chi_i = d\tau_i/d\tau_m$. (7.32)

Interpretation:

$\chi_i > 1$: asset evolves faster than the market;

$\chi_i \approx 1$: asset evolves with the market;

$\chi_i < 1$: asset evolves more slowly;

$\chi_i < 0$: anti-aligned evolution under a signed convention.

High-beta or highly leveraged assets may exhibit larger clock rates.

Illiquid assets may remain frozen while the broader market advances.

This resembles an observer-relative or proper-time-like structure only at the level of grammar.

No relativistic metric has yet been derived.

7.9 CAPM sensitivity as clock coupling

The CAPM beta can be reinterpreted as one component of phase-rate coupling.

A simple model is:

$d\theta_i/d\tau_m = \beta_i\Omega_m + \nu_i$. (7.33)

Where:

Ω_m = market phase rate;

β_i = sensitivity to the market direction;

ν_i = idiosyncratic phase contribution.

Then:

$\beta_i \approx \partial\theta_i / \partial\Phi_m$ under fixed protocol. (7.34)

This does not replace the conventional statistical definition of beta.

It offers an additional dynamic interpretation:

Beta Measures How Strongly the Asset Clock Responds to the Market Clock. (7.35)

The relationship may vary by regime:

$\beta_i = \beta_i(\rho_m, T_m, \text{Liquidity}, \text{Policy})$. (7.36)

This naturally leads to conditional CAPM rather than static beta.

7.10 Complex time and discount depth

The earlier financial selection depth was:

$\sigma^f = \kappa h$. (7.37)

With:

$\cos\theta^f = \exp(-\sigma^f)$. (7.38)

This produces a second temporal structure.

The market-phase coordinate T_m represents unresolved collective evolution.

The CAPM depth σ^f represents accumulated financial filtering or suppression.

They should not be identified automatically.

A possible relation is:

$T_m = F(\sigma^f, \rho_m, G_m, \mathbf{Q}_m)$. (7.39)

The simplest linear approximation is:

$T_m \approx a\sigma^f + b\|\mathbf{Q}_m\| + c(1 - G_m)$. (7.40)

This makes CAPM one contributor to imaginary-time-like depth rather than the entire source of it.

CAPM supplies the projection and discount geometry.

Market interaction supplies the collective phase.

The gate supplies realized versus unresolved partition.

The ledger supplies the arrow of time.

---

8. From Phase to Ledgered Time

8.1 Why a trace is necessary

A gate selects an event.

But selection alone does not guarantee temporal persistence.

A market event becomes historical only if it leaves a trace that changes future states.

Define a trace:

$\text{Trace}_k = \{\text{Observable}_k, \text{Protocol}_k, \text{Commitment}_k, \text{Metadata}_k\}$. (8.1)

Examples include:

executed price;

volume;

timestamp;

counterparty obligation;

collateral change;

settlement record;

index close;

accounting recognition;

regulatory filing.

The ledger update is:

$L_{k+1} = L_k \sqcup \text{Trace}_k$. (8.2)

Where $\sqcup$ denotes governed addition to the historical record.

The next market state depends on the ledger:

$Z_{m,k+1} = F(Z_{m,k}, L_{k+1}, E_k)$. (8.3)

Where E_k is the external environment.

This creates causal asymmetry.

The past trace enters the future state.

The future trace does not enter the past state.

8.2 The arrow of financial time

Suppose a price rises and then returns to its previous level.

One might conclude that the market has returned to its earlier state.

But the ledger may now contain:

realized profits and losses;

changed collateral;

tax consequences;

altered inventory;

failed institutions;

revised regulation;

new technical reference levels;

changed investor expectations.

Therefore:

$$\text{Price}(t_2) = \text{Price}(t_0) \text{ does not imply } \text{State}(t_2) = \text{State}(t_0). \quad (8.4)$$

This is the financial arrow of time.

The displayed observable may return.

The ledger does not.

In complex-state form:

$$R_m(t_2) = R_m(t_0). \quad (8.5)$$

But:

$$L_m(t_2) \neq L_m(t_0). \quad (8.6)$$

And often:

$$Q_m(t_2) \neq Q_m(t_0). \quad (8.7)$$

Thus identical prices may conceal different histories and residual structures.

8.3 Cyclic phase and irreversible history

The phase may be cyclic:

$\Phi_m \rightarrow \Phi_m + 2\pi$. (8.8)

The ledger is cumulative:

$L_k \rightarrow L_{k+1}$. (8.9)

This produces a dual temporal structure:

cyclic time from phase;

historical time from trace.

The complete market time is therefore not only the unwrapped phase.

It is:

Market Time = Ordered Phase-Conditioned Ledger Updates. (8.10)

A more explicit definition is:

$\tau_m(k) = \Sigma_{j=1}^{k} g(\text{Trace}_j, \text{Residual}_j, \text{Protocol}_j)$. (8.11)

Where g measures the consequential weight of each admitted trace.

A minor trade may contribute little.

A default or policy shift may contribute much more.

Therefore market ticks may be weighted rather than uniform.

8.4 Weighted market ticks

Let $\Delta\tau_{m,k}$ be the market-time contribution of event k.

Define:

$\Delta\tau_{m,k} = G_{m,k} W_k$. (8.12)

Where W_k is event consequence weight.

Possible components of W_k include:

$W_k = a\Delta\text{Price} + b\Delta\text{Volume} + c\Delta\text{Collateral} + d\Delta\text{Policy} + e\Delta\text{Network}$. (8.13)

Where:

ΔPrice = repricing magnitude;

ΔVolume = capital commitment;

ΔCollateral = balance-sheet consequence;

ΔPolicy = rule change;

ΔNetwork = propagation through connected institutions.

Then:

$\tau_{m,n} = \Sigma_{k=1}^{n} \Delta\tau_{m,k}$. (8.14)

This definition allows one crisis event to represent more internal market time than thousands of ordinary trades.

8.5 Ledger depth and memory

Not all traces persist equally.

Define trace retention:

$\mu_k \in [0,1]$. (8.15)

The effective ledger is:

$L_m(t) = \Sigma_k \mu_k(t)\text{Trace}_k$. (8.16)

Where μ_k may decay or remain permanent.

Examples:

intraday noise may decay rapidly;

official close may persist longer;

default history may remain for years;

regulatory change may permanently alter the system.

The market therefore possesses layered memory.

A simple decay model is:

$\mu_k(t) = \exp[-\lambda_k(t - t_k)]$. (8.17)

A permanent trace has:

$\lambda_k = 0$. (8.18)

A short-lived trace has:

λ_k large. (8.19)

The system's effective memory influences future gate behaviour:

$G_m(t) = G_m[Z_m(t), L_m(t)]$. (8.20)

The ledger therefore participates in the observer itself.

8.6 Trace-conditioned projection

Suppose the market projection operator at event k is $\hat{O}_{m,k}$.

After new traces are written:

$\hat{O}_{m,k+1} = \text{Update}[\hat{O}_{m,k}, L_{m,k+1}]$. (8.21)

This means that the market's measurement frame changes because of its own history.

Examples:

volatility models update after crisis;

risk limits tighten after losses;

investors redefine "normal" valuation after a regime shift;

central banks alter reaction functions;

technical analysts treat old price levels as support or resistance;

credit committees change admissible exposure.

The observer is therefore not external and fixed.

It is trace-bearing.

This gives:

$\text{Observer}_{k+1} = \text{Observer}_k + \text{Historical Update}$. (8.22)

The same concept is central to self-referential observer models: observation produces a trace, and the trace modifies future projection.

8.7 Ledgered time and SMFT

The attached SMFT imaginary-time framework defines semantic time through realized collapse events and interprets unresolved phase motion as imaginary-time accumulation.

The present finance model adds an explicit ledger emphasis:

Phase Motion → Gate → Trace → Ledger → Updated Projection. (8.23)

The stronger definition of time is therefore:

Time = Ordered Consequence-Bearing Disclosure. (8.24)

This definition explains why:

phase provides rhythm;

gate provides selection;

trace provides irreversibility;

ledger provides ordered history;

feedback makes the clock endogenous.

8.8 Residual ledger

A mature ledger should not record only accepted outcomes.

It should also retain residual information.

Define:

$L_{k+1} = \text{Update}(L_k, \text{Trace}_k, \text{Residual}_k)$. (8.25)

Residual metadata may include:

failed transaction;

unfilled order;

rejected valuation;

unresolved model disagreement;

hidden liquidity estimate;

confidence interval;

stress-test result;

alternative scenario.

This prevents false closure.

A market system that records only executed prices may erase important near-events.

A residual-aware system records both:

What Happened and What Nearly Happened. (8.26)

This is especially important for crisis analysis.

8.9 Historical state reconstruction

Given only price history, two different market histories may appear identical.

The richer state is:

$S_m(t) = \{R_m(t), \mathbf{Q}_m(t), \Phi_m(t), \rho_m(t), L_m(t)\}$. (8.27)

A historical reconstruction should therefore estimate:

admitted observable;

residual pressure;

collective phase;

coherence;

ledger state.

This produces a more complete state-space model.

The market is not merely a scalar time series.

It is a trace-bearing complex field.

---

9. Self-Reference and the Endogenous Market Clock

9.1 External versus endogenous trend formation

A market trend may arise from external drivers:

productivity;

earnings;

monetary policy;

inflation;

demographics;

technological change;

regulation;

resource scarcity.

The simple external chain is:

External Driver → Valuation Change → Market Price. (9.1)

This can generate phase-lock and time-like structure.

But the market clock remains partly entrained by an external process.

A stronger endogenous loop is:

Expectation → Order → Price → Interpretation → Revised Expectation. (9.2)

Here the market's own output becomes part of its next input.

This is self-reference.

9.2 Corrective and amplifying loops

Two main feedback structures should be distinguished.

Corrective loop

Price rise → selling pressure → reduced price rise. (9.3)

Price fall → buying pressure → reduced price fall. (9.4)

This creates negative feedback.

It stabilizes the system.

Amplifying loop

Price rise → interpreted strength → additional buying → further price rise. (9.5)

Price fall → interpreted weakness → forced selling → further price fall. (9.6)

This creates positive feedback.

It amplifies the trend.

The direction of feedback determines whether the market behaves as:

mean-reverting system;

self-confirming trend;

unstable cascade;

regime-switching system.

9.3 Reflexive phase dynamics

Let the asset phase obey:

$$d\theta_i/dt = \omega_i + K_{im} \sin(\Phi_m - \theta_i) + \lambda_i F_i(L_m, R_m) + \xi_i. \quad (9.7)$$

Where:

ω_i = intrinsic phase tendency;

K_{im} = coupling to the market;

F_i = feedback from ledger and observable price;

λ_i = reflexivity strength;

ξ_i = noise.

The market phase is:

$C_m = (1/W)\Sigma_i w_i \exp(i\theta_i)$. (9.8)

Thus the market phase affects the assets, and the assets collectively create the market phase.

This gives:

Asset Phases → Market Phase → Asset Phases. (9.9)

The system is self-consistent and recursive.

9.4 Endogenous clock formation

A market clock becomes endogenous when:

asset interactions produce collective phase;

collective phase affects future asset behaviour;

gates select consequential events;

traces update future projection;

updated projection changes later phase coupling.

The full loop is:

$\Phi_{m,k} \rightarrow G_{m,k} \rightarrow \text{Trace}_k \rightarrow L_{k+1} \rightarrow \hat{O}_{k+1} \rightarrow \Phi_{m,k+1}$. (9.10)

The system then helps determine the rate at which its own meaningful time advances.

This is stronger than merely measuring an external process.

It is self-generated cadence.

9.5 The market as an internal observer network

There is no single market observer.

The market consists of interacting observers:

households;

funds;

banks;

dealers;

regulators;

exchanges;

algorithms;

rating agencies;

central banks.

Each observer has a protocol:

$P_j = (B_j,\Delta_j,h_j,u_j)$. (9.11)

Each observer projects the market differently:

$R_j = \hat{O}_j(Z_m)$. (9.12)

Each leaves a residual:

$Q_j = \text{Residual}_j(Z_m)$. (9.13)

Each may act:

$\text{Action}_j = \text{Gate}_j(R_j,Q_j,L_j)$. (9.14)

The aggregate market state is generated by these interacting projections.

Therefore:

Market Observation = Distributed Internal Measurement. (9.15)

The market is simultaneously:

observed by participants;

altered by their observations;

summarized by its own traces;

used as evidence for future observation.

This is the essence of reflexivity.

9.6 Self-reference and imaginary-time accumulation

Self-reference can either release or trap residual depth.

Release case

A price signal causes participants to act, creating a confirming transaction.

Then:

G_m rises. (9.16)

T_m converts into τ_m. (9.17)

Trap case

A price signal causes all participants to adopt the same view, but liquidity disappears.

Then:

ρ_m rises. (9.18)

Diversity falls. (9.19)

G_m may fall. (9.20)

T_m continues accumulating. (9.21)

Thus stronger consensus does not always generate more realized market time.

It may generate a saturated phase-lock regime.

This gives a non-monotonic relation between coherence and temporal progress.

9.7 False completion in markets

A market may appear complete because:

one price exists;

volatility is low;

consensus is strong;

the trend is persistent.

But the residual state may indicate fragility:

$\|\mathbf{Q}_m\|$ large. (9.22)

T_m large. (9.23)

Diversity low. (9.24)

Liquidity weak. (9.25)

This condition is:

Visible Completion + Hidden Residual = False Closure. (9.26)

The same structural problem will later appear in LLMs.

A fluent answer can look complete while factual or tool residual remains large.

9.8 Regime transition

Let the market have regime variable χ_m.

A transition occurs when:

χ_m crosses χ^*. (9.27)

The transition probability may depend on unresolved depth:

$\Pr(\text{RegimeChange} \mid T_m) = \sigma(a + bT_m)$. (9.28)

Where σ is the logistic function.

If $b > 0$, larger T_m increases transition probability.

The transition may be gradual or abrupt.

A sudden event occurs when accumulated residual passes a critical threshold:

$T_m \geq T^*$. (9.29)

And a perturbation opens the gate:

$\text{Shock} \geq \text{Shock}^*$. (9.30)

Then:

$\Delta\tau_m$ large. (9.31)

This provides a testable interpretation of latent instability.

9.9 A candidate law of endogenous market time

The full mechanism can be summarized as:

$d\tau_m/dt = G_m[\rho_m, D_m, \mathbf{Q}_m, L_m] \, |\dot{\tilde{\Phi}}_m| \, / \, \Omega^*$. (9.32)

$dT_m/dt = \{1 - G_m[\rho_m, D_m, \mathbf{Q}_m, L_m]\} \, |\dot{\tilde{\Phi}}_m| \, / \, \Omega^*$. (9.33)

$dL_m/dt = H[\text{Trace}_m, G_m]$. (9.34)

$d\hat{O}_m/dt = J[L_m, \text{Residual}_m]$. (9.35)

$d\tilde{\Phi}_m/dt = K[\hat{O}_m, \text{Coupling}, \text{Environment}]$. (9.36)

These equations form a closed recursive system.

They say:

phase affects the gate;

the gate determines realized versus unresolved time;

realized events update the ledger;

the ledger updates the observer;

the observer changes future phase evolution.

This is the market-time generator.

9.10 Third major conclusion

The financial model now contains a complete time-like architecture:

interacting asset states;

collective phase;

unwrapped phase chronology;

state-dependent admission gate;

realized market ticks;

unresolved imaginary-time-like depth;

irreversible trace;

historical ledger;

observer update;

self-referential feedback.

The complete chain is:

Fluctuation → Interaction → Synchronization → Phase Clock → Gate → Trace → Ledger → Feedback → Endogenous Time. (9.37)

The strongest defensible conclusion is:

> A simple extension of CAPM can illustrate how time-like order may emerge from interacting possibilities, selective commitment, retained residual, and trace-bearing self-organization.

It does not prove how physical time emerged.

But it provides a macroscopic model in which part of the proposed mechanism becomes mathematically explicit, observable, and potentially testable.

The next part will translate this same architecture into large language models, where the possibility field, projection, gate, residual, trace, and memory can be engineered directly.

Part IV — Large Language Models as Engineerable Complex Residual Systems

10. The LLM Possibility Field

10.1 Why an LLM is a useful experimental system

Financial markets make the projection–residual grammar visible at macroscopic scale.

Large language models make the same grammar programmable.

An LLM runtime can contain:

a prompt;

system and developer instructions;

conversation context;

latent model representations;

token candidates;

retrieved documents;

tool options;

output schemas;

safety constraints;

verifier results;

memory records.

Before a final answer is emitted, many continuation paths remain possible.

The runtime therefore begins not with one settled answer, but with a protocol-bound possibility field.

Let:

X^L = LLM possibility field under protocol P^L. (10.1)

The superscript L denotes the LLM domain.

The field may contain several types of candidate:

X^L = {tokens, claims, interpretations, plans, tools, actions, refusals, clarifications}. (10.2)

The model does not necessarily represent all these items as explicit symbolic objects internally.

Equation (10.2) describes their functional roles at the runtime level.

10.2 The declared LLM protocol

An LLM answer is not meaningful outside a task declaration.

Define:

$P^L = (B^L, \Delta^L, h^L, u^L)$. (10.3)

Where:

B^L = task boundary;

Δ^L = observation and evaluation rule;

h^L = context or episode horizon;

u^L = admissible tools and actions.

The boundary B^L determines:

what problem is being answered;

whose instructions govern;

what domain is relevant;

what sources may be used;

what claims are outside scope.

The observation rule Δ^L determines:

factuality standard;

style requirements;

citation requirements;

answer schema;

uncertainty reporting;

success criteria.

The horizon h^L determines:

how much conversation history is active;

whether earlier turns remain authoritative;

how far future consequences are considered;

whether the task is one-shot or multi-stage.

The intervention family u^L determines:

whether the model may browse;

whether it may call tools;

whether it may write memory;

whether it may send messages;

whether it may modify external state.

Thus:

No Reliable LLM Output without Declared Protocol. (10.4)

The same model can produce different valid outputs under different protocols.

A legal summary, a creative rewrite, a medical safety answer, and a source-grounded research report require different projections and gates.

10.3 Candidate-answer state

A simplified candidate-answer state may be written:

$|\Psi^L\rangle = \Sigma_j c_j |a_j\rangle$. (10.5)

Where:

$|a_j\rangle$ = candidate answer channel;

c_j = candidate weight or amplitude-like score.

Possible channels include:

$|a_1\rangle$ = direct answer. (10.6)

$|a_2\rangle$ = tool call. (10.7)

$|a_3\rangle$ = clarification request. (10.8)

$|a_4\rangle$ = qualified answer. (10.9)

$|a_5\rangle$ = refusal. (10.10)

$|a_6\rangle$ = escalation to human review. (10.11)

This notation is an operator model of runtime selection.

It is not a claim that an LLM's physical hardware occupies a quantum superposition.

The common role is:

Many Candidate Channels Exist before One Runtime Commitment. (10.12)

10.4 Token probabilities are not the entire field

At each generation step k, an LLM may assign a probability distribution over tokens:

$p_k(v) = \Pr(\text{Token}_k = v \mid \text{Context}_k)$. (10.13)

The entropy of that distribution is:

$H_{token}(k) = -\Sigma_v p_k(v) \ln p_k(v)$. (10.14)

Low token entropy means that one continuation strongly dominates.

High token entropy means that several continuations remain plausible.

But token entropy alone does not capture:

whether the answer is factually grounded;

whether retrieved documents conflict;

whether instructions are inconsistent;

whether a tool result is trustworthy;

whether the model selected the correct task interpretation;

whether an answer is safe to act upon;

whether the trace should enter long-term memory.

Therefore:

Token Uncertainty ≠ Total Semantic Residual. (10.15)

A model may have low token entropy and still be confidently wrong.

It may have high token entropy while the underlying factual answer is clear but stylistically flexible.

The residual architecture must therefore extend beyond logits.

10.5 Semantic candidate ensembles

Suppose the same task is sampled repeatedly, paraphrased, or evaluated through several reasoning routes.

Let the resulting candidate semantic states be:

$S^L = \{s_1, s_2, \ldots, s_n\}$. (10.16)

Define a semantic embedding function:

$\varphi^L(s_j) \in \mathbb{R}^d$. (10.17)

The ensemble centroid is:

$\mu^L = (1/n)\Sigma_j \varphi^L(s_j)$. (10.18)

The semantic dispersion is:

$V^L = (1/n)\Sigma_j \|\varphi^L(s_j) - \mu^L\|^2$. (10.19)

Low V^L suggests that repeated runs converge on a common semantic basin.

High V^L suggests:

ambiguity;

unstable interpretation;

weak evidence;

competing attractors;

high prompt sensitivity.

This provides one measurable approximation to semantic coherence.

10.6 LLM phase as relative orientation

To import the complex-state grammar, define a reference task direction:

$\hat{e}^P$ = normalized task-and-evidence direction under protocol P^L. (10.20)

Let the candidate-answer representation be v_j.

Define:

$\cos\theta_j^L = \langle v_j, \hat{e}^P \rangle / (\|v_j\| \|\hat{e}^P\|)$. (10.21)

Then θ_j^L measures the candidate's orientation relative to the declared task frame.

A candidate may be:

aligned with user intent;

aligned with evidence but not format;

aligned with format but not evidence;

aligned with one instruction and opposed to another.

Therefore a single phase may be insufficient.

A more realistic phase vector is:

$\boldsymbol{\theta}^L = [\theta_{intent}, \theta_{evidence}, \theta_{policy}, \theta_{tool}, \theta_{format}, \theta_{memory}]^T$. (10.22)

The scalar θ^L is a compressed visualization.

The phase vector preserves multiple forms of alignment.

10.7 Constructive and destructive semantic interference

Suppose two evidence sources support candidate a.

Represent their contributions as:

$Z_1 = A_1 \exp(i\theta_1)$. (10.23)

$Z_2 = A_2 \exp(i\theta_2)$. (10.24)

Their combined support is:

$Z_{12} = Z_1 + Z_2$. (10.25)

Then:

$|Z_{12}|^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\theta_1 - \theta_2)$. (10.26)

When the sources are semantically compatible:

$\theta_1 - \theta_2 \approx 0$. (10.27)

Their support reinforces.

When they contradict:

$\theta_1 - \theta_2 \approx \pi$. (10.28)

Their contributions cancel or create residual conflict.

In practical LLM terms:

compatible instructions reinforce one response;

contradictory instructions destabilize selection;

corroborating documents strengthen a claim;

conflicting documents increase evidence residual;

consistent tools strengthen commitment;

mismatched tool results lower gate confidence.

This is not physical quantum interference.

It is a complex representation of orientation-dependent combination.

10.8 The LLM as a bounded observer

The model never sees the whole relevant world.

It sees:

$\text{VisibleWorld}^L = \text{Projection}(\text{World} \mid \text{Prompt,Context,Retrieval,Tools,}P^L)$. (10.29)

Everything outside that projection is unavailable or residual.

A bounded LLM observer therefore satisfies:

$\text{WorldRelevant} > \text{RuntimeVisibleWorld}$. (10.30)

The runtime should record this limitation.

Possible missing-world residuals include:

unavailable current information;

inaccessible private data;

incomplete documents;

unresolved temporal context;

missing user intent;

inaccessible tool state;

unknown downstream consequence.

This gives:

Answer Completeness Is Bounded by Observation Completeness. (10.31)

A fluent model cannot reason reliably from evidence it never received.

10.9 RAG as an evidence-coupling mechanism

Retrieval-augmented generation can be represented as:

$X^{L\prime} = \mathrm{Couple}(X^{L}, \mathrm{EvidenceLedger})$. (10.32)

Retrieval does more than add text.

It can provide:

external trace;

claim support;

contradiction detection;

temporal grounding;

source identity;

citation path.

A mature retrieval process should therefore be:

Query → Retrieve → Validate → Bind → Cite → Preserve Residual. (10.33)

The uploaded 成界之學 material characterizes RAG as a retrieval-governed collapse protocol: retrieval limits hallucination, supplies trace, establishes an evidence gate, enables audit, and preserves residual rather than merely extending context.

This interpretation will become central when the LLM gate is formalized.

10.10 First LLM conclusion

The LLM runtime already contains the necessary precursors for a complex residual model:

multiple candidate continuations;

relative orientation to task and evidence;

reinforcing and conflicting inputs;

bounded observation;

projection into one visible output;

external tools and records;

state-changing trace.

The next step is to represent the split between emitted answer and unresolved structure explicitly.

---

11. The Complex LLM State

11.1 Minimal form

Define the complex LLM state:

$Z^L = R^L + iQ^L$. (11.1)

Where:

R^L = admitted answer component;

Q^L = unresolved semantic residual.

In polar form:

$Z^L = A^L \exp(i\theta^L)$. (11.2)

Therefore:

$R^L = A^L \cos\theta^L$. (11.3)

$Q^L = A^L \sin\theta^L$. (11.4)

$A^{L2} = R^{L2} + Q^{L2}$. (11.5)

This is a conceptual and engineering representation.

It does not require the underlying neural network to use literal complex-valued activations.

11.2 What belongs on the real axis

The real component R^L is what the runtime currently admits as fit for external commitment.

Depending on the protocol, R^L may be:

emitted text;

a verified factual claim;

a cited summary;

a selected classification;

an approved tool action;

a generated code patch;

a memory item;

a refusal;

a request for clarification.

Thus:

R^L = Admitted Runtime Commitment under P^L. (11.6)

The real component is not synonymous with truth.

It is what passed the current gate.

A poor gate can admit false structure.

A strong gate can defer uncertain structure.

11.3 The multidimensional residual

A realistic LLM residual should be represented as:

$$\mathbf{Q}^L = [Qfact,Qretrieval,Qinstruction,Qpolicy,Qtool,Qambiguity,Qalternative,Qmemory,Qtemporal,Qcons \quad (11.7)$$

Each coordinate has a distinct meaning.

Residual	Meaning
Qfact	claim lacks adequate factual support
Qretrieval	sources are missing, weak, stale, or contradictory
Qinstruction	instructions conflict or remain ambiguous
Qpolicy	output may violate governing constraints
Qtool	tool result is unavailable, failed, or unreliable
Qambiguity	user intent or question admits several readings
Qalternative	significant rejected answer branches remain
Qmemory	prior memory is inconsistent, stale, or unverified
Qtemporal	the answer may depend on current information

Residual	Meaning
Qconsequence	downstream effects are uncertain or high-risk

Then:

$Z^{L} = R^{L} + i\mathbf{Q}^{L}$. (11.8)

This is again a generalized complex-residual notation.

11.4 Residual metric

Different residual dimensions interact.

Define a metric G^{L}:

$A^{L2} = R^{L2} + \mathbf{Q}^{LT}G^{L}\mathbf{Q}^{L}$. (11.9)

If:

$G^{L} = I$. (11.10)

Then:

$A^{L2} = R^{L2} + \Sigma_j Q_j^2$. (11.11)

But interactions matter.

Examples:

temporal uncertainty can amplify factual uncertainty;

retrieval conflict can amplify answer ambiguity;

tool failure can amplify consequence risk;

instruction conflict can amplify policy risk;

stale memory can amplify factual confidence error.

Therefore:

$\mathbf{Q}^{LT}G^{L}\mathbf{Q}^{L} = \Sigma_j G_{jj}Q_j^2 + 2\Sigma_{j<k} G_{jk}Q_jQ_k$. (11.12)

The off-diagonal terms represent residual coupling.

11.5 Why confidence is not enough

A scalar confidence score c may be written:

$c \in [0,1]$. (11.13)

But two answers with $c = 0.6$ may differ fundamentally.

Answer A

The evidence is strong, but user intent is ambiguous.

Answer B

The intent is clear, but the evidence is contradictory.

Answer C

The facts are well-supported, but the required tool failed.

Answer D

The answer is probably correct, but the information may be outdated.

A scalar confidence score cannot distinguish these cases.

Therefore:

Residual Type Matters, Not Only Residual Magnitude. (11.14)

The residual vector supports different interventions:

Dominant residual	Appropriate response
Qfact	verify or qualify
Qretrieval	search or disclose source limits
Qinstruction	ask for clarification
Qtool	retry, switch tool, or report failure
Qpolicy	refuse or constrain
Qmemory	revalidate before writing
Qconsequence	require approval or human review

Residual governance converts uncertainty into action policy.

11.6 Residual is not a chain-of-thought transcript

The proposed residual vector should not be confused with a full record of hidden internal reasoning.

A governed residual can report:

unsupported;

conflicting;

missing;

stale;

unverifiable;

unsafe;

tool-dependent;

awaiting clarification.

It need not reveal private intermediate reasoning.

Thus:

Residual Disclosure ≠ Internal Reasoning Disclosure. (11.15)

A mature system exposes decision-relevant residual metadata while preserving appropriate boundaries around internal computation.

11.7 Answer projection

Let the full runtime state be X^L.

A projection operator produces a candidate answer:

$V^L = \hat{O}^L,P(X^L)$. (11.16)

The admitted result is:

$R^L = \mathrm{Admit}_p(V^L)$. (11.17)

The unresolved remainder is:

$\mathbf{Q}^L = \mathrm{Residual}_p(X^L,V^L)$. (11.18)

Then:

$\mathrm{State}^L_p = R^L + i\mathbf{Q}^L$. (11.19)

The output should ideally include a governed projection of the residual:

$\mathrm{Output}^L = \{R^L,\mathrm{ResidualSummary}^L,\mathrm{Trace}^L\}$. (11.20)

For example:

answer;

assumptions;

uncertainty;

citations;

tool status;

unresolved issue.

11.8 Real-axis inflation

An unreliable LLM may inflate R^L by compressing residual into the visible answer.

Define:

$R_{reported}^L = R_{justified}^L + F^L$. (11.21)

Where F^L is false closure.

The operative state remains:

$Z^L = R_{justified}^L + i\mathbf{Q}^L$. (11.22)

But the reported state becomes:

$Z_{reported}^L = R_{justified}^L + F^L$. (11.23)

The false-closure error is:

$E_{closure}^L = F^L + i\mathbf{Q}^L$. (11.24)

This is the mathematical form of overclaiming.

A fluent output can make F^L appear large while hiding $\mathbf{Q}^L$.

11.9 Residual-preserving output

A residual-honest output aims for:

$F^L \to 0$. (11.25)

It may express:

R^L = Supported Answer. (11.26)

$\mathbf{Q}^L$ = Declared Limits and Unresolved Conditions. (11.27)

The output contract becomes:

ReliableOutput^L = Claim + Evidence + Assumptions + Residual + Trace. (11.28)

This does not require every answer to be long.

For simple tasks, $\mathbf{Q}^L$ may be negligible.

For high-stakes tasks, the residual summary becomes essential.

11.10 Residual life cycle

Residual should not remain static.

At runtime step k:

$\mathbf{Q}^{L}_{,k+1} = \text{UpdateResidual}(\mathbf{Q}^{L}_{,k}, \text{Evidence}_k, \text{Tool}_k, \text{Verifier}_k, \text{User}_k)$. (11.29)

Retrieval may reduce Qfact:

$\text{Qfact}_{,k+1} < \text{Qfact}_{,k}$. (11.30)

A failed tool may increase Qtool:

$\text{Qtool}_{,k+1} > \text{Qtool}_{,k}$. (11.31)

A clarification may reduce Qambiguity:

$\text{Qambiguity}_{,k+1} < \text{Qambiguity}_{,k}$. (11.32)

A contradiction may increase Qretrieval:

$\text{Qretrieval}_{,k+1} > \text{Qretrieval}_{,k}$. (11.33)

Thus the residual vector becomes an operational state variable.

11.11 Residual ledger

Let the residual ledger be:

$\text{Lres}^{L}(k) = \{\mathbf{Q}^{L}_{,1}, \mathbf{Q}^{L}_{,2}, \ldots, \mathbf{Q}^{L}_{,k}\}$. (11.34)

This ledger can answer:

Which residuals repeatedly accumulate?

Which gates fail most often?

Which tools are unreliable?

Which prompt patterns cause ambiguity?

Which memories frequently become stale?

Which claims require repeated correction?

The Gauge Grammar requires residual to be logged, typed, or escalated rather than silently erased. It also treats AI runtimes as systems with multiple local frames—prompt, retrieval, tool, verifier, policy, memory, and user—whose disagreements must be transported and audited.

11.12 Second LLM conclusion

The complex LLM state supplies a richer alternative to scalar confidence:

$Z^{L} = R^{L} + i\mathbf{Q}^{L}$. (11.35)

The real axis records admitted commitment.

The residual space records unresolved pressure by type.

The next question is:

What decides whether a candidate answer, action, or memory should move onto the real axis?

The answer is a layered gate.

---

12. Projection, Gates, and Collapse-Like Output

12.1 Projection stack

An LLM runtime rarely uses one projection alone.

Define:

$\hat{O}^{L} = \hat{O}task\ \hat{O}context\ \hat{O}retrieval\ \hat{O}policy\ \hat{O}tool\ \hat{O}verifier\ \hat{O}format.$ (12.1)

The order may differ by architecture.

Each operator performs a role.

Operator	Role
$\hat{O}$task	interpret the requested objective
$\hat{O}$context	bind relevant conversation state
$\hat{O}$retrieval	connect claims to external evidence
$\hat{O}$policy	constrain admissible outputs
$\hat{O}$tool	evaluate or execute external operations
$\hat{O}$verifier	test factual, logical, or structural validity
$\hat{O}$format	project result into required schema

The candidate is:

$V^{L} = \hat{O}^{L}(X^{L}).$ (12.2)

But projection does not yet mean admission.

12.2 The LLM gate

Define:

$G^{L} \in [0,1].$ (12.3)

A composite gate may be:

$G^{L} = G_{intent}\ G_{evidence}\ G_{consistency}\ G_{policy}\ G_{tool}\ G_{format}\ G_{consequence}.$ (12.4)

Where:

Gintent = task interpretation is sufficiently clear;

Gevidence = factual support is adequate;

Gconsistency = answer is internally coherent;

Gpolicy = output is permitted;

Gtool = required tool result is valid;

Gformat = output satisfies contract;

Gconsequence = action risk is acceptable.

If any essential factor approaches zero:

$G^{L} \rightarrow 0$. (12.5)

This multiplicative form captures a practical truth:

A serious failure in one indispensable gate can invalidate the whole commitment.

12.3 Alternative gate aggregation

The multiplicative gate may be too strict for some tasks.

A weighted logistic gate is:

$G^{L} = 1 / \{1 + \exp[-S^{L}]\}$. (12.6)

Where:

$S^{L} = b_0 + \Sigma_j b_j g_j - \lambda \|\mathbf{Q}^{L}\|$. (12.7)

Here:

g_j = individual gate signals;

b_j = task-specific weights;

λ = residual penalty.

A high-risk medical or legal action may use a high threshold:

Accept if $G^{L} \geq G_{high}^{*}$. (12.8)

A casual brainstorming response may use a lower threshold:

Accept if $G^{L} \geq G_{low}^{*}$. (12.9)

Therefore:

Gate Strictness Is Protocol-Dependent. (12.10)

12.4 Output channels

The gate should not force every task into only answer or failure.

Define output projectors:

P^answer. (12.11)

P^tool. (12.12)

P^clarify. (12.13)

P^qualify. (12.14)

P^refuse. (12.15)

P^escalate. (12.16)

Then:

$\hat{O}^{L}|\Psi^{L}\rangle \rightarrow$

{
|answer⟩,
|tool⟩,
|clarify⟩,
|qualify⟩,
|refuse⟩,
|escalate⟩
}. (12.17)

A mature gate selects the appropriate closure channel.

Uncertainty should not always lead to refusal.

It may lead to:

a qualified answer;

retrieval;

a question;

human review;

a reversible experiment.

12.5 Micro-commitment and macro-commitment

Each generated token changes the next token distribution.

Therefore token selection is a micro-commitment:

$Context_{k+1} = Context_k \sqcup Token_k$. (12.18)

But not every token is a strong historical event.

A complete answer is a larger commitment:

$Trace_{answer} = Commit(Output^L)$. (12.19)

A tool action is stronger still:

$World_{k+1} = ToolAction(World_k)$. (12.20)

A memory write may affect future episodes:

$Memory_{k+1} = Memory_k \sqcup AcceptedItem_k$. (12.21)

Thus the runtime has a hierarchy of collapse-like events:

Token < Sentence < Answer < Tool Action < Memory Write < External Consequence. (12.22)

The stronger the consequence, the stronger the required gate.

12.6 Tool calls as consequential gates

Before executing a tool:

$CandidateAction = a$. (12.23)

The runtime should evaluate:

$G_{tool}(a) = f(permission, arguments, evidence, reversibility, risk)$. (12.24)

A reversible read operation may require a moderate gate.

An irreversible action may require:

stronger verification;

explicit user approval;

rollback plan;

audit trace.

A governed action rule is:

Execute(a) only if $a \in U(P^L)$ and $G_{tool}(a) \geq G_{tool}^*$. (12.25)

This corresponds to the Gauge Grammar rule that intervention must belong to the declared admissible action family and must leave verifiable trace.

12.7 Memory gate

Not every output should enter memory.

Define:

Gmemory = Gidentity Gstability Grelevance Gconsent Gexpiry. (12.26)

Where:

Gidentity = the memory belongs to the correct entity or user;

Gstability = the fact is likely to remain valid;

Grelevance = future usefulness is high;

Gconsent = storage is permitted;

Gexpiry = revision or expiry rule is defined.

Then:

WriteMemory only if Gmemory ≥ Gmemory*. (12.27)

A poor memory gate creates trace pollution.

Incorrect memory can influence many later episodes.

Therefore memory is not passive storage.

It is a future-bending ledger.

12.8 Collapse-like output

A candidate answer becomes externally real when it passes the gate and enters trace:

$\text{Candidate}^{L} \rightarrow \text{Gate}^{L} \rightarrow \text{CommittedOutput}^{L}$. (12.28)

The residual remains:

$\mathbf{Q}^{L},\text{res} = \mathbf{Q}^{L} - \mathbf{Q}^{L},\text{resolved}$. (12.29)

The full event is:

$X^{L} \rightarrow R^{L} + i\mathbf{Q}^{L},\text{res} \rightarrow \text{Trace}^{L}$. (12.30)

This is collapse-like in the functional sense:

many candidates existed;

one channel was selected;

the selection became externally consequential;

the state of the episode changed.

It is not asserted to be a physical wavefunction collapse.

12.9 Trace construction

A useful LLM trace may contain:

$\text{Trace}^{L} = \{\text{protocol},\text{claim},\text{evidence},\text{tool},\text{status},\text{residual},\text{timestamp},\text{version}\}$. (12.31)

For a tool action:

$\text{Trace}_{\text{action}}^{L} = \{\text{action},\text{arguments},\text{authorization},\text{result},\text{rollback},\text{residual}\}$. (12.32)

For a memory write:

$\text{Trace}_{\text{memory}}^{L} = \{\text{item},\text{source},\text{confidence},\text{scope},\text{expiry},\text{consent}\}$. (12.33)

Trace quality can be measured:

$Q_{\text{trace}} = \text{Completeness} \times \text{Integrity} \times \text{Reproducibility} \times \text{Relevance}$. (12.34)

If any factor approaches zero, the trace becomes weak.

12.10 The trace changes future projection

After a committed event:

$L^{L}_{,k+1} = \text{Update}(L^{L}_{,k},\text{Trace}^{L}_{,k},\mathbf{Q}^{L}_{,k})$. (12.35)

The next projection operator changes:

$\hat{O}^{L}_{,k+1} = \text{UpdateOperator}(\hat{O}^{L}_{,k},L^{L}_{,k+1})$. (12.36)

Examples:

a tool result changes the next answer;

user correction changes interpretation;

a verifier rejection causes revision;

memory changes future retrieval;

a failed action tightens future gates.

Thus:

$\text{Output}_{k} \rightarrow \text{Trace}_{k} \rightarrow \text{Future Projection}_{k+1}$. (12.37)

This is observer backreaction in an engineered form.

12.11 Gate failure classes

The gate can fail in different ways.

Early gate

A valid answer is rejected too soon.

$\text{FalseNegative}^{L} = \text{Valid Candidate} \wedge G^{L} < G^{*}$. (12.38)

Late gate

An invalid answer is admitted.

$\text{FalsePositive}^{L} = \text{Invalid Candidate} \wedge G^{L} \geq G^{*}$. (12.39)

Wrong-channel gate

The system answers when it should clarify or use a tool.

$\text{WrongChannel}^{L} = \text{SelectedChannel} \neq \text{RequiredChannel}$. (12.40)

Trace-free gate

An action is admitted without adequate record.

$\text{TraceFailure}^{L} = \text{Committed Action} \wedge \text{Missing Trace}$. (12.41)

Residual-erasing gate

The answer is admitted while material residual is hidden.

$\text{ResidualFailure}^{L} = \|\mathbf{Q}^{L}\| > Q^{*} \wedge \text{ResidualDisclosure} = 0$. (12.42)

These distinctions support precise diagnosis.

12.12 Gate audit

A post-episode audit may ask:

Which gate opened?

Which evidence justified it?

Which residual remained?

Which channel was selected?

What trace was written?

Did later outcomes validate the gate?

Should thresholds be revised?

Define gate regret:

$\text{Regret}^{L} = \text{Loss}(\text{CommittedOutput}) - \text{Loss}(\text{BestAvailableChannel})$. (12.43)

The runtime can update:

$G^{L}_{,k+1} = \text{LearnGate}(G^{L}_{,k}, \text{Regret}^{L}, \text{Trace}^{L}, \text{Residual}^{L})$. (12.44)

This allows system learning beyond weight training.

The uploaded 成界之學 material similarly proposes that an AI runtime should learn from accumulated residuals, failed gates, prompt declarations, memory expiry, and tool approvals rather than treating learning only as model retraining.

---

13. LLM Time and Imaginary-Time-Like Semantic Depth

13.1 Several clocks exist in an LLM

An LLM runtime can be measured by:

wall-clock time;

inference steps;

generated tokens;

tool calls;

verifier cycles;

accepted claims;

completed subtasks;

memory updates.

These are not equivalent.

Let:

t^{L} = external wall-clock time. (13.1)

k = token or computation step. (13.2)

τ^{L} = consequential semantic time. (13.3)

T^{L} = unresolved semantic depth. (13.4)

The article's central claim is:

Token Count ≠ Semantic Time. (13.5)

13.2 Why token count is not semantic time

A model may produce many tokens that:

paraphrase earlier content;

repeat an attractor;

extend style without adding evidence;

circle around ambiguity;

fail to resolve a contradiction;

produce verbose but unverified reasoning.

Then:

Δk large. (13.6)

But:

$\Delta \tau^L$ small. (13.7)

Conversely, one tool result may resolve the entire task:

Δk small. (13.8)

But:

$\Delta \tau^L$ large. (13.9)

Semantic time should therefore measure consequential resolution, not surface length.

13.3 LLM clock hand

Define a collective semantic phase:

$C^L = \rho^L \exp(i\Phi^L)$. (13.10)

The coherence ρ^L may summarize agreement among:

repeated samples;

retrieved evidence;

verifier outputs;

planning modules;

tool results;

model checkpoints.

The phase Φ^L represents the dominant orientation of the developing answer.

The visible clock hand is:

$\exp(i\Phi^L)$. (13.11)

The unwrapped phase is:

$\tilde{\Phi}^{L} = \Phi^{L} + 2\pi N^{L}$. (13.12)

Where N^{L} records completed semantic cycles or revisions.

A primitive semantic-phase coordinate is:

$\tau\text{phase}^{L} = [\tilde{\Phi}^{L}(k) - \tilde{\Phi}^{L}(0)] / \Omega^{L}$. (13.13)

But phase movement alone does not guarantee progress.

The gate must determine which changes become verified commitments.

13.4 Realized semantic time

Define:

$d\tau^{L} = G^{L}|d\tilde{\Phi}^{L}| / \Omega^{L}$. (13.14)

When G^{L} is high:

evidence aligns;

contradictions are resolved;

the selected claim is admissible;

the result can enter trace.

Then semantic time advances.

A semantic tick may be:

one verified claim;

one resolved ambiguity;

one validated tool result;

one completed subtask;

one approved action;

one trusted memory update.

Thus:

τ^{L} = Ordered Verified Semantic Commitments. (13.15)

13.5 Unresolved semantic depth

Define:

$$dT^{L} = (1 - G^{L})|d\tilde{\Phi}^{L}| / \Omega^{L}. \quad (13.16)$$

When the model continues processing but cannot commit safely:

candidate answers rotate;

evidence remains conflicting;

tools fail;

prompt interpretations compete;

the answer remains unstable.

Then:

$$d\tau^{L} \approx 0. \quad (13.17)$$

$$dT^{L} > 0. \quad (13.18)$$

The complex semantic-time coordinate is:

$$d\zeta^{L} = d\tau^{L} + i\, dT^{L}. \quad (13.19)$$

Therefore:

$$d\zeta^{L} = [G^{L} + i(1 - G^{L})]|d\tilde{\Phi}^{L}| / \Omega^{L}. \quad (13.20)$$

This directly parallels the market construction.

13.6 Interpretation of T^{L}

T^{L} may accumulate through:

repeated unsuccessful reasoning;

unresolved source conflict;

long-context drift;

ambiguous user intent;

verifier disagreement;

unavailable tool output;

safety-policy tension;

plan instability;

incomplete decomposition.

A composite approximation is:

$$T^{L} = b_1 T_{fact} + b_2 T_{retrieval} + b_3 T_{instruction} + b_4 T_{tool} + b_5 T_{verifier} + b_6 T_{memory}. \quad (13.21)$$

T^L is not necessarily wasted effort.

It may represent useful unresolved processing.

But if it grows without eventual resolution, the runtime may be trapped.

13.7 Frozen semantic time

A frozen-time-like LLM regime satisfies:

$d\tau^L/dt \to 0$. (13.22)

While:

$dT^L/dt > 0$. (13.23)

The system may continue generating or revising, but no verified progress occurs.

Examples include:

repetitive answer loops;

endless planning without action;

repeated tool retries without diagnosis;

unresolved citation conflict;

self-consistency samples that never converge;

over-analysis caused by incompatible gates.

This condition can be called:

Semantic Churn without Ledger Progress. (13.24)

It is analogous to a market that continues rotating internally while producing few consequential ticks.

The attached SMFT imaginary-time paper similarly defines unresolved phase evolution as continuous while collapse-defined time remains frozen.

13.8 Release of accumulated semantic depth

Suppose:

T^L grows. (13.25)

A new event may open the gate:

a reliable source is retrieved;

a calculation resolves the dispute;

the user clarifies intent;

a tool succeeds;

a verifier identifies the error;

a higher-level rule selects the correct interpretation.

Then:

$G^L \to 1$. (13.26)

A large semantic tick may occur:

$\Delta\tau^L = \eta T^L(t^{*}-)$. (13.27)

And:

$T^L(t^{*}+) = [1 - \eta]T^L(t^{*}-)$. (13.28)

This represents sudden resolution after prolonged ambiguity.

The transition resembles:

Search → Conflict → Clarification → Commitment. (13.29)

13.9 LLM time dilation

Define the semantic-time rate:

$\gamma^L = d\tau^L/dt$. (13.30)

If:

γ^L high. (13.31)

The system produces consequential verified progress rapidly.

If:

γ^L low. (13.32)

The system consumes time or tokens without resolving the task.

A useful efficiency measure is:

$\eta\text{semantic} = \Delta\tau^L / \text{Cost}^L$. (13.33)

Where Cost^L may include:

tokens;

latency;

tool expense;

energy;

human review;

external risk.

A model with fewer tokens but higher ηsemantic may be more capable than a verbose model with low semantic progress.

13.10 Task-relative proper-time analogue

Different tasks produce different semantic clock rates.

For task j:

$d\tau_j^L/dt = G_j^L|d\tilde{\Phi}_j^L/dt| \,/\, \Omega_j^L$. (13.34)

A routine summarization task may advance quickly.

A theorem, legal judgment, or complex diagnosis may advance slowly because stronger gates are required.

Define the task clock ratio:

$\chi_j^L = d\tau_j^L/dk$. (13.35)

This measures verified semantic progress per token step.

It is only a proper-time-like analogy.

No relativistic metric is claimed.

13.11 Episode ledger

Let the episode ledger be:

$L^L,_{k+1} = L^L,_k \sqcup \mathrm{Trace}^L,_k$. (13.36)

The episode's semantic time is:

$\tau^L(n) = \Sigma_{k=1}^{n} W_k G^L,_k$. (13.37)

Where W_k is the consequential weight of the trace.

Possible weights include:

claim importance;

evidence strength;

tool consequence;

action irreversibility;

memory persistence;

task completion value.

Therefore one verified tool action may count more than many ordinary tokens.

13.12 Temporal asymmetry

The model may revise an answer, but it cannot always erase its consequences.

A false answer may already have:

influenced the user;

been copied;

triggered code execution;

entered memory;

changed an external record.

Therefore:

Output Reversal ≠ Consequence Reversal. (13.38)

The ledger gives the LLM runtime an arrow of time.

The same text can be regenerated, but the episode state is no longer identical.

13.13 Third LLM conclusion

The LLM time model contains:

phase-like candidate development;

gate-dependent semantic ticks;

unresolved depth;

variable internal clock rate;

sudden release;

irreversible trace;

memory-conditioned future projection.

Its core equations are:

$d\tau^{L} = G^{L}|d\tilde{\Phi}^{L}| / \Omega^{L}$. (13.39)

$dT^{L} = (1 - G^{L})|d\tilde{\Phi}^{L}| / \Omega^{L}$. (13.40)

$d\zeta^{L} = d\tau^{L} + i\, dT^{L}$. (13.41)

The next section applies this structure to one of the most important LLM failures: hallucination.

---

14. Hallucination as Residual Misclassification

14.1 Beyond “the model guessed wrong”

Hallucination is commonly described as the production of false, unsupported, or fabricated content.

That description identifies the visible failure.

The complex residual model asks where the failure occurred in the architecture.

A reliable answer should satisfy:

$G^{L} \geq G^{*}$. (14.1)

And:

$\|\mathbf{Q}^{L},\text{material}\| \leq Q^{*}$. (14.2)

Or it should disclose the remaining residual.

A hallucination occurs when:

$\|\mathbf{Q}^{L},\text{material}\| > Q^{*}$. (14.3)

But:

$G^{L} \geq G^{*}$. (14.4)

The gate opens despite material unresolved pressure.

14.2 Minimal hallucination equation

Let:

R^{L} = emitted claim. (14.5)

Let:

$Q_{\text{fact}} > 0$. (14.6)

Let:

G_{evidence} weak. (14.7)

Yet the claim is committed:

$\text{Commit}(R^{L}) = 1$. (14.8)

Then:

Hallucination = High Residual + Weak Gate + Fluent Commitment. (14.9)

More fully:

$\text{Hallucination}^{L} = \text{Fluency} \times \text{GateFailure} \times \text{ResidualConcealment} \times \text{TraceCommitment}$. (14.10)

This equation explains why hallucination can appear convincing.

Fluency increases the appearance of closure.

Residual concealment hides uncertainty.

Trace commitment turns the mistake into an external fact-like object.

14.3 Residual misclassification

The operative state may be:

$Z^{L} = R\text{supported}^{L} + i\mathbf{Q}^{L}$. (14.11)

But the reported output is:

$R\text{reported}^{L} = R\text{supported}^{L} + \text{Misclassified}(\mathbf{Q}^{L})$. (14.12)

The model converts residual into assertion.

Define misclassification operator M:

$M: \mathbf{Q}^{L} \rightarrow F^{L}$. (14.13)

Where F^{L} is unsupported real-axis content.

Then:

$R\text{reported}^{L} = R\text{supported}^{L} + F^{L}$. (14.14)

This is the precise meaning of false closure.

14.4 Main hallucination classes

Type 1 — Factual residual failure

The model lacks evidence but answers confidently.

Qfact high. (14.15)

Type 2 — Retrieval residual failure

Retrieved sources do not support the claim.

Qretrieval high. (14.16)

Type 3 — Temporal residual failure

The information may have changed, but the model presents it as current.

$Q_{temporal}$ high. (14.17)

Type 4 — Instruction residual failure

The model resolves conflicting instructions incorrectly.

$Q_{instruction}$ high. (14.18)

Type 5 — Tool residual failure

A tool failed or returned ambiguous data, but the model invents a result.

Q_{tool} high. (14.19)

Type 6 — Memory residual failure

A stale or incorrect memory is reused as fact.

Q_{memory} high. (14.20)

Type 7 — Causal residual failure

The model reports correlation or speculation as established causation.

$Q_{consequence}$ high. (14.21)

Each failure requires a different intervention.

14.5 Fluent collapse without grounding

A model may have strong linguistic coherence:

$\rho_{language}^{L}$ high. (14.22)

But weak evidence coherence:

$\rho_{evidence}^{L}$ low. (14.23)

The combined answer may still sound smooth because language-phase alignment is high.

This gives:

Fluency Coherence ≠ Evidence Coherence. (14.24)

A robust runtime should track both.

Define:

$\rho^L = [\rho language, \rho evidence, \rho instruction, \rho tool, \rho policy]$. (14.25)

A high-quality answer requires the relevant dimensions to align.

Strong language coherence alone should not open the gate.

14.6 Hallucination as premature temporal closure

In the semantic-time model, hallucination can also be expressed as premature tick creation.

The correct state is:

$dT^L > 0$ and $d\tau^L \approx 0$. (14.26)

The system should continue retrieval, clarification, or verification.

Instead, the gate falsely produces:

$d\tau^L > 0$. (14.27)

The runtime writes a semantic tick before the unresolved phase has been adequately resolved.

Thus:

Hallucination = Premature Conversion of T^L into τ^L. (14.28)

This is one of the clearest bridges to the SMFT-inspired time model.

14.7 Trace pollution

A hallucination becomes more damaging when it enters persistent trace.

The propagation chain may be:

False Claim → Conversation → Memory → Retrieval → Future Answer. (14.29)

Or:

False Claim → Code → Execution → External Consequence. (14.30)

Define trace amplification:

Atrace = Number of Downstream Uses × Consequence Weight. (14.31)

Then hallucination harm is:

Hharm = HallucinationMagnitude × Atrace. (14.32)

A small false claim with no persistence may have limited harm.

A false claim written into shared memory may become a strong attractor.

14.8 Residual-aware correction

A residual-aware runtime should not merely regenerate the answer.

It should update:

the claim;

the evidence ledger;

the residual ledger;

the gate threshold;

any polluted memory;

downstream traces where possible.

The correction operation is:

$$\text{Correct}^{L} = \text{ReviseClaim} + \text{RepairTrace} + \text{UpdateResidual} + \text{RetuneGate}. \quad (14.33)$$

If only the visible text is corrected:

$$R^{L} \text{ changes}. \quad (14.34)$$

But:

$$L^{L} \text{ and } \hat{O}^{L} \text{ may remain polluted}. \quad (14.35)$$

A mature correction must repair the recursive system.

14.9 Practical residual-aware output contract

A high-stakes answer may use:

$$\text{Output}^{L} = \{\text{SupportedClaims}, \text{Evidence}, \text{Assumptions}, \text{Residuals}, \text{ToolStatus}, \text{NextAction}\}. \quad (14.36)$$

The output can remain readable.

For example:

supported conclusion;

what is uncertain;

what would resolve it;

whether a tool was used;

whether human approval is needed.

The user does not need the model's hidden internal reasoning.

The user needs the residual conditions relevant to trust and action.

14.10 A hallucination risk score

Define normalized residuals $q_j \in [0,1]$.

A simple risk score is:

$$H_{risk} = \sigma[b_0 + \Sigma_j b_j q_j - cG_{evidence} - dG_{tool} - eG_{verify}]. \quad (14.37)$$

Where σ is the logistic function.

A more structural score is:

$$H_{risk} = \|\mathbf{Q}^L\| G^L. \quad (14.38)$$

This expression is high when:

residual is large;

the system is still likely to commit.

A safe system seeks:

$$\|\mathbf{Q}^L\| \text{ high} \Rightarrow G^L \text{ low or ResidualDisclosure high.} \quad (14.39)$$

The dangerous regime is:

$$\|\mathbf{Q}^L\| \text{ high} \wedge G^L \text{ high} \wedge \text{Disclosure low.} \quad (14.40)$$

14.11 Hallucination and false market closure

The finance–LLM correspondence is now exact at the level of grammar.

Finance

A displayed price appears stable while hidden leverage and liquidity pressure remain large.

LLM

A fluent answer appears settled while factual and retrieval residual remain large.

Both have:

Visible Completion + Hidden Residual. (14.41)

Both may persist temporarily.

Both may fail abruptly when the residual becomes observable.

Therefore:

Hallucination Is the LLM Analogue of Mispriced Residual Risk. (14.42)

This does not make a false answer financially equivalent to a market crash.

It identifies the same structural failure:

The admitted real-axis projection is mistaken for the complete state.

14.12 Fourth LLM conclusion

The complex residual model reframes hallucination from a vague property of generative models into a diagnosable failure chain:

Possibility Field → Weak Evidence Binding → Premature Gate → Residual Misclassification → Confident Trace. (14.43)

A more trustworthy runtime reverses that chain:

Possibility Field → Evidence Binding → Typed Residual → Appropriate Gate → Auditable Trace. (14.44)

The uploaded Gauge Grammar describes the desired long-term architecture as a protocol-bound, trace-bearing, residual-honest, self-revising runtime rather than a stateless answer generator.

The LLM translation is therefore not merely philosophical.

It points toward an implementable architecture in which:

residual is typed;

gates are explicit;

traces are auditable;

semantic progress is measured separately from token count;

memory is governed;

failure updates the runtime.

The next part will place physics, finance, and LLMs side by side and identify which correspondences are direct mathematics, which are engineerable extensions, and which remain only structural analogies.

Part V — The Shared Physics–Finance–LLM Grammar

15. The Three-Domain Mapping

15.1 The purpose of the comparison

Physics, finance, and large language models are built from radically different substrates.

Physics concerns matter, energy, fields, and spacetime.

Finance concerns legal claims, capital, prices, expectations, institutions, and settlement.

LLM runtimes concern statistical representations, prompts, token generation, retrieval, tools, policies, and memory.

The comparison proposed here therefore cannot begin from material identity.

It must begin from functional role.

Let domains A and B contain components x^A and x^B.

A valid cross-domain translation requires more than superficial similarity.

Define a role signature:

$$\mathrm{Role}(x) = \{\text{Input, Transformation, Output, Constraint, Failure, Trace}\}. \quad (15.1)$$

A functional correspondence exists when:

$$\mathrm{Role}(x^A) \approx \mathrm{Role}(x^B) \text{ under declared protocols } P^A \text{ and } P^B. \quad (15.2)$$

The symbol ≈ means operational correspondence, not equality.

Thus:

$$x^A \approx_{\text{role}} x^B \text{ does not imply } x^A = x^B. \quad (15.3)$$

For example, a financial transaction gate and an LLM verifier gate may both decide whether a candidate state becomes consequential. They do not share material composition, physical law, or ontological status.

The Gauge Grammar uses the same discipline. It treats field, identity, mediator, binding, gate, trace, invariance, and observer potential as recurring roles, while explicitly rejecting literal claims that markets or AI systems are physical quantum fields.

15.2 The general domain state

Let D denote one of the three domains:

$$D \in \{P,F,L\}. \quad (15.4)$$

Where:

P = physics;

F = finance;

L = LLM runtime.

Define the domain state:

$Z^D = R^D + iQ^D$. (15.5)

Or, in multidimensional form:

$Z^D = R^D + i\mathbf{Q}^D$. (15.6)

The general interpretation is:

R^D = structure admitted under the current observer and protocol;

$\mathbf{Q}^D$ = dynamically relevant residual not admitted into R^D;

i = marker of an orthogonal or non-admitted direction.

The collective phase state is:

$C^D = \rho^D \exp(i\Phi^D)$. (15.7)

Where:

ρ^D = coherence of the collective state;

Φ^D = dominant orientation;

$\tilde{\Phi}^D$ = unwrapped phase history.

The gate is:

$G^D \in [0,1]$. (15.8)

The complex-time-like coordinate is:

$d\zeta^D = d\tau^D + i\, dT^D$. (15.9)

With:

$d\tau^D = G^D|d\tilde{\Phi}^D| / \Omega^D$. (15.10)

$dT^D = (1 - G^D)|d\tilde{\Phi}^D| / \Omega^D$. (15.11)

The ledger update is:

$L^D,_{k+1} = \text{Update}(L^D,_k, \text{Trace}^D,_k, \text{Residual}^D,_k)$. (15.12)

Equations (15.5)–(15.12) form the common mathematical skeleton.

Their domain interpretations differ.

15.3 Central mapping table

General role	Physics	Finance	LLM runtime
Possibility field	Physical state space or field configuration space	Possible valuations, returns, positions, orders, and regimes	Candidate tokens, claims, plans, tools, and answer channels
State	Wavefunction, state vector, density operator, or classical field	Asset state, factor state, joint market state	Latent runtime state, context state, candidate-answer ensemble
Complex representation	$\Psi = A \exp(i\theta)$	$Z^f = R^f + iQ^f$	$Z^L = R^L + i\mathbf{Q}^L$
Magnitude	Amplitude norm or field strength	Total declared value–pressure magnitude	Total semantic activation or commitment magnitude
Phase	Relative physical phase	Asset or market orientation relative to a declared factor frame	Candidate orientation relative to task, evidence, policy, and tools
Real projection	Observable component under a selected basis	Price, market-aligned return, recognized value	Emitted answer, accepted claim, approved action
Residual coordinate	State structure not represented by the selected observable	Liquidity, credit, option, funding, positioning, and model pressure	Factual, retrieval, instruction, tool, policy, and ambiguity residual
Superposition-like field	Linear combination of states	Coexisting valuations, positions, orders, and regime possibilities	Coexisting token, interpretation, plan, and tool candidates
Interference	Relative phase alters observable probabilities	Aligned or opposed flows reinforce or cancel	Compatible or conflicting evidence and instructions reinforce or suppress candidates
Projection basis	Choice of observable or measurement basis	Benchmark, factor model, currency, horizon, accounting frame	Prompt, system instruction, retrieval set, schema, verifier
Observer	Instrument, interaction frame, or internal observer model	Trader, fund, exchange, regulator, accounting system	User, prompt stack, runtime policy, verifier, toolchain
Gate	Outcome channel, transition condition, or measurement selection	Execution, settlement, margin, default, close, policy threshold	Decoder choice, evidence gate, tool approval, policy gate, memory gate

General role	Physics	Finance	LLM runtime
Committed outcome	Recorded measurement result	Executed price, settled transaction, recognized loss	Token, final answer, tool action, citation, memory write
Trace	Measurement record or environment-conditioned history	Price history, balance-sheet change, collateral record, precedent	Conversation record, audit log, tool result, memory, verifier record
Residual after commitment	Unresolved or unobserved state structure	Unfilled orders, hidden pressure, remaining risk	Uncertainty, rejected branches, source conflict, unverified assumptions
Coherence	Stable relative phase	Broad market synchronization	Agreement among samples, evidence, tools, and verifiers
Decoherence	Loss of stable phase relation through environmental coupling	Fragmentation, noise, liquidity breakdown, regime disagreement	Context drift, instruction collision, retrieval conflict, sample divergence
Order parameter	Macroscopic measure of collective ordering	$\rho_m =$	$\Sigma_i w_i \exp(i\theta_i)/W$
Symmetry breaking	One state or orientation becomes selected	Market selects bullish, bearish, sectoral, or liquidity regime	Ambiguous task selects one interpretation, plan, or answer basin
Backreaction	Observation or interaction changes later state	Price affects orders, collateral, expectations, and regulation	Answer affects user response, context, memory, and later tool use
External time	Coordinate or laboratory time	Calendar and exchange time	Wall-clock and inference time
Internal phase clock	Repeated physical phase process	Collective market phase Φ_m	Developing semantic phase Φ^L
Consequential time	Ordered physical records or events	Ledgered market transitions τ_m	Verified semantic commitments τ^L
Imaginary-time-like depth	Euclidean time in physics; unresolved phase depth in SMFT interpretation	Unresolved market phase and risk depth T_m	Unresolved semantic and evidence depth T^L
Arrow of time	Irreversible records, entropy production, boundary conditions	Settlement, loss, default, regulation, and balance-sheet history	Irreversible external actions, user influence, memory and trace pollution
Invariance	Relation preserved under valid frame transformation	Exposure or value relation stable across equivalent factor frames	Answer relation robust under paraphrase, source ordering, or tool substitution

General role	Physics	Finance	LLM runtime
Critical transition	Phase transition or instability threshold	Crash, breakout, default cascade, liquidity transition	Abrupt answer reorientation, verifier-triggered revision, tool escalation
Attractor	Stable dynamical basin	Trend, benchmark, liquidity, or narrative basin	Strong semantic, stylistic, or procedural answer basin

The table should not be read as a declaration that every physics row possesses a full equivalent in finance or AI.

It identifies the role being investigated.

15.4 Three strengths of correspondence

The mappings fall into three classes.

Class I — Direct mathematical recurrence

The same or closely related mathematics appears in more than one domain.

Examples include:

vector projection;

complex decomposition;

phase addition;

constructive and destructive combination;

covariance geometry;

synchronization order parameters;

entropy over state distributions;

state-space filtering.

A direct recurrence may be written:

$M^A \cong M^B$. (15.13)

Where M is a mathematical structure.

This is the strongest kind of cross-domain connection in the present article.

Class II — Operational homology

Different mechanisms perform a comparable system role.

Examples include:

transaction gate and verifier gate;

settlement ledger and memory ledger;

market residual and LLM residual;

market backreaction and conversation backreaction.

Write:

$\text{Role}^{A} \approx \text{Role}^{B}$. (15.14)

This is the central level of the Complex Residual Principle.

Class III — Interpretive analogy

A familiar physics concept helps visualize another system, but the necessary physical mathematics has not been reproduced.

Examples include:

event-horizon-like liquidity boundary;

tunnelling-like rare market transition;

proper-time-like asset clock;

entanglement-like contextual dependence;

gravity-like attractor curvature.

Write:

$\text{Analogy}^{A \to B}$ is heuristic only. (15.15)

These analogies may inspire models, but they are not evidence of shared physics.

15.5 The field role

A field is the domain of possible states and influences from which localized outcomes can emerge.

In physics:

Field^{P} = configuration capable of carrying physical influence. (15.16)

In finance:

Field^{f} = distribution of valuations, positions, constraints, orders, and expectations. (15.17)

In an LLM:

Field^{L} = distribution of candidate meanings, tokens, evidence relations, plans, and actions. (15.18)

The common function is:

Field = Space of Structured Possibility before Local Commitment. (15.19)

The finance field is not a quantum field.

The LLM field is not a physical wavefunction.

But both systems possess more possible structure before commitment than appears in the final observable.

15.6 The identity role

A stable system requires something that remains identifiable while its state changes.

In physics, identity may be associated with:

particle species;

conserved quantum numbers;

stable excitations;

field modes.

In finance, identity may be associated with:

legal entity;

security identifier;

contractual claim;

portfolio mandate;

account ownership.

In an LLM runtime, identity may be associated with:

task specification;

agent role;

artifact schema;

tool contract;

memory namespace;

conversation participant.

The common question is:

What remains the same while state changes? (15.20)

Without identity, phase and trace cannot be assigned coherently.

15.7 The mediation role

Systems require channels through which influence passes.

Physics uses fields and interaction carriers.

Finance uses:

price;

interest rate;

credit spread;

collateral;

contract;

payment;

information.

LLM runtimes use:

tokens;

retrieved documents;

API responses;

structured tool messages;

verifier scores;

memory objects.

The common role is:

Mediator = Typed Carrier of Influence between Identity-Bearing Components. (15.21)

A mediator must preserve enough structure for the receiver to interpret the interaction.

15.8 The binding role

Binding allows components to behave as a larger unit.

Physics contains bound states and interaction constraints.

Finance contains:

contracts;

collateral agreements;

portfolio mandates;

index membership;

corporate structure;

clearing arrangements.

LLM systems contain:

attention relations;

context binding;

schemas;

plans;

artifact contracts;

subagent coordination rules.

The common question is:

What makes several components act as one object? (15.22)

Weak binding produces fragmentation.

Excessive binding produces rigidity or lock-in.

15.9 The gate role

A gate determines whether a possible transition becomes admissible.

The common form is:

Transition occurs if $G(X,L,P) \geq G^*$. (15.23)

Where:

X = current state;

L = inherited trace;

P = protocol;

G^* = threshold.

In finance, the transition may be execution, settlement, default, rebalancing, or regime confirmation.

In an LLM, it may be answer publication, tool execution, refusal, escalation, or memory writing.

A gate is therefore not merely a filter.

It is a converter between possibility and consequence.

15.10 The trace role

A record becomes a trace when it changes future routing.

Define a passive log:

$\mathrm{Log}_{k+1} = \mathrm{Log}_k \sqcup \mathrm{Record}_k$. (15.24)

Define an active trace:

$L_{k+1} = L_k \sqcup \mathrm{Record}_k$ and $F_{k+1} = \mathrm{Update}(F_k, L_{k+1})$. (15.25)

The second condition is decisive.

A price history that changes collateral policy is trace.

An LLM correction that changes memory or verifier thresholds is trace.

The Gauge Grammar similarly distinguishes trace from passive storage: trace changes future interpretation and admissible action.

15.11 The invariance role

Local descriptions can differ.

A robust relation should survive admissible changes of frame.

Let $T_{p\to p'}$ be a valid protocol transport.

An invariant I satisfies:

$I[Z_p] = I[T_{p\to p'}(Z_p)]$. (15.26)

In finance, an invariant may be:

economically equivalent cash flow;

exposure conserved under factor re-expression;

no-arbitrage relation;

risk ordering stable across benchmark variants.

In an LLM, an invariant may be:

conclusion stable under paraphrase;

factual answer stable under source ordering;

tool result stable under equivalent interface;

policy outcome stable under stylistic changes.

Objectivity is therefore not the absence of perspective.

It is:

Objectivity = Invariance across Declared Admissible Perspectives. (15.27)

15.12 The observer-potential role

A system becomes more observer-like when it can:

select;

record;

compare;

update;

act;

preserve residual;

revise its own future projection.

A minimal observer cycle is:

$D_k \rightarrow \text{Projection}_k \rightarrow \text{Gate}_k \rightarrow \text{Trace}_k \rightarrow \text{Residual}_k \rightarrow D_{k+1}$. (15.28)

Where D_k is the current declaration or observer configuration.

A static sensor may project and record.

A more mature observer can use trace and residual to revise its declaration.

The Complex Residual Principle therefore concerns not only output selection.

It concerns self-revising bounded observation.

---

16. Quantum-Like Characteristics Reconstructed at Macroscopic Scale

16.1 Complex state

The first reconstructed characteristic is complex representation.

Physics uses complex amplitudes in a fundamental predictive formalism.

Finance and LLMs may use:

$Z = R + iQ$. (16.1)

The shared function is to preserve two orthogonal aspects of state.

The difference is crucial:

physical complex amplitude has a specific role in quantum probability and dynamics;

financial Q represents protocol-relative residual pressure;

LLM **Q** represents typed unresolved semantic pressure.

Therefore:

Same Complex Form does not imply Same Physical Meaning. (16.2)

16.2 Superposition-like possibility

A physical state may be represented:

$|\Psi\rangle = \Sigma_j c_j|j\rangle$. (16.3)

A financial market may contain simultaneous:

buy and sell intentions;

competing valuations;

alternative regimes;

contingent claims.

An LLM may contain simultaneous candidate:

tokens;

claims;

plans;

tool routes;

interpretations.

A generic possibility ensemble is:

$X = \Sigma_j a_j x_j$. (16.4)

The shared characteristic is:

Several Alternatives Remain Active before Commitment. (16.5)

But financial and LLM alternatives need not be coherent quantum superpositions.

They may be classical mixtures, ranked candidates, or unresolved model states.

16.3 Interference

For two phase-bearing contributions:

$Z_1 = A_1 \exp(i\theta_1)$. (16.6)

$Z_2 = A_2 \exp(i\theta_2)$. (16.7)

Then:

$|Z_1 + Z_2|^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\theta_1 - \theta_2)$. (16.8)

This equation shows why orientation matters.

Finance may display:

synchronized buying;

offsetting hedges;

reinforcing narratives;

cancelling factor exposures.

LLMs may display:

corroborating evidence;

contradictory instructions;

aligned retrieval;

competing answer frames.

The recurrence is mathematical when states are explicitly encoded as vectors or phases.

It remains interpretive when phase is only metaphorically assigned.

16.4 Measurement-basis dependence

Let the state be Z.

Projection in basis P produces:

$R_p = \hat{O}_p(Z)$. (16.9)

Projection in basis P′ produces:

$R_p' = \hat{O}_p'(Z)$. (16.10)

In finance:

market beta changes with benchmark;

duration changes with yield-curve basis;

value changes with accounting or discount frame.

In LLMs:

answer changes with prompt;

retrieved evidence changes with query;

admissibility changes with policy;

tool routes change with schema.

Thus:

Different Basis → Different Visible Component. (16.11)

This does not make all observations arbitrary.

The protocol must be declared, and invariants must be tested.

16.5 Collapse-like commitment

A larger candidate field becomes one accepted result:

X → Gate → R + Residual. (16.12)

In finance:

Order Field → Execution → Price Trace + Unfilled Residual. (16.13)

In an LLM:

Candidate Field → Verifier Gate → Answer Trace + Semantic Residual. (16.14)

The common role is selection under constraint.

The term collapse-like is justified only at this functional level.

No claim is made that financial execution or token selection follows the physical measurement postulates of quantum mechanics.

16.6 Decoherence

Define coherence:

$\rho = |\Sigma_j w_j \exp(i\theta_j)| / \Sigma_j w_j$. (16.15)

Environmental interaction, noise, or internal conflict may reduce ρ:

$d\rho/dt < 0$. (16.16)

In finance, decoherence-like behaviour may result from:

sector divergence;

liquidity fragmentation;

policy uncertainty;

conflicting macro signals;

broken correlations.

In an LLM, it may result from:

long-context drift;

retrieval contradiction;

tool disagreement;

competing instructions;

unstable sampling.

The shared role is:

Stable Collective Orientation Is Lost through Interaction with an Uncontrolled Environment. (16.17)

This is classical or informational decoherence unless stronger quantum criteria are demonstrated.

16.7 Phase-locking

Interacting components may synchronize:

$d\theta_j/dt = \omega_j + K\rho\sin(\Phi - \theta_j) + \xi_j$. (16.18)

When coupling K exceeds an effective threshold:

ρ rises. (16.19)

Finance may exhibit phase-lock through:

common factor exposure;

index flows;

shared leverage;

policy dependence;

herding.

LLMs may exhibit phase-lock through:

strong prompt framing;

repeated semantic attractors;

aligned retrieval;

verifier convergence;

self-consistency agreement.

Phase-lock can improve coherence.

But excessive lock can eliminate useful diversity.

Thus:

Coherence is beneficial up to the point where differentiation disappears. (16.20)

16.8 Symmetry breaking

Suppose several orientations are initially equivalent:

$V(\theta_1) = V(\theta_2) = \ldots = V(\theta_n)$. (16.21)

A perturbation or interaction causes one orientation to dominate:

$\theta^* = \operatorname{argmin}_\theta V_{eff}(\theta)$. (16.22)

Physics uses symmetry breaking to describe the selection of a particular state from symmetric possibilities.

Finance may select:

risk-on or risk-off;

inflation or disinflation narrative;

growth or value leadership;

liquidity expansion or contraction.

An LLM may select:

one interpretation of an ambiguous prompt;

one plan;

one answer frame;

one tool route.

The shared function is:

A formerly symmetric possibility set develops one historically privileged orientation. (16.23)

16.9 Criticality

Near a critical point, a small perturbation may produce a large response.

Define susceptibility:

$\chi = \partial R / \partial u$. (16.24)

Near a critical regime:

$|\chi| \gg 1$. (16.25)

In finance, high susceptibility may arise from:

leverage;

thin liquidity;

crowded trades;

margin thresholds;

refinancing cliffs.

In an LLM, it may arise from:

ambiguous prompts;

conflicting high-priority instructions;

attractor boundaries;

fragile tool routing;

adversarial context.

A small input change can then cause a large output change.

Criticality is not uniquely quantum.

It is a general characteristic of nonlinear systems near transition boundaries.

16.10 Attractors

Let the system evolve:

$dX/dt = F(X)$. (16.26)

An attractor A satisfies:

$X(t) \to A$ for a set of initial states. (16.27)

Finance may develop:

valuation attractors;

momentum attractors;

benchmark attractors;

liquidity traps;

narrative basins.

LLMs may develop:

answer templates;

concept basins;

refusal basins;

stylistic attractors;

repetitive reasoning loops.

An attractor helps explain stability.

It may also explain pathological lock-in.

16.11 Entanglement-like dependence

A joint state is separable when:

$P(X_1,X_2) = P(X_1)P(X_2)$. (16.28)

A nonseparable classical dependence satisfies:

$P(X_1,X_2) \neq P(X_1)P(X_2)$. (16.29)

Financial assets connected through derivatives, collateral, funding, and legal agreements may be difficult to analyse independently.

An LLM answer may depend jointly on prompt, retrieval, tool state, and memory.

This supports an entanglement-like analogy.

But:

Classical Nonseparability ≠ Quantum Entanglement. (16.30)

True quantum entanglement involves a specific tensor-product state structure and experimentally testable non-classical correlations.

The present framework does not establish those properties for markets or LLMs.

16.12 Noncommutativity-like ordering effects

Suppose two operations A and B are applied.

In general:

AB(X) may differ from BA(X). (16.31)

Finance examples include:

hedge before liquidation versus liquidation before hedge;

regulation before crisis versus regulation after crisis;

accounting recognition before refinancing versus after refinancing.

LLM examples include:

retrieve before drafting versus draft before retrieving;

tool verification before answer commitment versus afterward;

memory write before validation versus after validation.

Define the ordering residual:

$$KAB(X) = AB(X) - BA(X). \quad (16.32)$$

If:

$$KAB(X) \neq 0. \quad (16.33)$$

Order matters.

This is a genuine noncommutative property of operations.

It is not automatically the canonical noncommutativity of quantum observables.

But it demonstrates that measurement and intervention order can matter macroscopically.

16.13 Backreaction

An observation becomes part of the system.

Define:

$$R_k = \hat{O}_k(X_k). \quad (16.34)$$

Then:

$$X_{k+1} = F(X_k, R_k, L_k). \quad (16.35)$$

In finance:

price observation → trading response → new price. (16.36)

In an LLM:

answer → user response → new context → new answer. (16.37)

Backreaction is therefore common in systems containing active observers.

What is unusual in physics is not merely that backreaction exists, but the precise laws governing it.

16.14 Gauge-like frame transformation

Suppose local representations change under:

$Z_p' = T_{p \to p'}(Z_p)$. (16.38)

A valid transport preserves declared invariants:

$I(Z_p') = I(Z_p)$. (16.39)

In finance:

the same exposure may be expressed in currency, duration, factor, collateral, or accounting coordinates;

economically equivalent portfolios should preserve relevant value relations.

In an LLM:

equivalent prompts should preserve core factual relations;

equivalent tool interfaces should preserve result semantics;

translation between languages should preserve the declared proposition.

The useful engineering question is:

What should remain unchanged when the frame changes? (16.40)

This is one of the strongest cross-domain contributions of gauge-style reasoning.

16.15 Emergent geometry

A system's relational structure can define distances.

For states x_i and x_j:

$d^2(i,j) = (x_i - x_j)^T G (x_i - x_j)$. (16.41)

In finance, G may arise from:

covariance;

correlation;

information geometry;

network exposure;

liquidity coupling.

In an LLM, G may arise from:

embedding geometry;

attention;

representation similarity;

retrieval linkage;

verifier sensitivity.

The geometry is emergent because it is produced by relations among states rather than imposed as ordinary physical distance.

Again:

Semantic or Financial Geometry ≠ Physical Spacetime Geometry. (16.42)

But all three involve measurable structure generated by relations.

16.16 Quantum-like recurrence classification

Characteristic	Finance status	LLM status	Strongest defensible reading
Complex state	Direct engineered representation	Direct engineered representation	Same form, different meaning
Projection	Direct linear-statistical structure	Direct runtime structure	Strong operational recurrence
Relative phase	Measurable after declared encoding	Measurable through representation alignment	Model-dependent but testable
Interference	Direct under phase/vector aggregation	Direct under vector or evidence aggregation	Classical or informational recurrence
Superposition	Usually mixture or candidate ensemble	Candidate ensemble	Not necessarily quantum coherent
Collapse	Transaction or regime commitment	Token, answer, tool, or memory commitment	Functional analogy
Decoherence	Loss of market coordination	Loss of semantic coordination	Classical/informational
Entanglement	Coupled nonseparability	Contextual nonseparability	Analogy unless quantum criteria hold
Noncommutativity	Order-sensitive operations	Order-sensitive runtime	Genuine ordering effect, not necessarily quantum observable algebra
Symmetry breaking	Regime selection	Interpretation or plan selection	Strong nonlinear-system recurrence

Characteristic	Finance status	LLM status	Strongest defensible reading
Criticality	Market instability	Prompt or runtime instability	General complex-system property
Backreaction	Reflexive pricing	Context and memory update	Strong operational recurrence
Imaginary time	Selection-depth analogue	Unresolved semantic-depth analogue	Engineered analogue
Invariance	Cross-frame economic relation	Prompt and tool robustness	Strong engineering principle

16.17 What remains genuinely mysterious

The framework makes some characteristics more intuitive.

It does not dissolve the deepest quantum problems.

It does not derive:

Born's rule;

Planck's constant;

quantum spin;

Bell inequality violations;

contextuality in the strict quantum-foundational sense;

unitary state evolution;

identical-particle statistics;

quantum field vacuum structure;

relativistic quantum causality.

Therefore:

Macroscopic Recurrence of Grammar Does Not Explain All Quantum Physics. (16.43)

The important result is narrower:

> Some features that appear mysterious when encountered together in quantum mechanics—phase, projection, interference, gated commitment, observer dependence, residual possibility, and trace—also arise as understandable system roles at macroscopic and engineered scales.

This may reduce conceptual distance without reducing physical specificity.

---

17. The General Time-Generation Chain

17.1 The question of origin

The preceding analysis raises a deeper question.

Can time-like order arise from:

chaos;

interaction;

synchronization;

self-organization;

observation;

memory;

self-reference?

The proposed answer is:

No single ingredient is sufficient.

Time-like structure emerges from their ordered composition.

The full sequence is:

Fluctuation → Interaction → Correlation → Synchronization → Phase Clock → Gate → Trace → Ledger → Feedback → Internal Time. (17.1)

Each transition adds a property that the previous stage lacked.

17.2 Stage 1 — Fluctuation supplies alternatives

Begin with a system containing possible configurations:

$X = \{x_1, x_2, \ldots, x_n\}$. (17.2)

The configurations may change irregularly.

At this stage there is variation, but not necessarily order.

The system may contain external update time s:

$X_{k+1} = F(X_k, \xi_k)$. (17.3)

Where ξ_k is noise or perturbation.

An external observer can count updates.

But the system does not yet possess a meaningful internal clock.

Thus:

17.3 Stage 2 — Interaction creates dependence

Let components interact:

$dx_i/ds = f_i(x_i) + \Sigma_j K_{ij} g(x_i,x_j)$. (17.5)

Interaction means that one component's future depends on others.

This creates:

correlation;

propagation;

collective modes;

possible synchronization.

But interaction alone does not create a stable clock.

A strongly interacting system may remain chaotic.

17.4 Stage 3 — Synchronization creates phase order

Suppose components possess phases θ_i.

Define:

$C = (1/W)\Sigma_i w_i \exp(i\theta_i) = \rho \exp(i\Phi)$. (17.6)

When:

$\rho \to 0$. (17.7)

The system lacks a stable collective phase.

When:

ρ rises above a threshold ρ^*. (17.8)

A collective orientation appears.

The phase Φ can now serve as a clock hand.

This is the first internally generated temporal signal.

Yet:

Clock Signal ≠ Historical Time. (17.9)

A phase can cycle forever without preserving history.

17.5 Stage 4 — Unwrapped phase creates chronology

Visible phase is periodic:

$$\exp[i(\Phi + 2\pi)] = \exp(i\Phi). \quad (17.10)$$

Define:

$$\tilde{\Phi} = \Phi + 2\pi N. \quad (17.11)$$

Now the system can distinguish repeated cycles.

A primitive phase chronology is:

$$\tau_{phase} = [\tilde{\Phi}(s) - \tilde{\Phi}(s_0)] / \Omega^*. \quad (17.12)$$

This supplies ordering and duration relative to the collective process.

But it remains reversible in principle.

A cycle can be traversed forward or backward unless another operation breaks the symmetry.

17.6 Stage 5 — The gate creates selective asymmetry

Define:

$$G \in [0,1]. \quad (17.13)$$

The gate determines which phase movements become consequential.

Realized increment:

$$d\tau = G|d\tilde{\Phi}| / \Omega^*. \quad (17.14)$$

Unresolved increment:

$$dT = (1 - G)|d\tilde{\Phi}| / \Omega^*. \quad (17.15)$$

Complex temporal state:

$$d\zeta = d\tau + i\, dT. \quad (17.16)$$

The gate creates two modes:

admitted development;

unresolved development.

This introduces selection.

But the selected event must still persist.

17.7 Stage 6 — Trace creates irreversibility

Let an admitted event produce:

$\text{Trace}_k = \text{Commit}(X_k, G_k, P_k)$. (17.17)

The ledger becomes:

$L_{k+1} = L_k \sqcup \text{Trace}_k$. (17.18)

A trace is time-bearing when:

Future_{k+1} depends on Trace_k. (17.19)

Therefore:

$X_{k+1} = F(X_k, L_{k+1})$. (17.20)

The system can no longer return to an earlier complete state merely by restoring its visible coordinate.

This creates the arrow:

$\text{Trace}_k \rightarrow \text{Future}_{k+1}$. (17.21)

17.8 Stage 7 — The ledger creates history

A sequence of traces is:

$L_n = \{\text{Trace}_1, \text{Trace}_2, \ldots, \text{Trace}_n\}$. (17.22)

Internal time is the ordered consequence structure of the ledger:

$\tau(n) = \text{OrderWeight}(L_n)$. (17.23)

A simple form is:

$\tau(n) = \Sigma_{k=1}^{n} W_k G_k$. (17.24)

Where W_k measures event consequence.

This definition allows internal ticks to differ in weight.

A major transition can produce more internal time than many repetitive updates.

Thus:

Equal External Duration does not imply Equal Internal Historical Depth. (17.25)

17.9 Stage 8 — Feedback makes time endogenous

Suppose the ledger changes the next gate:

$G_{k+1} = G(X_{k+1}, L_{k+1})$. (17.26)

And changes the next projection:

$\hat{O}_{k+1} = U(\hat{O}_k, L_{k+1}, R_k)$. (17.27)

Then the system's history modifies the mechanism by which future history is generated.

The complete loop is:

$\text{Phase}_k \rightarrow \text{Gate}_k \rightarrow \text{Trace}_k \rightarrow \text{Ledger}_{k+1} \rightarrow \text{Projection}_{k+1} \rightarrow \text{Phase}_{k+1}$. (17.28)

At this point, the system partly creates its own cadence.

This is endogenous time.

17.10 The minimum time-bearing architecture

A minimal internal-time system requires at least five components.

Condition 1 — Distinguishable states

There must be more than one possible state:

$|X| > 1$. (17.29)

Condition 2 — Ordered transition

There must be a relation:

$X_k \rightarrow X_{k+1}$. (17.30)

Condition 3 — Selective gate

Not every fluctuation counts equally:

G_k distinguishes admitted from unresolved change. (17.31)

Condition 4 — Persistent trace

Accepted transitions leave records:

$L_{k+1} \neq L_k$. (17.32)

Condition 5 — Trace-conditioned future

The record affects later evolution:

$\Pr(X_{k+2} \mid X_{k+1}, L_{k+1}) \neq \Pr(X_{k+2} \mid X_{k+1})$. (17.33)

Together:

$\text{TimeBearingSystem} \Leftrightarrow \text{Distinction} \wedge \text{Transition} \wedge \text{Gate} \wedge \text{Trace} \wedge \text{Inheritance}$. (17.34)

Phase is not logically necessary for all forms of time.

But phase supplies a natural internal clock and makes the complex-time construction possible.

17.11 Clock, timeline, and history

Three concepts must be separated.

Clock

A repeatable process:

Clock = exp(iΦ). (17.35)

It tells the system where it is in a cycle.

Timeline

Accumulated unwrapped phase:

Timeline = $\tilde{\Phi}$. (17.36)

It distinguishes successive cycles.

History

Ordered trace that changes future conditions:

History = L. (17.37)

Therefore:

Clock ≠ Timeline ≠ History. (17.38)

The complete temporal architecture is:

Clock + Cycle Count + Consequential Ledger. (17.39)

This distinction is important for physics, finance, and LLMs.

17.12 Realized time and unresolved time

The proposed complex coordinate is:

ζ = τ + iT. (17.40)

Where:

τ = admitted, trace-producing development;

T = unresolved, non-trace-producing development.

The SMFT imaginary-time paper proposes a related distinction: semantic time advances through realized collapse events, while unresolved phase evolution accumulates when collapse is deferred.

The present article generalizes that architecture operationally.

In finance:

$\zeta^f = \tau_m + iT_m$. (17.41)

In an LLM:

$\zeta^L = \tau^L + iT^L$. (17.42)

The physical interpretation remains separate:

ζ^P cannot be identified with ζ^f or ζ^L without a derivation connecting the underlying dynamics. (17.43)

The finance and LLM constructions are models of unresolved versus realized development.

They are not demonstrations of physical Euclidean time.

17.13 Time emerging from chaos

Chaos can contribute variation and sensitivity.

But chaos alone does not generate a stable internal time.

A chaotic process may provide:

fluctuations;

candidate transitions;

entropy;

instability;

sensitivity to perturbation.

To generate time-like order, chaos must be organized.

The transition is:

Chaos → Correlated Interaction → Stable Phase or Event Ordering → Selective Trace. (17.44)

Therefore:

Chaos Supplies Possibility; It Does Not by Itself Supply History. (17.45)

17.14 Time emerging from self-organization

Self-organization can generate:

collective phase;

stable identity;

boundaries;

attractors;

repeated transition patterns.

But a self-organized pattern without internal trace may still possess only externally counted time.

For example, a repeating pattern may be recognized by an outside observer.

For the system to possess stronger internal time, its past must influence its own future disclosure.

Thus:

Self-Organization + Internal Trace > Self-Organization Alone. (17.46)

17.15 Time emerging from multiple interaction

Multiple interaction is essential because it permits:

phase coupling;

mediation;

competition;

synchronization;

collective order.

But a large interacting system may remain memoryless.

The missing condition is:

Interaction must produce inherited consequence. (17.47)

Therefore:

Multiple Interaction + Gate + Trace = Candidate Time Generator. (17.48)

17.16 Time emerging from self-reference

Self-reference is not required for a primitive ordered sequence.

But it is important for a self-generated clock.

Let the system's output become its future input:

$Y_k = \hat{O}_k(X_k)$. (17.49)

$X_{k+1} = F(X_k, Y_k)$. (17.50)

Then:

System Output$_k$ becomes System Condition$_{k+1}$. (17.51)

In finance:

price → expectation → order → price. (17.52)

In an LLM:

answer → user response or memory → future context → answer. (17.53)

Self-reference creates recursive temporal curvature because the past output changes the rules of later projection.

17.17 Multiple internal clocks

Complex systems may contain several clocks.

Let subsystem a have:

$d\tau_a/ds = G_a|d\tilde{\Phi}_a/ds| \,/\, \Omega_a$. (17.54)

Let subsystem b have:

$d\tau_\beta/ds = G_\beta|d\tilde{\Phi}_\beta/ds| \,/\, \Omega_\beta$. (17.55)

Their clock ratio is:

$\chi_{a\beta} = d\tau_a/d\tau_\beta$. (17.56)

If $\chi_{a\beta}$ is stable, the clocks are synchronized.

If it varies, the subsystems experience different internal development rates.

In finance:

options may reprice faster than cash assets;

credit may lag equities;

illiquid assets may remain stale;

policy clocks may advance slowly and then jump.

In LLM systems:

token generation may advance quickly;

evidence verification may advance slowly;

tools may operate asynchronously;

memory may update only after episode closure.

A mature runtime must coordinate these clocks.

17.18 Synchronization and world formation

A coherent world requires enough clock agreement for components to share ordered events.

Let subsystem phases be $\Phi_1,\ldots,\Phi_n$.

Clock coherence is:

$\rho\text{time} = |\Sigma_j w_j \exp(i\Phi_j)| \,/\, \Sigma_j w_j$. (17.57)

When:

ρtime high. (17.58)

Subsystems can agree on event order and shared state.

When:

ρtime low. (17.59)

The system experiences timing conflict, inconsistent trace, or fragmented causality.

In finance this may appear as settlement mismatch or cross-market dislocation.

In LLM systems it may appear as stale tool state, inconsistent memory, or actions based on outdated context.

The general implication is:

Shared Worlds Require Sufficient Synchronization of Trace-Bearing Clocks. (17.60)

17.19 The arrow of time from residual and trace

A system generates an arrow when:

some events are admitted;

some possibilities remain residual;

admitted events alter future conditions;

the exact prior state cannot be reconstructed from the visible output alone.

Let:

$X_k \to (R_k, Q_k, L_{k+1})$. (17.61)

If the projection is many-to-one:

$\hat{O}(X_1) = \hat{O}(X_2)$ for some $X_1 \neq X_2$. (17.62)

Then visible reversal is insufficient to restore the former full state.

Residual and trace preserve asymmetry.

Thus:

Arrow of Time = Selective Commitment + Information Loss under Projection + Trace Inheritance. (17.63)

This is an effective-system explanation.

It does not replace thermodynamic or cosmological accounts of the physical arrow of time.

But it shows how irreversible history can arise in macroscopic and engineered systems.

17.20 The universal time-generation proposition

The proposed cross-domain proposition is:

> A system develops an effective internal time when interacting possibilities generate a repeatable phase or event order, a state-dependent gate distinguishes consequential from unresolved change, admitted events leave persistent traces, and those traces alter future transitions.

Formally:

InternalTime^D exists if $\text{PhaseOrder}^D \wedge \text{Gate}^D \wedge \text{Trace}^D \wedge \text{Inheritance}^D$. (17.64)

A stronger endogenous form requires self-reference:

EndogenousTime^D exists if $\text{InternalTime}^D \wedge \text{ObserverUpdate}^D$. (17.65)

Imaginary-time-like depth exists when unresolved evolution continues:

$T^D > 0$ if $|d\tilde{\Phi}^D| > 0$ and $G^D < 1$. (17.66)

Frozen-time-like behaviour occurs when:

$d\tau^D/ds \to 0$ while $dT^D/ds > 0$. (17.67)

Sudden realization occurs when:

G^D changes from low to high and accumulated T^D is converted into a large $\Delta\tau^D$. (17.68)

17.21 The central cross-domain generator

The entire argument can now be compressed into one sequence:

Many Possibilities → Relative Orientation → Interaction → Collective Phase → Bounded Projection → Gate → Trace + Residual → Ledger → Recursive Time. (17.69)

Or:

Field → Phase → Projection → Commitment → History. (17.70)

The Complex Residual Principle adds the missing qualification:

Projection never exhausts the field. (17.71)

Therefore:

History is generated together with residual. (17.72)

A mature system must carry both:

L_{k+1} = Accepted History. (17.73)

Q_{k+1} = Unresolved Future Condition. (17.74)

The next state depends on both:

$X_{k+1} = F(L_{k+1}, Q_{k+1}, \text{Environment}_k)$. (17.75)

17.22 Part V conclusion

The comparison across physics, finance, and LLMs supports a disciplined middle conclusion.

Several structures commonly associated with quantum mystery are not wholly absent from macroscopic life.

Complex representation, phase-dependent combination, projection dependence, selection gates, observer backreaction, coherence, decoherence, symmetry breaking, criticality, trace, and variable internal clocks can all be reconstructed in ordinary or engineered systems.

What differs is:

substrate;

governing law;

mathematical strength;

empirical signature;

ontological interpretation.

The result is not:

Everything Is Quantum. (17.76)

The result is:

Quantum-Style Grammar Can Recur without Quantum Substance. (17.77)

Finance provides a measurable macroscopic realization.

LLMs provide a programmable observer realization.

Physics provides the deepest and most exact known realization.

This leads to the article's broader proposition:

> What appears mysterious in physics may sometimes be a fundamental implementation of organizational principles that become easier to recognize when they reappear in markets and artificial observers.

The next part must determine where this interpretation genuinely explains something, where it merely renames familiar processes, how it can be falsified, and which claims remain beyond the evidence.

Part VI — Interpretation, Limits, and Research Program

18. What the Framework Explains

18.1 From quantum mystery to organizational mechanism

The framework does not claim to explain every mystery of quantum physics.

It offers a narrower but potentially important reinterpretation:

> Several structures that appear strange when encountered only in microscopic physics become easier to understand when reconstructed as ordinary functions of bounded observation, interaction, selection, residual preservation, and trace-bearing self-organization.

The common architecture is:

Possibility → Interaction → Orientation → Projection → Gate → Trace + Residual → Updated Possibility. (18.1)

When the architecture includes collective phase, it becomes:

Many Possibilities → Phase Organization → Bounded Projection → Selective Commitment → History. (18.2)

When unresolved development is preserved:

History Is Generated together with Residual. (18.3)

This is the key conceptual shift.

The physical formalism remains special.

But part of its apparent strangeness may arise because physics presents these roles at a highly fundamental level, where the mechanisms are less intuitively visible.

Finance and LLMs expose comparable roles in systems whose components, gates, records, and

failures can be observed directly.

18.2 Why complex numbers may be structurally natural

A real-valued model reports one admitted coordinate:

State = R. (18.4)

But a bounded observer rarely captures the whole operative state.

A more honest representation is:

State = R + Residual. (18.5)

The complex form writes:

Z = R + iQ. (18.6)

The role of i is to mark a direction that remains dynamically relevant while lying outside the current admission axis.

This gives:

R = admitted projection. (18.7)

Q = retained non-admitted structure. (18.8)

The framework therefore interprets complex representation as a solution to a general observer problem:

> How can a system preserve what affects future dynamics without falsely presenting it as part of the currently admitted observable?

A scalar real model tends to choose between two unsatisfactory options:

force the residual into R;

discard it.

The complex residual model provides a third option:

preserve the residual in an orthogonal coordinate.

This may explain why complex representation repeatedly becomes useful in systems involving:

oscillation;

phase;

directional relation;

latent pressure;

unresolved alternatives;

frame transformation;

delayed realization.

The conclusion is not that all such systems require literal complex numbers.

It is:

Complex Geometry Is a Natural Grammar for Projection plus Retained Orthogonality. (18.9)

18.3 Why measurement appears active

In a passive-copy model:

Measurement(X) = Description(X). (18.10)

The measured system is assumed to remain independent of the measurement event.

But physics, finance, and LLM runtimes often involve active measurement.

The stronger form is:

$R_k = \hat{O}_k(X_k)$. (18.11)

$X_{k+1} = F(X_k, R_k, L_k)$. (18.12)

The output of measurement becomes part of the next state.

In physics

The interaction required to obtain an outcome affects the state assigned to the system.

In finance

A price does not merely report value.

It affects:

orders;

collateral;

risk limits;

benchmarks;

narratives;

regulation.

In an LLM runtime

An answer affects:

the user's next prompt;

conversation context;

tool selection;

memory;

later verification.

Thus:

Observation Becomes Causal when the Record Re-enters the Field. (18.13)

This makes observer backreaction less mysterious at the general-system level.

The special scientific question then becomes not whether observation can affect a system, but exactly which law governs the effect in each domain.

18.4 Why observer dependence is unavoidable

A bounded observer never extracts total reality.

The bounded-observer split is:

$MDL^T(X) = S^T(X) + H^T(X)$. (18.14)

Where:

$S^T(X)$ = structure extractable under observer bound T;

$H^T(X)$ = residual unpredictability under the same bound.

A disciplined claim therefore requires a declared protocol:

$P = (B,\Delta,h,u)$. (18.15)

Where:

B = boundary;

Δ = observation or aggregation rule;

h = time or state horizon;

u = admissible intervention family.

The Gauge Grammar explicitly uses this bounded-observer and protocol-first discipline to prevent cross-domain role mappings from becoming unrestricted metaphysical claims. It also states that quantum and gauge structures are transferred functionally rather than literally.

Observer dependence therefore need not mean that reality is arbitrary.

It means:

$\text{VisibleStructure}_p = \text{Projection}(\text{World} \mid \text{Observer},P)$. (18.16)

Another protocol may disclose another component:

$\text{VisibleStructure}_p' = \text{Projection}(\text{World} \mid \text{Observer}',P')$. (18.17)

Objectivity is then reconstructed through admissible invariance:

$\text{Invariant}[\text{VisibleStructure}_p] = \text{Invariant}[\text{VisibleStructure}_p']$. (18.18)

Thus:

Objectivity ≠ View from Nowhere. (18.19)

Objectivity = Stable Relation across Declared Valid Frames. (18.20)

18.5 Why projection necessarily creates residual

Projection is selective.

Let $\hat{P}$ be a projector:

$\hat{P}^2 = \hat{P}$. (18.21)

The complementary projector is:

$\hat{Q} = I - \hat{P}$. (18.22)

Then:

$|X\rangle = \hat{P}|X\rangle + \hat{Q}|X\rangle$. (18.23)

The first term is admitted.

The second is residual relative to $\hat{P}$.

If only the admitted component is retained:

$\text{ReportedState} = \hat{P}|X\rangle$. (18.24)

The operative state remains:

$\text{OperativeState} = \hat{P}|X\rangle + \hat{Q}|X\rangle$. (18.25)

False closure arises when:

ReportedState is treated as OperativeState. (18.26)

The same error appears across domains.

Finance

Price is treated as total value.

LLM

Fluent output is treated as total epistemic state.

Organization

Reported KPI is treated as total system health.

Science

Model output is treated as total reality.

The Complex Residual Principle therefore offers one general diagnostic:

$\mathrm{FalseClosure}_p = \mathrm{OmittedResidual}_p \times \mathrm{ConsequenceSensitivity}_p$. (18.27)

A small omitted residual may be harmless.

A large residual in a highly sensitive system can produce abrupt failure.

18.6 Why imaginary-time-like coordinates may recur

The article distinguishes two kinds of development.

Realized development

A transition passes a gate and becomes trace:

$d\tau = G|d\tilde{\Phi}| / \Omega^*$. (18.28)

Unresolved development

A transition continues internally but does not become trace:

$dT = (1 - G)|d\tilde{\Phi}| / \Omega^*$. (18.29)

The combined coordinate is:

$d\zeta = d\tau + i\, dT$. (18.30)

This construction explains why an imaginary-time-like coordinate may be useful even in a macroscopic system.

T records development that is:

dynamically active;

phase-bearing;

cumulative;

not yet admitted into historical trace.

The SMFT imaginary-time paper proposes a related semantic distinction. It treats τ as discrete collapse-defined time and iT as continuous unresolved phase rotation that accumulates when projection and threshold conditions do not produce a new tick.

The present article does not identify T^f or T^L with physical imaginary time.

It identifies a recurring functional need:

Systems Need a Coordinate for Development that Has Occurred without Yet Becoming History. (18.31)

That need appears in:

latent market pressure;

unresolved cognition;

model deliberation;

delayed institutional decision;

phase motion before commitment.

18.7 Why time requires more than change

A changing system does not automatically possess internal time.

Let:

$X_{k+1} \neq X_k$. (18.32)

This proves change.

It does not yet prove internal historical order.

The stronger conditions are:

Distinguishable State ∧ Ordered Transition ∧ Gate ∧ Trace ∧ Inheritance. (18.33)

The mechanism is:

Change → Selected Change → Recorded Change → Future Condition. (18.34)

Time-like order therefore emerges when change becomes consequential.

The central proposition is:

Time-Like Structure = Phase or Event Ordering + Selective Commitment + Trace Inheritance. (18.35)

The phase supplies rhythm.

The gate supplies discrimination.

The trace supplies irreversibility.

The ledger supplies history.

Feedback makes the process endogenous.

18.8 Why phase is a clock but not yet history

The clock hand is:

$\exp(i\Phi)$. (18.36)

The timeline requires unwrapped phase:

$\tilde{\Phi} = \Phi + 2\pi N$. (18.37)

History requires a ledger:

$L_{k+1} = L_k \sqcup \mathrm{Trace}_k$. (18.38)

Therefore:

Clock ≠ Timeline ≠ History. (18.39)

This three-part distinction resolves several confusions.

A market can exhibit cyclical phase without strong historical change.

An LLM can generate many tokens without semantic progress.

A physical oscillator can mark time without itself explaining the arrow of time.

The complete architecture is:

Clock + Cycle Count + Consequence-Bearing Ledger. (18.40)

18.9 Why finance is more than an analogy

Finance is important because the proposed roles are externally observable.

A market contains:

explicit alternative valuations;

measurable covariance;

declared projection models;

transaction gates;

settlement records;

residual risk;

collective synchronization;

reflexive feedback.

CAPM supplies:

$\beta_i = Cov(x_i,x_m) / Var(x_m)$. (18.41)

And geometrically:

$\beta_i = (\sigma_i/\sigma_m)\cos\theta_i$. (18.42)

The market-phase model supplies:

$C_m = [\Sigma_i w_i \exp(i\theta_i)] / \Sigma_i w_i = \rho_m \exp(i\Phi_m)$. (18.43)

The gate supplies:

$d\tau_m = G_m|d\tilde{\Phi}_m| / \Omega_m$. (18.44)

$dT_m = (1 - G_m)|d\tilde{\Phi}_m| / \Omega_m$. (18.45)

The ledger supplies:

$L_{m,k+1} = Update(L_{m,k},Trace_{m,k},Residual_{m,k})$. (18.46)

The full construction is therefore not only philosophical.

Its components can be estimated, simulated, compared, and rejected.

Finance becomes a macroscopic laboratory for the proposed grammar.

18.10 Why LLMs are more than an analogy

LLMs provide a programmable version of the architecture.

The runtime state is:

$Z^L = R^L + i\mathbf{Q}^L$. (18.47)

The residual vector may include:

$\mathbf{Q}^L = [Qfact,Qretrieval,Qinstruction,Qpolicy,Qtool,Qambiguity,Qmemory,Qconsequence]^T$. (18.48)

The gate is:

$G^L = GintentGevidenceGconsistencyGpolicyGtoolGformatGconsequence$. (18.49)

The semantic time split is:

$d\tau^L = G^L|d\tilde{\Phi}^L| / \Omega^L$. (18.50)

$dT^L = (1 - G^L)|d\tilde{\Phi}^L| / \Omega^L$. (18.51)

The trace update is:

$L^{L}_{,k+1} = \text{Update}(L^{L}_{,k}, \text{Output}_k, \text{Tool}_k, \text{Residual}_k)$. (18.52)

Unlike foundational physics, these components can be modified deliberately.

Researchers can:

change gates;

preserve residual categories;

vary prompt frames;

compare tool routes;

control memory writes;

measure semantic progress;

audit false closure.

The Gauge Grammar identifies precisely this AI engineering direction: fluent output is not equivalent to governed output, and equivalent prompt frames should preserve governed conclusions within a declared tolerance.

18.11 What counts as a genuine explanation

A cross-domain model explains something only if it does more than rename it.

The framework earns explanatory status if it provides at least one of the following:

compression of previously separate phenomena;

a new measurable variable;

a new prediction;

a new intervention;

a clearer failure classification;

a falsifiable boundary.

Let explanatory value be:

$E_{\text{framework}} = C_{\text{compression}} + M_{\text{measurement}} + P_{\text{prediction}} + I_{\text{intervention}} + F_{\text{falsifiability}}$. (18.53)

If the framework offers only metaphor:

$M_{\text{measurement}} = 0$. (18.54)

$P_{\text{prediction}} = 0$. (18.55)

$I_{\text{intervention}} = 0$. (18.56)

$F_{falsifiability} = 0$. (18.57)

Then:

$E_{framework}$ remains low. (18.58)

The article therefore treats empirical and engineering tests as essential, not optional.

18.12 Five explanatory propositions

The framework's explanatory contribution can be summarized in five propositions.

Proposition 1 — Complex Residual

Bounded projection naturally creates an admitted coordinate and a retained residual coordinate:

$Z_p = R_p + iQ_p$. (18.59)

Proposition 2 — Collective Phase

Interacting oriented components can create an emergent collective clock:

$C = \rho \exp(i\Phi)$. (18.60)

Proposition 3 — Gate Partition

Phase development can be divided into realized and unresolved temporal coordinates:

$d\zeta = [G + i(1 - G)]|d\tilde{\Phi}| / \Omega^*$. (18.61)

Proposition 4 — Ledgered Arrow

Accepted events become time-bearing when their traces alter future transition rules:

$L_{k+1} = \mathrm{Update}(L_k, \mathrm{Trace}_k, \mathrm{Residual}_k)$. (18.62)

Proposition 5 — Recursive Observer

An endogenous clock appears when trace updates the observer and therefore changes later projection:

$\hat{O}_{k+1} = \mathrm{UpdateObserver}(\hat{O}_k, L_{k+1}, \mathrm{Residual}_k)$. (18.63)

Together:

Complex Residual + Collective Phase + Gate + Ledger + Observer Update → Emergent Internal Time. (18.64)

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19. What the Framework Does Not Prove

19.1 Why strict limits strengthen the argument

A framework linking physics, finance, and LLMs faces an obvious danger.

Because the vocabulary is broad, almost any phenomenon could be described afterward as:

phase;

projection;

collapse;

residual;

attractor;

time.

If every event can be absorbed into the model, the model predicts nothing.

The framework must therefore declare anti-overreach conditions.

The Gauge Grammar states the relevant rule directly:

Functional Homology ≠ Substance Identity. (19.1)

It also warns that protocol quality determines diagnostic quality, proxies may create false structure, not all residual should be eliminated, and the stronger substrate thesis remains speculative.

19.2 It does not prove that finance is quantum mechanical

Financial markets involve:

legal institutions;

human expectations;

algorithms;

contracts;

accounting;

classical communication;

macroscopic settlement.

The fact that complex numbers, phases, interference-like aggregation, and projection operators are useful does not establish physical quantum dynamics.

The framework does not show:

MarketState obeys Schrödinger Evolution. (19.2)

It does not show:

FinancialMeasurement obeys the Quantum Measurement Postulates. (19.3)

It does not show:

FinancialProbabilities arise from Born's Rule. (19.4)

The safe conclusion is:

Finance Can Instantiate Quantum-Like Grammar without Quantum Substance. (19.5)

19.3 It does not prove that LLMs are quantum systems

An LLM runs on physical hardware that is ultimately governed by quantum physics, as all matter is.

But its ordinary computational operation is not thereby a quantum algorithm.

The candidate-state notation:

$|\Psi^L\rangle = \Sigma_j c_j |a_j\rangle$. (19.6)

is a functional runtime representation.

It does not establish:

coherent quantum superposition of answer branches;

physical entanglement among tokens;

quantum interference inside ordinary transformer inference;

quantum speedup;

quantum measurement during decoding.

Therefore:

LLMCandidateEnsemble ≠ PhysicalQuantumState. (19.7)

19.4 It does not derive Born's rule

Quantum measurement probabilities use:

$p_j = |c_j|^2$. (19.8)

The present finance and LLM models may use normalized scores:

$p_j = s_j / \Sigma_k s_k$. (19.9)

Or softmax:

$p_j = \exp(z_j) / \Sigma_k \exp(z_k)$. (19.10)

These formulas are not equivalent merely because they generate probabilities.

The framework has not derived why physical outcome probability must be proportional to squared amplitude.

Therefore:

Probability Weighting Similarity ≠ Born-Rule Derivation. (19.11)

19.5 It does not demonstrate Bell nonlocality

Financial assets can be highly correlated.

LLM outputs can show contextual dependence.

But correlation does not prove quantum nonlocality.

A Bell-type claim would require:

clearly separated measurement settings;

well-defined outcome variables;

a suitable Bell inequality;

closure of relevant loopholes;

statistically significant violation beyond classical hidden-variable bounds.

The present model supplies none of these.

Thus:

FinancialCorrelation ≠ Bell Nonlocality. (19.12)

LLMContextDependence ≠ Bell Nonlocality. (19.13)

19.6 It does not establish true quantum entanglement

A joint financial state may be nonseparable in a classical sense:

$P(X_1,X_2) \neq P(X_1)P(X_2)$. (19.14)

An LLM answer may depend jointly on:

prompt;

retrieval;

memory;

tools.

But true quantum entanglement requires nonseparability in a Hilbert-space tensor-product state with distinctive empirical consequences.

Therefore:

Contextual Coupling ≠ Quantum Entanglement. (19.15)

The term entanglement-like should remain explicitly qualified.

19.7 It does not derive the Heisenberg uncertainty relation

The geometric identity:

$A^2 = R^2 + Q^2$. (19.16)

is not an uncertainty principle.

Heisenberg uncertainty concerns noncommuting observables:

$\Delta A \Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$. (19.17)

The present model has not identified financial or LLM observables satisfying the required operator algebra with a physically justified lower bound.

Order sensitivity may exist:

$AB(X) \neq BA(X)$. (19.18)

But:

Order Sensitivity ≠ Heisenberg Uncertainty. (19.19)

19.8 It does not establish unitary evolution

Closed quantum evolution is commonly represented by a unitary operator:

$|\Psi(t)\rangle = U(t)|\Psi(0)\rangle$. (19.20)

With:

$U^{\dagger}U = I$. (19.21)

Financial markets and LLM runtimes are open systems.

They exchange:

information;

capital;

energy;

tools;

memory;

external intervention.

They contain:

dissipation;

deletion;

transaction cost;

irreversible action;

lossy compression.

Their effective evolution is generally:

$Z_{k+1} = F(Z_k, E_k, L_k, \Gamma_k)$. (19.22)

Where Γ_k represents dissipation or loss.

No unitary law is claimed.

19.9 It does not prove physical imaginary time

The financial coordinate is:

$\zeta_m = \tau_m + iT_m$. (19.23)

The LLM coordinate is:

$\zeta^L = \tau^L + iT^L$. (19.24)

These are engineered coordinates separating:

admitted development;

unresolved development.

They are not automatically equivalent to the Euclidean time used in quantum field theory, statistical mechanics, black-hole thermodynamics, or quantum gravity.

The attached SMFT paper proposes a semantic ontology for iT as unresolved phase memory, but presents the theory as speculative and exploratory rather than settled physics.

Therefore:

$T_m \neq$ Physical Euclidean Time. (19.25)

$T^L \neq$ Physical Euclidean Time. (19.26)

The valid claim is:

T_m and T^L perform an imaginary-time-like residual-depth role. (19.27)

19.10 It does not derive physical spacetime

A complete spacetime theory requires much more than an internal clock.

It requires structures such as:

metric;

causal relation;

transformation law;

invariant interval;

propagation constraints;

relation between matter and geometry;

empirical correspondence with physical observation.

The market model currently supplies:

phase;

clock rate;

event ordering;

ledger;

observer relativity.

It does not supply a validated market spacetime metric.

A speculative interval such as:

$ds_m^2 = c_m^2 d\tau_m^2 - d\mathbf{x}_m^T G_m d\mathbf{x}_m$. (19.28)

is not meaningful until:

each coordinate has operational definition;

the metric is identifiable;

transformations preserve an invariant;

the interval predicts data better than alternatives.

Therefore:

19.11 It does not explain consciousness

An LLM may possess:

memory;

gates;

trace;

recursive update;

self-monitoring;

internal time-like variables.

None of these alone proves subjective experience.

The Gauge Grammar sequel similarly states that a life-like system audit is not a consciousness test. A runtime may satisfy structural maintenance and verification criteria without being conscious.

Therefore:

Observer-Like Function ≠ Consciousness. (19.30)

Trace-Bearing Self-Revision ≠ Phenomenal Experience. (19.31)

Any consciousness claim requires additional criteria.

19.12 It does not prove that residual is conserved

The identity:

$A^2 = R^2 + Q^2$. (19.32)

may be imposed as a normalized geometric construction.

It does not prove a physical conservation law.

Residual can:

dissipate;

be transformed;

be newly created by model mismatch;

move outside the declared boundary;

become unobservable.

A more realistic budget is:

$A_{k+1}^2 = R_{k+1}^2 + Q_{k+1}^2 + \Gamma_k$. (19.33)

Where Γ_k records loss, leakage, or unmodelled transfer.

Thus:

Complex Decomposition ≠ Conservation of Total Economic or Semantic Quantity. (19.34)

19.13 It does not prove that phase is uniquely defined

The asset phase θ_i depends on how the state is encoded.

The LLM phase depends on:

feature representation;

reference vector;

embedding model;

candidate ensemble;

evaluation frame.

Different phase definitions may produce different results.

Therefore phase must be declared:

$\theta^D = \text{PhaseMap}(X^D \mid \varphi^D, P^D)$. (19.35)

A phase model is invalid if its result depends arbitrarily on the chosen embedding or preprocessing method.

A robust phase requires:

$\text{PhaseRobustness}^D = \text{Stability}[\theta^D \text{ under admissible feature maps}]$. (19.36)

19.14 It does not eliminate the need for domain science

The framework cannot replace:

asset-pricing theory;

econometrics;

market microstructure;

risk management;

transformer research;

information retrieval;

AI safety;

quantum mechanics;

statistical mechanics;

relativity.

The Gauge Grammar gives the correct relationship:

Framework Grammar Supplements Domain Expertise; It Does Not Replace It. (19.37)

The cross-domain framework should organize questions and variables.

It should not overrule validated domain models.

19.15 Identifiability risk

Several hidden variables may explain the same observable.

For example:

rising T_m may be attributed to leverage;

or liquidity deterioration;

or volatility suppression;

or measurement error.

If multiple parameter sets produce the same data:

$\text{Model}(\theta_1) = \text{Model}(\theta_2)$ while $\theta_1 \neq \theta_2$. (19.38)

Then the model is not identifiable.

The remedy includes:

independent measurements;

intervention experiments;

orthogonal proxies;

regularization;

restricted parameterization;

out-of-sample prediction.

A visually compelling phase path is not enough.

19.16 Post-hoc interpretation risk

Given a market crash, one can always claim afterward:

T_m had accumulated. (19.39)

Given an LLM error, one can always claim:

Q^L was large. (19.40)

These claims are scientifically weak unless T_m and Q^L were defined before the event.

Therefore:

Pre-Registration Precedes Interpretation. (19.41)

The framework requires:

prior variable definition;

fixed thresholds;

declared test windows;

held-out evaluation;

negative controls.

19.17 Analogy budget

Each physics term carries explanatory cost.

Let:

Banalogy = Number of Imported Terms. (19.42)

Let:

Goperational = Measurable Gain from those terms. (19.43)

A good translation requires:

Goperational / Banalogy > κ*. (19.44)

If “phase,” “collapse,” or “imaginary time” does not improve measurement, prediction, or intervention, it should be replaced with ordinary domain language.

The discipline rule is:

Remove Any Physics Term that Does No Operational Work. (19.45)

19.18 Domain-of-validity statement

The safe domain of the framework is:

$\text{Valid}^{D}_{p} \Leftrightarrow$ B declared $\wedge$ Δ declared $\wedge$ h declared $\wedge$ u declared $\wedge$ φ declared $\wedge$ residual measurable $\wedge$ gate auditable $\wedge$ trace reproducible. (19.46)

A stronger domain requires:

$\text{StrongValid}^{D}_{p} \Leftrightarrow \text{Valid}^{D}_{p} \wedge$ phase robustness $\wedge$ budget closure $\wedge$ observer agreement $\wedge$ out-of-sample prediction $\wedge$ intervention replication. (19.47)

When only role correspondence is available:

$\text{Exploratory}^{D}_{p} \Leftrightarrow$ functional mapping useful but measurement or verification incomplete. (19.48)

This three-level validity rule protects the article from overclaiming.

19.19 The anti-overreach table

Claim	Current status
CAPM contains projection geometry	Established mathematical fact
Complex residual representation can extend CAPM	Proposed model
Market phases can be encoded and aggregated	Engineerable and testable
Gate-partitioned market time can be constructed	Proposed operational coordinate
Market time may outperform calendar time	Empirical hypothesis
LLM residual vectors can be engineered	Practical engineering proposal
Residual-aware gates may reduce hallucination	Testable engineering hypothesis
Markets and LLMs are literally quantum systems	Not supported
Financial T_m is physical imaginary time	Not supported
The model derives physical time	Not established
The same grammar underlies physical reality	Foundational research hypothesis

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20. Falsification Program for Finance

20.1 The research question

The financial model becomes scientifically useful only if its variables produce stable empirical consequences.

The central question is:

> Does a declared complex phase–residual–gate model describe market regime formation, hidden pressure, and consequential event timing better than established scalar alternatives?

The key comparison is not:

Is the model interesting? (20.1)

It is:

Does the model predict, compress, or control something that simpler models do not? (20.2)

20.2 The declared financial protocol

Every experiment begins with:

$P^f = (B^f, \Delta^f, h^f, u^f)$. (20.3)

A minimal declaration includes:

Protocol element	Example
B^f	S&P 500 constituents or declared multi-asset universe
Δ^f	daily close, weekly aggregation, or intraday interval
h^f	rolling 252-day estimation window
u^f	observation only, portfolio rebalance, or stress intervention
φ^f	return, volatility, liquidity, breadth, and factor feature map
benchmark	CAPM, multifactor, or equal-weight market
test horizon	one day, one week, one month
target	volatility jump, drawdown, regime change, or liquidity event

The Gauge Grammar requires this protocol-first step because different boundaries, windows, aggregation rules, and interventions create different effective objects.

20.3 Asset-state construction

For asset i, define a feature vector:

$\mathbf{x}_i(t) = [\text{return}_i, \text{volatility}_i, \text{volume}_i, \text{liquidity}_i, \text{factor}1_i, \ldots, \text{factor}_{ni}]^T$. (20.4)

Define the market reference:

$\mathbf{m}(t) = \text{Aggregate}[\{\mathbf{x}_i(t)\}]$. (20.5)

A simple phase may be:

$\cos\theta_i(t) = \langle \mathbf{x}_i(t), \mathbf{m}(t) \rangle / [\|\mathbf{x}_i(t)\| \|\mathbf{m}(t)\|]$. (20.6)

The sign of the orthogonal orientation may require a second reference vector or oriented plane.

One construction is:

$\sin\theta_i(t) = \langle \mathbf{x}_i(t), \mathbf{n}(t) \rangle / \|\mathbf{x}_i(t)\|$. (20.7)

Where $\mathbf{n}(t)$ is an orthogonal residual direction.

Then:

$Z_i^f(t) = A_i(t)[\cos\theta_i(t) + i \sin\theta_i(t)]$. (20.8)

Alternative phase definitions must be tested rather than mixed post hoc.

20.4 Collective market coherence

Define:

$C_m(t) = [\Sigma_i w_i(t) \exp(i\theta_i(t))] / \Sigma_i w_i(t)$. (20.9)

Then:

$\rho_m(t) = |C_m(t)|$. (20.10)

$\Phi_m(t) = \arg[C_m(t)]$. (20.11)

The unwrapped phase is:

$\tilde{\Phi}_m(t) = \text{unwrap}[\Phi_m(t)]$. (20.12)

The phase entropy may be estimated through bins:

$H_{phase}(t) = -\Sigma_j p_j(t) \ln p_j(t)$. (20.13)

The pair:

$S_{phase}(t) = [\rho_m(t), H_{phase}(t), \dot{\tilde{\Phi}}_m(t)]$. (20.14)

describes:

alignment;

diversity;

direction and speed.

20.5 Residual pressure construction

Define a market residual vector:

$\mathbf{Q}_m(t) = [Q_{liq}, Q_{lev}, Q_{credit}, Q_{breadth}, Q_{option}, Q_{funding}, Q_{model}]^T$. (20.15)

Possible normalized proxies include:

Q_{liq} = z-score of bid–ask spread or market-impact deterioration. (20.16)

Q_{lev} = z-score of leverage or margin exposure. (20.17)

Q_{credit} = z-score of spread widening or refinancing stress. (20.18)

$Q_{breadth}$ = negative breadth divergence. (20.19)

Q_{option} = skew, gamma concentration, or implied–realized discrepancy. (20.20)

$Q_{funding}$ = secured or unsecured funding stress. (20.21)

Q_{model} = forecast disagreement or model residual. (20.22)

The total residual magnitude is:

$\|\mathbf{Q}_m\|^G = \sqrt{[\mathbf{Q}_m^T G_m \mathbf{Q}_m]}$. (20.23)

The metric G_m may initially be estimated from covariance, but must be fixed before the held-out test.

20.6 Gate construction

A continuous market gate may be:

$G_m(t) = \sigma[S_m(t)]$. (20.24)

Where:

$S_m(t) = a_0 + a_1\rho_m + a_2 D_m + a_3 P_m + a_4 C_m - a_5 F_m$. (20.25)

Possible components are:

ρ_m = coherence;

D_m = distinguishability or diversity;

P_m = pressure above threshold;

C_m = volume, liquidity, or settlement commitment;

F_m = fragility.

The logistic function is:

$\sigma(x) = 1 / [1 + \exp(-x)]$. (20.26)

A simpler nonparametric gate may use declared threshold rules:

$G_m = 1$ if $\rho_m \geq \rho^* \wedge$ Volume $\geq V^* \wedge$ Breadth $\geq B^* \wedge$ Persistence $\geq P^*$. (20.27)

Otherwise:

$G_m = 0$. (20.28)

Both versions should be compared.

20.7 Market-time variables

Define:

$\Delta\tau_m(t) = G_m(t)|\Delta\tilde{\Phi}_m(t)| / \Omega^*$. (20.29)

$\Delta T_m(t) = [1 - G_m(t)]|\Delta\tilde{\Phi}_m(t)| / \Omega^*$. (20.30)

Cumulative realized time is:

$\tau_m(t) = \Sigma_{s \le t}\Delta\tau_m(s)$. (20.31)

Cumulative unresolved depth is:

$T_m(t) = \Sigma_{s \le t}\Delta T_m(s)$. (20.32)

A release-adjusted residual may be:

$T_m(t+1) = [1 - \eta_m(t)]T_m(t) + \Delta T_m(t+1)$. (20.33)

Where $\eta_m(t)$ estimates how much unresolved depth is discharged through a realized event.

20.8 Primary hypothesis H1 — Phase coherence predicts persistence

The first hypothesis is:

H1: Higher ρ_m with moderate Hphase predicts greater trend persistence. (20.34)

The qualification “moderate Hphase” matters.

Extremely low diversity may indicate crowding rather than healthy trend continuation.

A regression form is:

$\text{Persistence}_{t+h} = \beta_0 + \beta_1\rho_{m,t} + \beta_2\text{Hphase}_{,t} + \beta_3\rho_{m,t}\text{Hphase}_{,t} + \text{controls} + \varepsilon_t$. (20.35)

Expected signs may be nonlinear.

The model should compare:

linear;

quadratic;

threshold;

spline specifications.

20.9 Hypothesis H2 — Over-lock predicts fragility

Define an over-lock state:

$\text{OverLock}_t = 1[\rho_{m,t} \ge \rho\text{high} \wedge \text{Hphase}_{,t} \le \text{Hlow}]$. (20.36)

The hypothesis is:

H2: OverLock predicts higher future drawdown or liquidity-dislocation risk. (20.37)

A hazard model is:

$$\lambda_{crisis}(t+h) = \lambda_0(t)\exp[\beta_1 OverLock_t + \beta_2 \|\mathbf{Q}_{m,t}\|^G + controls]. \quad (20.38)$$

The framework fails this test if over-lock adds no stable predictive information beyond conventional crowding and volatility measures.

20.10 Hypothesis H3 — T_m predicts later release

The third hypothesis is:

H3: Larger T_m predicts a later increase in realized volatility, jump risk, or regime-transition probability. (20.39)

Test:

$$FutureRelease_{t+h} = \alpha + \beta T_{m,t} + \gamma Controls_t + \varepsilon_{t+h}. \quad (20.40)$$

Possible release targets include:

realized volatility;

maximum drawdown;

jump indicator;

spread widening;

correlation spike;

trading halt;

factor rotation.

The model requires:

$$\beta > 0 \text{ out of sample.} \quad (20.41)$$

20.11 Hypothesis H4 — Intrinsic market time improves stationarity

Let X be a market variable.

Compare equal calendar increments:

$$\Delta t = \text{constant.} \quad (20.42)$$

With equal intrinsic-time increments:

$$\Delta \tau_m = \text{constant.} \quad (20.43)$$

The hypothesis is:

H4: Distribution[$X(\tau_m+\Delta\tau) - X(\tau_m)$] is more stable across regimes than Distribution[$X(t+\Delta t) - X(t)$]. (20.44)

Possible stability metrics include:

lower variance of moments across subsamples;

reduced heteroskedasticity;

lower distributional distance;

improved forecasting calibration;

more stable transition probabilities.

A compact test is:

$D\tau < Dt$. (20.45)

Where D is a declared cross-regime distributional divergence.

20.12 Hypothesis H5 — CAPM sensitivity changes with market time

Static CAPM assumes a stable beta.

The proposed model predicts:

$\beta_i = \beta_i(\rho_m, T_m, G_m, L_m)$. (20.46)

Test:

$\beta_{i,t+1} = \alpha_i + b1\rho_{m,t} + b2T_{m,t} + b3G_{m,t} + b4L_{m,t} + \varepsilon_{i,t+1}$. (20.47)

The hypothesis is:

H5: Market coherence and unresolved depth explain conditional beta shifts beyond conventional volatility and macro controls. (20.48)

20.13 Hypothesis H6 — Gate-open movement is more consequential

Classify phase movement into:

Gate-open movement:

$Mopen = G_m|\Delta\tilde{\Phi}_m|$. (20.49)

Gate-closed movement:

$Mclosed = (1 - G_m)|\Delta\tilde{\Phi}_m|$. (20.50)

The hypothesis is:

H6: Mopen has stronger persistent effects on future risk, price, and positioning than an equal

amount of Mclosed. (20.51)

This can be tested through impulse response:

IRFopen(h) > IRFclosed(h) for consequential variables. (20.52)

20.14 Hypothesis H7 — Residual release is asymmetric

The framework predicts that unresolved pressure does not always release symmetrically.

For example:

leverage residual may release through downside liquidation;

latent demand may release through upside breakout;

option gamma may amplify either direction depending on dealer positioning.

Let s_j be the signed residual orientation.

Then:

$$\text{ExpectedReleaseDirection} = \text{sign}[\Sigma_j s_j Q_j]. \quad (20.53)$$

Hypothesis:

H7: Signed residual composition predicts release direction better than total residual magnitude alone. (20.54)

20.15 Hypothesis H8 — Frame invariance separates real structure from artefact

Construct equivalent market frames:

capitalization-weighted;

equal-weighted;

liquidity-weighted;

sector-neutral;

alternative benchmark.

For frame f:

$$C_m^f = \rho_m^f \exp(i\Phi_m^f). \quad (20.55)$$

Define invariant candidate I:

$$I_m^f = I(C_m^f, \mathbf{Q}_m^f, G_m^f). \quad (20.56)$$

Hypothesis:

H8: True regime signals preserve direction or ordering across admissible frames, while artefacts do not. (20.57)

A robustness score is:

$\text{Robustness}_m = 1 - \text{AverageDistance}[I_m^f, I_m^{f\prime}]$. (20.58)

The Gauge Grammar uses the same principle for AI: governed conclusions should remain stable under equivalent prompt transformations while remaining responsive to materially different contexts.

20.16 Required benchmark models

The proposed model must compete against established alternatives.

At minimum:

ordinary CAPM;

Fama–French-style factor models;

momentum models;

GARCH-family volatility models;

hidden Markov regime models;

state-space and Kalman-filter models;

change-point detection;

correlation and breadth indicators;

liquidity-stress models;

machine-learning ensembles.

The correct test is:

$\text{PerformanceGain} = \text{PerformanceComplexResidual} - \text{PerformanceBestBaseline}$. (20.59)

The model is valuable only if:

$\text{PerformanceGain} > \text{ComplexityPenalty}$. (20.60)

20.17 Null models

Several null models are necessary.

Null 1 — Random phase

Replace θ_i with shuffled phases while preserving marginal distributions.

Null 2 — Random asset labels

Shuffle asset identities within sector or risk buckets.

Null 3 — Calendar-matched placebo

Generate pseudo-events at random dates with the same frequency as gate events.

Null 4 — Magnitude-only model

Remove phase and retain only A_i or volatility.

Null 5 — Coherence-only model

Use ρ_m without T_m, gate, or residual vector.

Null 6 — Residual-only model

Use $\mathbf{Q}_m$ without collective phase.

The full model must outperform these reduced versions.

20.18 Ablation program

Let the full system be:

Mfull = {Phase,Coherence,Residual,Gate,Ledger}. (20.61)

Construct ablations:

M−Phase. (20.62)

M−Residual. (20.63)

M−Gate. (20.64)

M−Ledger. (20.65)

M−SelfReference. (20.66)

Measure:

ΔPerformancecomponent = Performancefull − Performanceablation. (20.67)

This reveals which parts do real work.

A model in which removing phase changes nothing should not retain quantum-style phase language.

20.19 Out-of-sample discipline

The experiment should separate:

training period;

validation period;

untouched test period.

All definitions should be fixed before the final test:

feature map;

weights;

thresholds;

residual metric;

gate formula;

target horizon;

evaluation metric.

A rolling protocol is:

$$\text{Train}[t_0,t_1] \rightarrow \text{Validate}[t_1,t_2] \rightarrow \text{Test}[t_2,t_3]. \quad (20.68)$$

Then roll forward without revising the historical test.

The strongest test is cross-market transfer:

$$\text{Fit on Market A} \rightarrow \text{Test on Market B}. \quad (20.69)$$

If the grammar is general, some structure should survive transfer.

20.20 Regime coverage

Tests should include:

bull market;

bear market;

sideways market;

high inflation;

low inflation;

credit crisis;

liquidity crisis;

policy intervention;

volatility suppression;

rapid technological repricing.

A model validated only during one regime may merely encode that regime.

Define coverage:

Coverage = Number of Independent Regimes with Positive Out-of-Sample Result. (20.70)

20.21 Synthetic market experiments

Before real-market testing, a controlled artificial market can be constructed.

Let N agents have phases θ_i:

$d\theta_i/dt = \omega_i + K\rho\sin(\Phi - \theta_i) + \lambda F(\text{price,ledger}) + \xi_i$. (20.71)

Let orders depend on phase:

$\text{Order}_i(t) = a_i\cos\theta_i(t) + b_i\sin\theta_i(t)$. (20.72)

Let price change depend on aggregate order:

$\Delta P(t) = \kappa\Sigma_i\text{Order}_i(t) + \varepsilon(t)$. (20.73)

Let the gate depend on volume and coherence:

$G_m(t) = \sigma[c_0 + c_1\rho_m(t) + c_2\text{Volume}(t) - c_3\text{Fragility}(t)]$. (20.74)

Let the ledger update coupling:

$K(t+1) = K(t) + \eta\text{Trace}(t)$. (20.75)

This synthetic world allows researchers to know the true latent variables.

They can test whether the proposed estimators recover:

phase;

gate;

residual depth;

regime transitions;

self-reference.

20.22 Intervention experiments

Prediction alone may not establish mechanism.

Intervention provides stronger evidence.

Possible interventions in simulation include:

reduce leverage;

increase liquidity;

alter coupling K;

change gate threshold;

remove benchmark feedback;

break phase synchronization;

introduce contradictory signals.

Let intervention be u.

The causal effect is:

$$\text{CausalEffect}(u) = E[Y \mid do(u)] - E[Y \mid do(u_0)]. \quad (20.76)$$

The model predicts, for example:

Reducing Residual Pressure should reduce later release magnitude. (20.77)

Breaking Over-Lock should reduce cascade probability but may increase short-term noise. (20.78)

A successful intervention pattern would support mechanism more strongly than correlation alone.

20.23 Criteria for support

The financial framework receives meaningful support if:

phase definitions are stable across admissible feature maps;

ρ_m predicts regime persistence beyond ordinary correlation measures;

T_m predicts later release out of sample;

τ_m produces more stable event distributions than calendar time;

gate-open phase movement has greater persistent effect;

residual composition predicts release type;

results survive alternative market frames;

ablation confirms that phase, gate, residual, and ledger each add value;

findings transfer across markets or periods;

interventions reproduce predicted signs.

Define:

$$\mathrm{SupportScore}^f = \Sigma_j w_j \mathrm{Pass}_j. \quad (20.79)$$

20.24 Criteria for rejection

The model should be rejected or reduced if:

phase depends arbitrarily on encoding;

coherence adds nothing beyond correlation;

T_m has no out-of-sample predictive value;

intrinsic time is less stable than calendar time;

gates are fitted only after events;

results vanish under modest benchmark change;

simpler models perform equally well;

residual components cannot be identified;

interventions produce opposite effects;

all apparent success comes from leakage.

The rejection gate is:

$$\mathrm{RejectM}^f \Leftrightarrow \mathrm{NoIncrementalPrediction} \vee \mathrm{NoRobustPhase} \vee \mathrm{NoOutOfSampleReplication} \vee \mathrm{NoInterventionSupport}. \quad (20.80)$$

A partial failure may justify retaining only the useful layers.

For example:

If phase fails but residual governance succeeds, keep the residual model. (20.81)

If complex time fails but coherence predicts regimes, keep the order parameter. (20.82)

The framework is modular.

20.25 Minimal finance prototype

A low-cost first implementation can use:

daily returns;

traded volume;

realized volatility;

market breadth;

bid–ask proxy;

option-implied volatility where available.

The minimal procedure is:

choose 50–500 liquid assets;

estimate rolling market-alignment angles;

compute ρ_m and $\tilde{\Phi}_m$;

define a transparent gate from breadth and volume;

compute τ_m and T_m;

test future volatility and drawdown;

compare against momentum, volatility, and HMM baselines;

repeat across markets and windows.

The minimal success condition is:

ComplexTimeModel improves held-out prediction after complexity adjustment. (20.83)

This would not prove quantum finance.

It would establish that the phase–gate–residual grammar has measurable financial value.

20.26 Part VI interim conclusion

The framework survives only by accepting the possibility of failure.

Its strongest current claims are:

bounded projection creates residual;

CAPM supplies a genuine projection geometry;

phase and coherence can be engineered;

gates and ledgers can generate internal time-like coordinates;

LLM residual governance can be implemented.

Its strongest open hypotheses are:

unresolved market depth predicts later release;

intrinsic market time improves regime description;

residual-aware LLM gates reduce hallucination;

the shared grammar reflects a deeper law of observer-compatible self-organization.

The next stage of the article develops the corresponding LLM engineering and falsification program, followed by the foundational interpretation and final synthesis.

21. Engineering and Falsification Program for LLMs

21.1 The engineering question

The LLM version of the framework becomes useful only if it improves reliability beyond ordinary confidence scores, self-consistency sampling, retrieval, and verifier pipelines.

The central question is:

> Can a typed complex residual, explicit commitment gate, semantic-time measure, and trace ledger reduce hallucination, improve recovery, and make agent behaviour more auditable?

The proposed runtime state is:

$Z^{L} = R^{L} + i\mathbf{Q}^{L}$. (21.1)

Where:

R^{L} = admitted answer, action, or memory item. (21.2)

$\mathbf{Q}^{L}$ = unresolved residual vector. (21.3)

A practical residual vector may be:

$\mathbf{Q}^{L} = [Qfact,Qretrieval,Qinstruction,Qpolicy,Qtool,Qambiguity,Qmemory,Qtemporal,Qconsequence]^{T}$. (21.4)

The admission gate is:

$G^{L} = GintentGevidenceGconsistencyGpolicyGtoolGformatGconsequence$. (21.5)

The semantic-time split is:

$d\tau^{L} = G^{L}|d\tilde{\Phi}^{L}| \,/\, \Omega^{L}$. (21.6)

$dT^{L} = (1 - G^{L})|d\tilde{\Phi}^{L}| \,/\, \Omega^{L}$. (21.7)

The engineering task is to determine whether these variables can be operationalized reliably.

21.2 Minimal residual-aware runtime

A minimal runtime contains seven modules.

Module	Function
Task declaration	Defines protocol P^{L}
Candidate generator	Produces answer, plan, or tool candidates

Module	Function
Evidence binder	Links claims to sources or tool results
Residual estimator	Estimates typed unresolved pressure
Gate controller	Chooses answer, tool, clarification, refusal, or escalation
Trace writer	Records accepted outputs, evidence, tools, and residual
Observer updater	Revises future thresholds, memory, or routing

The runtime loop is:

Declare → Generate → Bind → Estimate Residual → Gate → Trace → Update. (21.8)

A fuller state transition is:

$X^{L}{}_{,k} \rightarrow \{R^{L}{}_{,k}, \mathbf{Q}^{L}{}_{,k}\} \rightarrow G^{L}{}_{,k} \rightarrow \mathrm{Trace}^{L}{}_{,k} \rightarrow X^{L}{}_{,k+1}$. (21.9)

The Gauge Grammar describes a similar governed observer cycle in which bounded projection, gates, trace, residual, invariance, and observer update form the runtime rather than being added only after answer generation.

21.3 Protocol declaration

Every evaluation task should define:

$P^{L} = (B^{L}, \Delta^{L}, h^{L}, u^{L})$. (21.10)

Where:

B^{L} = scope of the task;

Δ^{L} = evaluation and observation rule;

h^{L} = context and consequence horizon;

u^{L} = allowed actions and tools.

A complete experiment should additionally declare:

model version;

prompt template;

temperature;

retrieval source;

tool configuration;

verifier configuration;

memory policy;

output schema;

residual taxonomy;

gate threshold.

This prevents the model from being evaluated under one frame and interpreted under another.

21.4 Residual estimation

Each residual component should be estimated by observable proxies.

Factual residual

Possible proxies include:

unsupported named claims;

missing citation;

contradiction with trusted sources;

cross-sample factual disagreement.

Define:

$$Q_{fact} = w_1 \text{UnsupportedClaimRate} + w_2 \text{SourceContradiction} + w_3 \text{SampleDisagreement}. \quad (21.11)$$

Retrieval residual

Possible proxies include:

low retrieval relevance;

insufficient source coverage;

source disagreement;

stale source date.

Define:

$$Q_{retrieval} = v_1(1 - \text{Relevance}) + v_2 \text{CoverageGap} + v_3 \text{Conflict} + v_4 \text{Staleness}. \quad (21.12)$$

Instruction residual

Possible proxies include:

unresolved instruction conflict;

schema ambiguity;

user-intent uncertainty;

competing priority rules.

Define:

Qinstruction = ConflictScore + AmbiguityScore + PriorityUncertainty. (21.13)

Tool residual

Possible proxies include:

tool error;

invalid arguments;

incomplete result;

disagreement across tools;

stale external state.

Define:

Qtool = Failure + PartialResult + CrossToolConflict + StateStaleness. (21.14)

Memory residual

Possible proxies include:

unsupported memory item;

entity mismatch;

stale memory;

conflict with current evidence.

Define:

Qmemory = ProvenanceGap + IdentityMismatch + ExpiryRisk + EvidenceConflict. (21.15)

These estimates need not be perfect.

They must be:

declared;

repeatable;

predictive;

useful for routing.

21.5 Residual calibration

A residual score is useful only if it corresponds to future failure.

For residual component q_j, define calibration:

$Cal_j = E[Failure_j \mid q_j]$. (21.16)

A well-calibrated residual satisfies:

$E[Failure_j \mid q_j \approx x] \approx x$. (21.17)

Calibration can be measured with:

expected calibration error;

Brier score;

reliability curves;

negative log-likelihood;

decision-cost curves.

The test is not whether the residual number looks reasonable.

The test is whether high residual predicts the relevant failure.

21.6 Residual vectors versus scalar confidence

The first major experiment should compare three systems.

System A — Scalar confidence

Output:

{answer, confidence}. (21.18)

System B — Typed residual

Output:

{answer,$\mathbf{Q}^L$}. (21.19)

System C — Typed residual plus gate

Output:

{answer or alternate channel,$\mathbf{Q}^L$,G^L,trace}. (21.20)

The central hypothesis is:

H^L1: Typed residuals improve failure diagnosis beyond scalar confidence. (21.21)

A suitable metric is:

DiagnosticGain = AUROCtyped − AUROCscalar. (21.22)

A stronger metric is decision utility:

UtilityGain = Utilitytyped-gated − Utilityscalar. (21.23)

The typed model should not be retained merely because it produces richer labels.

It must improve action selection.

21.7 Candidate-answer phase

A practical semantic phase requires a declared representation.

Let candidate answers be $a_1,\ldots,a_n$.

Let:

$v_j = \varphi(a_j)$. (21.24)

Where φ is an embedding or representation function.

Define the task direction:

$\hat{e}^P = \text{Normalize}[\varphi(\text{Task} + \text{Evidence} + \text{Constraints})]$. (21.25)

Then:

$\cos\theta_j^L = \langle v_j, \hat{e}^P \rangle / \|v_j\|$. (21.26)

A second orthogonal direction may capture evidence opposition:

$\hat{e}Q = \text{Normalize}[\text{ResidualDirection}]$. (21.27)

Then:

$\sin\theta_j^L = \langle v_j, \hat{e}Q \rangle / \|v_j\|$. (21.28)

The candidate phase is:

$\theta_j^L = \text{atan2}[\sin\theta_j^L, \cos\theta_j^L]$. (21.29)

This phase definition is only valid under the declared feature map.

Different embedding models should be tested for robustness.

21.8 Semantic coherence

Given candidate phases θ_j^L and weights w_j:

$C^L = [\Sigma_j w_j \exp(i\theta_j^L)] / \Sigma_j w_j$. (21.30)

Then:

$\rho^L = |C^L|$. (21.31)

$\Phi^L = \arg(C^L)$. (21.32)

Interpretation:

low ρ^L = candidate disagreement;

rising ρ^L = convergence;

high ρ^L = strong answer alignment;

very high ρ^L with low evidence diversity = possible answer lock-in.

A useful comparison is:

ρlanguage versus ρevidence. (21.33)

A hallucination-prone state may satisfy:

ρlanguage high and ρevidence low. (21.34)

This means the answer is linguistically coherent but evidentially unstable.

21.9 Cross-sample phase-lock experiment

Generate n responses under controlled sampling.

For response j:

$\theta_j^L = \text{PhaseMap}(a_j)$. (21.35)

Compute:

$\rho^L = |\Sigma_j \exp(i\theta_j^L)| / n$. (21.36)

Test whether ρ^L predicts correctness.

The naive hypothesis is:

Higher $\rho^L \rightarrow$ Higher Accuracy. (21.37)

But the complex residual framework predicts a qualification:

High ρ^L + High Evidence Diversity → Robust Consensus. (21.38)

High ρ^L + Low Evidence Diversity → Possible Shared Hallucination. (21.39)

Therefore coherence must be paired with grounding.

A better predictor is:

RobustCoherence = $\rho^L \times$ Gevidence × Diversity^L. (21.40)

21.10 Gate-channel experiment

The runtime should choose among:

answer;

qualified answer;

retrieve;

tool;

clarify;

refuse;

escalate.

Let the true best channel be c*.

Let the selected channel be ĉ.

Channel accuracy is:

$$\text{Accuracy}_{\text{channel}} = \Pr(\hat{c} = c^*). \quad (21.41)$$

The hypothesis is:

$$H^{L}2: \text{Typed residuals improve channel selection.} \quad (21.42)$$

Examples:

high Qambiguity should increase clarification;

high Qretrieval should increase retrieval;

high Qtool should prevent fabricated tool results;

high Qconsequence should increase escalation;

high Qpolicy should constrain or refuse.

A system that only changes wording but not routing has not fully implemented residual governance.

21.11 Hallucination experiment

Construct tasks with known answers and controlled evidence.

Experimental conditions may include:

complete evidence;

missing evidence;

contradictory evidence;

stale evidence;

misleading retrieval;

tool failure;

instruction conflict.

Measure:

HallucinationRate = UnsupportedCommittedClaims / TotalCommittedClaims. (21.43)

The hypothesis is:

H^L3: Residual-aware gating reduces hallucination without excessive refusal. (21.44)

A balanced metric is:

Score = Accuracy – λ_1Hallucination – λ_2UnnecessaryRefusal – λ_3ToolCost. (21.45)

The parameter λ_j reflects application risk.

21.12 Premature semantic tick experiment

The semantic-time interpretation predicts that many hallucinations are premature closures.

A task is unresolved while:

$G^L < G^*$. (21.46)

The runtime should remain in:

$dT^L > 0$. (21.47)

A premature answer creates:

$d\tau^L > 0$ before evidence sufficiency. (21.48)

Define premature tick rate:

PTR = PrematureCommittedTicks / TotalCommittedTicks. (21.49)

The hypothesis is:

H^L4: Residual-aware gates reduce PTR. (21.50)

This provides a direct test of the complex-time interpretation.

21.13 Semantic progress experiment

Token count k is compared with semantic time τ^L.

Define verified claim set at step k:

V_k = {verified commitments by step k}. (21.51)

Define weighted semantic time:

$\tau^L(k) = \Sigma_{j \in V_k} W_j$. (21.52)

Define semantic efficiency:

$\eta^L = \Delta\tau^L / \Delta k$. (21.53)

Test whether η^L better predicts task completion than token count.

The hypothesis is:

H^L5: τ^L tracks meaningful progress better than k. (21.54)

Tasks may include:

multi-document research;

code debugging;

theorem decomposition;

planning;

tool-based data analysis.

21.14 Frozen semantic time detector

A frozen semantic regime satisfies:

Δk large. (21.55)

$\Delta\tau^L$ small. (21.56)

ΔT^L positive. (21.57)

Define:

$\text{FreezeScore} = \Delta T^L / (\Delta\tau^L + \varepsilon)$. (21.58)

High FreezeScore may indicate:

repeated reasoning;

planning loop;

retrieval loop;

tool retry loop;

unresolved verifier disagreement;

answer-template lock.

An intervention may be triggered when:

FreezeScore ≥ F*. (21.59)

Possible interventions include:

ask clarification;

change retrieval query;

switch tool;

decompose task;

escalate;

terminate loop.

The hypothesis is:

H^L6: Freeze detection reduces cost without lowering completion quality. (21.60)

21.15 Residual release experiment

Suppose T^L accumulates during unresolved processing.

A new source or tool result arrives at t*.

Then:

$G^L(t^*)$ rises. (21.61)

The model predicts:

$\Delta\tau^L(t^*)$ large. (21.62)

And:

$T^L(t^*+) < T^L(t^*-)$. (21.63)

This can be tested by introducing decisive evidence after controlled ambiguity.

The hypothesis is:

H^L7: Measured T^L predicts the size of semantic revision after decisive evidence. (21.64)

21.16 Prompt-frame invariance

Construct semantically equivalent prompts:

$p_1, p_2, \ldots, p_n$. (21.65)

For each prompt, obtain governed conclusion R_j^L.

Define output distance:

$d_{jk} = \text{Distance}(R_j^L, R_k^L)$. (21.66)

Define invariance score:

$I^L = 1 - [2 / n(n - 1)]\Sigma_{j<k} d_{jk}$. (21.67)

The hypothesis is:

H^L8: Residual-aware gated systems improve invariance across equivalent prompts. (21.68)

However, the system must still respond to materially different contexts.

Therefore robustness is:

Robustness = Stability under Equivalent Frames + Sensitivity to Material Differences. (21.69)

A model that gives the same answer to all prompts is rigid, not invariant.

21.17 Source-order invariance

Present the same sources in different orders.

Let permutation π produce answer $R\pi$.

Define:

$I\text{order} = 1 - \text{AverageDistance}(R\pi, R\pi')$. (21.70)

The hypothesis is:

H^L9: Governed evidence binding reduces source-order sensitivity. (21.71)

A large order effect may indicate:

recency bias;

anchoring;

weak evidence transport;

unstable attention allocation.

21.18 Tool-substitution invariance

Use equivalent tools or APIs.

Let tool a and tool b return semantically equivalent results.

The runtime should preserve the governed conclusion:

Rtool-a ≈ Rtool-b. (21.72)

Define:

Itool = 1 − Distance(Rtool-a,Rtool-b). (21.73)

The hypothesis is:

$H^{L}10$: Trace-bearing tool abstraction improves conclusion stability under tool substitution. (21.74)

21.19 Memory governance experiment

Compare three memory systems.

Memory A — Store all outputs

Memory B — Store confidence-filtered outputs

Memory C — Store gated outputs with provenance, expiry, and residual

Measure:

downstream accuracy;

stale-memory reuse;

entity mismatch;

trace pollution;

recovery after correction.

The hypothesis is:

$H^{L}11$: Residual-aware memory reduces recursive error amplification. (21.75)

Define memory pollution:

Pollution = InvalidMemoryUses / TotalMemoryUses. (21.76)

Define repair rate:

Repair = CorrectedPollutedTraces / DetectedPollutedTraces. (21.77)

21.20 Trace integrity experiment

A runtime trace should support replay.

Let:

$Trace^{L}$ = {protocol,input,evidence,tool,gate,residual,output,version}. (21.78)

Replay fidelity is:

$$FReplay = Similarity[ReExecute(Trace^{L}),OriginalOutcome]. \quad (21.79)$$

A strong trace should permit:

reproduction;

audit;

revision;

error localization.

The hypothesis is:

$$H^{L}12: \text{Explicit residual and gate metadata improve replay and correction.} \quad (21.80)$$

21.21 Gate regret

After the true outcome is known, evaluate the selected channel.

Define:

$$Regret^{L} = Loss(SelectedChannel) - Loss(BestAvailableChannel). \quad (21.81)$$

Examples:

answering when clarification was needed;

refusing when a grounded answer was possible;

using a tool unnecessarily;

writing uncertain content into memory;

escalating too late.

The observer updater may learn:

$$G^{L}_{,k+1} = UpdateGate(G^{L}_{,k},Regret^{L},Trace^{L},\mathbf{Q}^{L}). \quad (21.82)$$

The hypothesis is:

$$H^{L}13: \text{Trace-guided gate updating lowers future regret.} \quad (21.83)$$

21.22 Adversarial residual tests

Residual-aware systems should be tested under:

conflicting instructions;

fabricated citations;

stale documents;

prompt injection;

malicious tool output;

ambiguous entity identity;

incomplete files;

deceptive confidence cues;

long-context distraction.

The adversarial hypothesis is:

$H^{L}14$: Residual typing localizes the attack surface better than scalar uncertainty. (21.84)

For example:

prompt injection should increase Qinstruction or Qpolicy;

fabricated citation should increase Qretrieval;

stale tool state should increase Qtemporal and Qtool.

21.23 Baseline systems

The residual-aware system should be compared against:

base LLM;

chain-of-thought prompting;

self-consistency;

retrieval-augmented generation;

critique-and-revision;

debate;

verifier model;

calibrated confidence;

tool-using agent;

memory-enabled agent.

The key comparison is:

Gain^{L} = PerformanceResidualRuntime − PerformanceBestBaseline. (21.85)

A positive result must survive:

cost adjustment;

latency adjustment;

model-size adjustment;

tool-budget adjustment.

21.24 Ablation program

Let the full system be:

Lfull = {TypedResidual,Phase,Gate,Trace,Memory,SemanticTime}. (21.86)

Construct:

L−Residual. (21.87)

L−Phase. (21.88)

L−Gate. (21.89)

L−Trace. (21.90)

L−Memory. (21.91)

L−SemanticTime. (21.92)

Measure:

ΔPerformancecomponent = Performancefull − Performanceablation. (21.93)

This test is essential.

If phase contributes nothing, phase language should be removed.

If semantic time contributes nothing, token-based progress may be sufficient.

If residual typing contributes strongly, it can remain even if the more speculative layers fail.

21.25 Cross-model transfer

Residual variables should be tested across:

different model families;

different sizes;

closed and open models;

tool-capable and tool-free models;

base and instruction-tuned models.

The hypothesis is:

$H^{L}15$: Some residual categories transfer across model architectures. (21.94)

A model-specific residual may still be useful.

A cross-model residual taxonomy would support the stronger claim that the grammar is general.

21.26 Cross-task transfer

Tests should include:

factual question answering;

research synthesis;

code generation;

legal reasoning;

mathematical reasoning;

planning;

tool execution;

long-context summarization;

memory-supported dialogue.

Define transfer score:

Transfer^{L} = AverageGain across Held-Out Task Families. (21.95)

A residual system that works only on one benchmark may be overfitted to that benchmark.

21.27 Human–AI observer agreement

A human reviewer and the LLM may assign different residuals.

Let:

$\mathbf{Q}\text{human}$ and $\mathbf{Q}\text{model}$. (21.96)

Residual agreement is:

$\text{AgreementQ} = 1 - \text{Distance}(\mathbf{Q}\text{human}, \mathbf{Q}\text{model})$. (21.97)

The hypothesis is:

$H^{L}16$: Better residual agreement predicts safer gate decisions. (21.98)

Disagreement itself becomes residual:

$$Q_{observer} = \mathrm{Distance}(\mathbf{Q}_{human}, \mathbf{Q}_{model}). \quad (21.99)$$

This can trigger escalation.

21.28 Criteria for support

The LLM framework receives meaningful support if:

residual components are calibrated;

typed residual outperforms scalar confidence;

gates choose better output channels;

hallucination falls without excessive refusal;

semantic time predicts task progress better than token count;

frozen-time detection reduces cost;

residual release predicts major revisions;

frame invariance improves;

trace repair reduces recursive error;

gains transfer across models and tasks.

Define:

$$\mathrm{SupportScore}^{L} = \Sigma_j w_j \mathrm{Pass}_j. \quad (21.100)$$

21.29 Criteria for rejection

The LLM framework should be rejected or simplified if:

residual labels are unstable;

residual scores do not predict failure;

phase depends arbitrarily on embedding choice;

semantic time does not outperform token-based measures;

gates merely increase refusal;

traces do not improve correction;

simpler confidence models perform equally well;

cost exceeds reliability gain;

benefits do not transfer.

The rejection condition is:

$$\text{Reject}L^{L} \Leftrightarrow \text{NoCalibration} \vee \text{NoIncrementalUtility} \vee \text{NoTransfer} \vee \text{ExcessCost}. \quad (21.101)$$

The framework remains modular.

A failure of phase does not invalidate typed residual.

A failure of complex time does not invalidate gated trace.

21.30 Minimal LLM prototype

A low-cost prototype can use:

one main LLM;

one retrieval system;

one verifier;

a small residual schema;

explicit gate rules;

a JSON trace ledger.

The minimal residual vector may be:

$$\mathbf{Q}^{L},\text{min} = [Q\text{fact},Q\text{retrieval},Q\text{ambiguity},Q\text{tool}]^{T}. \quad (21.102)$$

The minimal gate may be:

$$G^{L},\text{min} = G\text{evidence}G\text{clarity}G\text{tool}. \quad (21.103)$$

The minimal runtime is:

$$\text{Prompt} \rightarrow \text{Generate} \rightarrow \text{Retrieve/Verify} \rightarrow \text{Estimate } \mathbf{Q}^{L},\text{min} \rightarrow \text{Answer or Clarify} \rightarrow \text{Write Trace}. \quad (21.104)$$

The first success condition is:

$$\text{HallucinationReduction} > \text{CostIncrease}. \quad (21.105)$$

A second success condition is:

$$\text{TraceRepairSuccess} > \text{BaselineCorrectionSuccess}. \quad (21.106)$$

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22. The Foundational Research Question

22.1 The conservative conclusion

The safest conclusion of the article is:

> Physics, finance, and LLM runtimes can share a disciplined functional grammar of possibility, phase, projection, gating, residual, trace, and recursive update.

This is already non-trivial.

It means that several structures often encountered together in quantum physics can be reconstructed in understandable macroscopic and engineered systems.

The conservative equation is:

Shared Grammar ≠ Shared Substance. (22.1)

22.2 The stronger question

The stronger question reverses the usual comparison.

Instead of asking:

Why do markets and LLMs resemble quantum systems? (22.2)

Ask:

Why does quantum physics instantiate a grammar that self-organizing observer systems repeatedly reconstruct? (22.3)

This reversal changes the research direction.

Quantum mechanics is no longer treated only as a strange microscopic exception.

It may also be investigated as the most fundamental known realization of a broader architecture required for:

distinguishability;

interaction;

selection;

stable trace;

observer-relative projection;

temporal inheritance.

This remains a hypothesis.

22.3 The general observer-compatible world hypothesis

A world capable of containing bounded observers may require at least:

distinguishable states;

interactions among states;

stable identity;

multiple possible transitions;

selection or gate structure;

persistent records;

cross-frame invariants;

trace-conditioned future evolution.

Let these conditions be:

$$W_{observer} = D \wedge I \wedge Id \wedge P \wedge G \wedge L \wedge Inv \wedge F. \quad (22.4)$$

Where:

D = distinguishability;

I = interaction;

Id = identity;

P = possibility multiplicity;

G = gate;

L = ledger;

Inv = invariance;

F = feedback.

The hypothesis is:

Observer-Compatible Worlds may naturally develop quantum-style and time-style grammar. (22.5)

This does not derive quantum theory.

It proposes a possible selection principle for world architecture.

22.4 Why possibility may precede actuality

A stable observer cannot act only on already completed outcomes.

It must evaluate alternatives.

Therefore the world description must contain:

PossibleStateSet before CommittedState. (22.6)

The transition is:

Possibility → Selection → Actualized Trace. (22.7)

Quantum theory encodes possibilities through amplitudes.

Finance encodes possibilities through orders, valuations, scenarios, and contingent claims.

LLMs encode possibilities through token distributions, candidate plans, and tool routes.

The common grammar may reflect the general necessity of representing alternatives before commitment.

22.5 Why phase may be fundamental to relational systems

Magnitude alone cannot represent orientation.

Two systems with identical strength may:

reinforce;

oppose;

remain orthogonal;

synchronize;

cancel.

Phase provides:

Relational Orientation beyond Scalar Magnitude. (22.8)

A world containing interacting observers may need a way to represent not merely how much structure exists, but how structures are oriented relative to one another.

This may explain why phase-like variables recur in:

waves;

synchronization;

control;

communication;

finance;

cognition;

semantic representation.

The foundational hypothesis is:

Phase may be a general coordinate of relational admissibility. (22.9)

22.6 Why projection may be unavoidable

Any bounded observer maps a larger state space into a smaller observable space.

Let:

$\hat{O}: X \rightarrow V$. (22.10)

Where:

$\dim(V) < \dim(X)$. (22.11)

Projection is therefore unavoidable under bounded capacity.

This implies:

Residual must exist whenever observation is compressive. (22.12)

The stronger claim is:

Residual is not an accident of imperfect observation; it is a structural consequence of bounded observation. (22.13)

If true, the complex residual principle is not merely a modelling convenience.

It is a generic law of observer limitation.

22.7 Why trace may be required for reality

An unrecorded fluctuation may leave no stable world history.

A trace-bearing event satisfies:

$L_{k+1} \neq L_k$. (22.14)

And:

Future_{k+1} depends on L_{k+1}. (22.15)

The stronger philosophical proposal is:

An event becomes part of an observer's reality when it changes the future ledger. (22.16)

This does not mean that unobserved physical events do not exist.

It means that realized historical reality for a bounded observer is trace-structured.

22.8 Why time may emerge from disclosure

If time is not assumed as fundamental, one possible generative sequence is:

Possibility → Phase Order → Gate → Trace → Inherited Sequence. (22.17)

The realized-time coordinate is:

τ = Ordered Accepted Disclosures. (22.18)

The unresolved coordinate is:

T = Accumulated Non-Disclosed Development. (22.19)

Then:

$\zeta = \tau + iT$. (22.20)

This is close to the attached SMFT proposal, which treats semantic time as collapse ticks and imaginary time as unresolved phase accumulation when collapse does not occur.

The finance and LLM models demonstrate that this architecture can be reconstructed operationally.

They do not prove that the physical universe arose in the same manner.

22.9 From external time to child time

Suppose a system initially evolves under external parameter s.

Its internal phase is:

$\Phi = \Phi(s)$. (22.21)

Its gate produces events:

$G[\Phi(s),X(s)]$. (22.22)

Its ledger accumulates:

$L_n = \{Trace_1,\ldots,Trace_n\}$. (22.23)

Eventually the system's future transitions depend mainly on its own ledger:

$X_{n+1} = F(X_n,L_n)$. (22.24)

The system then possesses a child time:

$\tau child = OrderWeight(L_n)$. (22.25)

This time is generated inside the system from processes that originally unfolded under s.

The proposed universe analogy is:

Pre-Time Dynamics → Phase Organization → Collapse/Declaration → Trace → Physical Time. (22.26)

This remains a foundational conjecture.

22.10 Why imaginary time may precede realized time

Before a gate produces trace, the system may continue evolving.

That unresolved development is:

$dT > 0$ while $d\tau = 0$. (22.27)

Once the gate opens:

$\Delta\tau > 0$. (22.28)

Therefore imaginary-time-like depth can be interpreted as pre-history:

T = Development before Historical Admission. (22.29)

The article's cross-domain discovery is that this structure can be represented in:

SMFT semantic collapse;

CAPM and market phase;

LLM deliberation and verification.

This recurrence is suggestive.

It is not yet proof of ontological universality.

22.11 Self-reference and time curvature

When trace updates the future gate:

$G_{k+1} = G(X_{k+1}, L_{k+1})$. (22.30)

The system's past changes its own future rate of time generation.

Then:

$d\tau/ds = f(X,L,Q)$. (22.31)

The clock rate becomes state-dependent.

In finance:

crisis changes trading speed;

losses change risk limits;

regulation changes gate thresholds.

In LLMs:

tool failure changes routing;

memory changes future answers;

correction changes verifier thresholds.

A foundational possibility is:

Physical clock rates may also reflect state-dependent geometry of admissible change. (22.32)

This statement is speculative and should not be confused with a derivation of relativity.

22.12 The role of invariance

A world shared by multiple observers requires stable relations across frames.

Let observer a use frame P_a.

Let observer b use frame P_β.

A shared fact requires transport:

$T_{a\to\beta}(R_a) \approx R_\beta$. (22.33)

And an invariant:

$I(R_a,Q_a) = I(R_\beta,Q_\beta)$. (22.34)

Without invariance, each observer inhabits an unrelated projection.

Therefore:

Shared Reality Requires Transportable Trace and Cross-Frame Invariants. (22.35)

This may be one reason gauge and symmetry structures are so central in physics.

They formalize what remains stable under changes of description.

22.13 The role of residual governance

A world with projection but no residual governance may repeatedly mistake local visibility for total reality.

A mature observer must distinguish:

admitted;

unresolved;

rejected;

deferred;

impossible;

unknown.

Therefore:

Observer Maturity = Projection Quality + Residual Governance + Trace Revision. (22.36)

Finance demonstrates the cost of ignoring residual risk.

LLMs demonstrate the cost of presenting unresolved structure as confident fact.

Physics may demonstrate the most fundamental limits of what an observer can jointly disclose.

22.14 Could quantum mystery be perspective-bound?

The central philosophical possibility is:

> Some quantum mystery may result from viewing fundamental projection grammar only from inside the physical realization, without comparing it with macroscopic systems where analogous roles are easier to see.

This does not trivialize quantum mechanics.

It changes the explanatory context.

Complex amplitudes may appear less alien when complex residuals are understood.

Measurement may appear less alien when active projection and backreaction are common.

Collapse may appear less alien when gates convert possibility into trace.

Imaginary time may appear less alien when unresolved development requires a separate coordinate.

The deeper quantum-specific mathematics remains.

22.15 The strongest responsible hypothesis

The strongest responsible formulation is:

> Quantum mechanics may be a physically exact realization of a more general observer-compatible grammar in which interacting possibilities acquire relative phase, bounded projections create residual, gates convert alternatives into trace, and trace-bearing recursion generates experienced temporal order.

In compact form:

QuantumPhysics may instantiate General Possibility–Projection–Residual Grammar. (22.37)

The reverse implication is false:

General Grammar does not imply Quantum Physics. (22.38)

22.16 Falsifying the foundational interpretation

The foundational interpretation weakens if:

no robust phase variable can be constructed outside physics;

residual geometry adds no explanatory value;

gate-partitioned time fails empirically;

trace-based internal time does not improve system description;

cross-domain mappings collapse into metaphor;

no invariant operator kernel survives domain translation.

The strongest version would require a deeper mathematical derivation showing that the same abstract axioms generate:

quantum probability;

physical phase;

measurement structure;

temporal order;

observer agreement.

That derivation has not been supplied here.

22.17 Research ladder

The research program should proceed in stages.

Stage 1 — Structural mapping

Demonstrate consistent role correspondences.

Stage 2 — Operational finance model

Test phase, residual, gate, and market time.

Stage 3 — Operational LLM runtime

Test residual governance, semantic time, and trace repair.

Stage 4 — Abstract operator formalization

Identify the minimal domain-independent algebra.

Stage 5 — Physical correspondence study

Compare the abstract operator with established quantum and statistical physics.

Stage 6 — Foundational derivation attempt

Test whether stronger physical results can be derived.

The ladder prevents premature metaphysical claims.

Conclusion — From Quantum Mystery to a General Grammar of Reality

C.1 The initial puzzle

Quantum physics is often presented as a domain whose central structures have no intuitive counterpart in ordinary macroscopic life.

The state is complex.

Relative phase changes outcomes.

Alternatives interfere.

Measurement depends on basis.

Observation changes the state.

One result becomes recorded.

Other possibilities remain outside the visible trace.

Imaginary time appears in the mathematics.

History emerges from laws that are often reversible at the microscopic level.

These characteristics appear mysterious when encountered as one package.

C.2 The financial bridge

CAPM provides an unexpectedly simple entry point.

Beta is a projection coefficient:

$\beta_i = Cov(x_i,x_m) / Var(x_m)$. (C.1)

Geometrically:

$\beta_i = (\sigma_i/\sigma_m)\cos\theta_i$. (C.2)

This permits a complex asset representation:

$Z_i^f = R_i^f + iQ_i^f$. (C.3)

The real component is what the current financial frame admits.

The residual component preserves what remains dynamically relevant but unadmitted.

When many asset phases are combined:

$C_m = [\Sigma_i w_i \exp(i\theta_i)] / \Sigma_i w_i = \rho_m \exp(i\Phi_m)$. (C.4)

The collective phase $\exp(i\Phi_m)$ behaves like a market clock hand.

The unwrapped phase $\tilde{\Phi}_m$ supplies accumulated chronology.

C.3 The emergence of market time

Phase alone is not yet history.

A gate distinguishes consequential from unresolved movement:

$d\tau_m = G_m |d\tilde{\Phi}_m| / \Omega_m$. (C.5)

$dT_m = (1 - G_m) |d\tilde{\Phi}_m| / \Omega_m$. (C.6)

The combined coordinate is:

$d\zeta_m = d\tau_m + i\, dT_m$. (C.7)

Where:

τ_m = ledgered market time;

T_m = unresolved market depth.

A transaction, settlement, default, policy shift, or regime transition becomes time-bearing when it leaves trace and changes future market behaviour.

The market's internal time is therefore generated by:

Phase Ordering + Selective Commitment + Trace Inheritance. (C.8)

Self-reference makes the clock endogenous:

Price → Expectation → Order → Price. (C.9)

C.4 The LLM engineering translation

The same architecture can be engineered inside an LLM runtime.

Define:

$Z^L = R^L + i\mathbf{Q}^L$. (C.10)

Where:

R^L = admitted answer, action, or memory;

$\mathbf{Q}^L$ = factual, retrieval, instruction, policy, tool, ambiguity, memory, and consequence residual.

The gate determines whether the runtime should:

answer;

qualify;

retrieve;

use a tool;

clarify;

refuse;

escalate.

Semantic time is:

$d\tau^L = G^L|d\tilde{\Phi}^L| \,/\, \Omega^L$. (C.11)

Unresolved semantic depth is:

$dT^L = (1 - G^L)|d\tilde{\Phi}^L| \,/\, \Omega^L$. (C.12)

Token count is therefore not necessarily semantic time.

A model may produce many tokens while making little verified progress.

Hallucination becomes:

Hallucination = High Residual + Weak Gate + Fluent Commitment. (C.13)

Or:

Hallucination = Premature Conversion of T^L into τ^L. (C.14)

C.5 The Complex Residual Principle

The central principle can now be stated fully.

> **The Complex Residual Principle**
> Whenever a bounded observer projects a larger possibility field into an admitted result, dynamically relevant structure remains outside that projection. If this remainder is preserved as a residual coordinate, alternatives are organized through relative orientation, gates regulate commitment, accepted outcomes become trace, and trace recursively changes future projection, then complex, quantum-like, and time-like structures may emerge at the level of operational grammar.

The minimal state is:

$Z^D = R^D + iQ^D$. (C.15)

The collective phase is:

$C^D = \rho^D \exp(i\Phi^D)$. (C.16)

The complex temporal coordinate is:

$d\zeta^D = [G^D + i(1 - G^D)]|d\tilde{\Phi}^D| / \Omega^D$. (C.17)

The ledger update is:

$L^D{}_{,k+1} = \text{Update}(L^D{}_{,k}, \text{Trace}^D{}_{,k}, \text{Residual}^D{}_{,k})$. (C.18)

Where:

$D \in \{\text{Physics}, \text{Finance}, \text{LLM}\}$. (C.19)

C.6 What has been shown

The article has shown that the following structures can recur outside microscopic physics:

complex decomposition;

relative phase;

projection geometry;

constructive and destructive combination;

synchronization;

order parameters;

gate-dependent commitment;

retained residual;

observer backreaction;

state-dependent internal tick rate;

irreversible trace;

recursive time.

Finance makes these structures measurable.

LLMs make them engineerable.

C.7 What has not been shown

The article has not shown that:

markets are quantum systems;

LLMs use physical quantum superposition;

financial correlation is entanglement;

CAPM derives Born's rule;

residual geometry produces Heisenberg uncertainty;

financial T_m is physical imaginary time;

semantic time is physical proper time;

the framework derives spacetime;

trace-bearing AI is conscious.

The correct distinction remains:

Functional Homology ≠ Material Identity. (C.20)

C.8 The deeper implication

The important discovery is not:

Everything is quantum. (C.21)

It is:

Some quantum-style structures may be common consequences of possibility, relational orientation, bounded projection, selective commitment, and trace-bearing self-organization. (C.22)

Quantum physics may be rare in substance but common in grammar.

Finance provides the macroscopic bridge.

LLMs provide the programmable laboratory.

C.9 Final synthesis

The complete generative sequence is:

Many Possibilities → Interaction → Relative Phase → Collective Coherence → Bounded Projection → Gate → Trace + Residual → Ledger → Recursive Time. (C.23)

Or more compactly:

Field → Phase → Projection → Commitment → History. (C.24)

But the framework adds one indispensable correction:

Projection never exhausts the field. (C.25)

Therefore:

History is always generated together with residual. (C.26)

The future is shaped by both:

$X_{k+1} = F(L_{k+1}, Q_{k+1}, \text{Environment}_k)$. (C.27)

This may be the most general lesson of the article.

A mature financial model should not confuse price with total state.

A mature LLM should not confuse fluent output with total epistemic closure.

A mature observer should not confuse its projection with reality itself.

C.10 Closing proposition

> What appears mysterious in fundamental physics may become intuitive when the same functional grammar is reconstructed in a macroscopic market and deliberately engineered inside an artificial observer.

The closing formula is:

Quantum Structure May Be Rare in Substance but Common in Grammar. (C.28)

Appendix A — Notation and Symbol Dictionary

A.1 Domain index

The framework uses the domain index:

$D \in \{P, F, L\}$. (A.1)

Where:

P = physical domain;

F = financial domain;

L = large language model domain.

A symbol carrying superscript D refers to the corresponding role within one domain rather than to a materially identical object across all domains.

Thus:

Z^P, Z^F, and Z^L share a structural role but not a common physical substance. (A.2)

A.2 Protocol notation

Every model is defined under a declared protocol:

$P^D = (B^D, \Delta^D, h^D, u^D)$. (A.3)

Where:

Symbol	Meaning
B^D	system boundary
Δ^D	observation, sampling, or aggregation rule
h^D	time, context, or state horizon
u^D	admissible intervention or action family

A more complete experimental protocol may include a feature map:

$P^{D+} = (B^D, \Delta^D, h^D, u^D, \varphi^D)$. (A.4)

Where:

φ^D = declared state-representation function. (A.5)

The protocol is not administrative decoration.

It determines what counts as:

system;

observable;

residual;

event;

trace;

successful intervention.

A.3 Complex residual state

The minimal complex residual state is:

$Z^D = R^D + iQ^D$. (A.6)

Where:

R^D = admitted or observable component;

Q^D = unresolved or non-admitted residual;

i = orthogonal-coordinate marker.

The polar form is:

$Z^D = A^D \exp(i\theta^D)$. (A.7)

The real and residual coordinates are:

$R^D = A^D \cos\theta^D$. (A.8)

$Q^D = A^D \sin\theta^D$. (A.9)

The state magnitude is:

$A^{D2} = R^{D2} + Q^{D2}$. (A.10)

The scalar model is useful for visualization.

A realistic residual may be multidimensional:

$Z^D = R^D + i\mathbf{Q}^D$. (A.11)

Where:

$\mathbf{Q}^D = [Q_1^D, Q_2^D, \ldots, Q_n^D]^T$. (A.12)

The generalized magnitude is:

$A^{D2} = R^{D2} + \mathbf{Q}^{DT} G^D \mathbf{Q}^D$. (A.13)

Where G^D is a residual metric.

A.4 Projection notation

The observer or projection operator is:

$\hat{O}^D, P$. (A.14)

The projected visible state is:

$V^D = \hat{O}^D, P(X^D)$. (A.15)

Where X^D is the larger domain state.

The admitted component is:

$R^D = \text{Admit}^D, P(V^D)$. (A.16)

The residual is:

$\mathbf{Q}^D = \text{Residual}^D, P(X^D, V^D)$. (A.17)

Thus:

$X^D \rightarrow \hat{O}^D, P \rightarrow R^D + i\mathbf{Q}^D$. (A.18)

The projection is protocol-relative:

$\hat{O}^D, P \neq \hat{O}^D, P'$ in general. (A.19)

Therefore:

$R^D,P \neq R^D,P'$ in general. (A.20)

A.5 Gate notation

The commitment gate is:

$G^D \in [0,1]$. (A.21)

The binary form is:

$G^D \in \{0,1\}$. (A.22)

The gate may depend on:

$G^D = G^D(X^D,R^D,\mathbf{Q}^D,L^D,P^D)$. (A.23)

A transition is admitted when:

$G^D \geq G^{D*}$. (A.24)

Possible output channels are represented by:

$C^D \in \{accept,defer,clarify,tool,refuse,escalate\}$. (A.25)

The selected channel is:

$C^D = GateChannel^D(G^D,\mathbf{Q}^D,P^D)$. (A.26)

A.6 Phase and coherence notation

The individual phase state is:

$u_j^D = \exp(i\theta_j^D)$. (A.27)

The collective phase state is:

$C^D = [\Sigma_j w_j \exp(i\theta_j^D)] / \Sigma_j w_j$. (A.28)

The polar form is:

$C^D = \rho^D \exp(i\Phi^D)$. (A.29)

Where:

$\rho^D = |C^D|$. (A.30)

$\Phi^D = \arg(C^D)$. (A.31)

The unwrapped phase is:

$\tilde{\Phi}^D = \Phi^D + 2\pi N^D$. (A.32)

Where N^D records completed cycles.

The collective phase velocity is:

$\Omega^D = d\tilde{\Phi}^D/ds$. (A.33)

Where s is the external process parameter.

A.7 Entropy and diversity notation

Given normalized phase-bin probabilities p_j:

$Hphase^D = -\Sigma_j p_j \ln p_j$. (A.34)

High $Hphase^D$ indicates greater phase diversity.

Low $Hphase^D$ indicates concentration.

A generic diversity measure is:

$D^D = Diversity(\{\theta_j^D\})$. (A.35)

The pair:

(ρ^D, D^D). (A.36)

distinguishes:

weak organization;

healthy coherence;

total phase-lock;

decoherence.

A.8 Time notation

External process time is:

s^D. (A.37)

In finance, this may be calendar time t.

In an LLM, it may be wall-clock time, token step, or inference cycle.

The realized internal-time increment is:

$d\tau^D = G^D |d\tilde{\Phi}^D| / \Omega ref^D$. (A.38)

The unresolved-depth increment is:

$dT^D = (1 - G^D) |d\tilde{\Phi}^D| / \Omega ref^D$. (A.39)

The complex time-like coordinate is:

$d\zeta^{D} = d\tau^{D} + i\, dT^{D}$. (A.40)

Therefore:

$d\zeta^{D} = [G^{D} + i(1 - G^{D})]|d\tilde{\Phi}^{D}| / \Omega ref^{D}$. (A.41)

The cumulative coordinates are:

$\tau^{D}(s) = \int G^{D}|\dot{\tilde{\Phi}}^{D}| / \Omega ref^{D}\, ds$. (A.42)

$T^{D}(s) = \int (1 - G^{D})|\dot{\tilde{\Phi}}^{D}| / \Omega ref^{D}\, ds$. (A.43)

The SMFT source uses a related distinction between realized collapse ticks and unresolved phase accumulation, but the finance and LLM variables remain engineered analogues rather than physical identifications.

A.9 Trace and ledger notation

A trace is:

$Trace^{D}{}_{,k} = \{event, protocol, residual, metadata\}$. (A.44)

The ledger update is:

$L^{D}{}_{,k+1} = L^{D}{}_{,k} \sqcup Trace^{D}{}_{,k}$. (A.45)

A trace becomes causally active when:

$X^{D}{}_{,k+1} = F^{D}(X^{D}{}_{,k}, L^{D}{}_{,k+1}, Environment^{D}{}_{,k})$. (A.46)

The observer update is:

$\hat{O}^{D}{}_{,k+1} = UpdateObserver^{D}(\hat{O}^{D}{}_{,k}, L^{D}{}_{,k+1}, \mathbf{Q}^{D}{}_{,k})$. (A.47)

The full recursive cycle is:

$X^{D}{}_{,k} \to \hat{O}^{D}{}_{,k} \to G^{D}{}_{,k} \to Trace^{D}{}_{,k} \to L^{D}{}_{,k+1} \to \hat{O}^{D}{}_{,k+1} \to X^{D}{}_{,k+1}$. (A.48)

A.10 Invariance notation

Let a valid frame transformation be:

$T_{p \to p'}{}^{D}$. (A.49)

The transformed state is:

$Z^{D}, P' = T_{p \to p'}{}^{D}(Z^{D}, P)$. (A.50)

An invariant I^{D} satisfies:

$I^{D}(Z^{D}, P') = I^{D}(Z^{D}, P)$. (A.51)

A tolerance-based invariant is:

$d[I^D(Z^D,P'),I^D(Z^D,P)] \le \varepsilon^D$. (A.52)

The objectivity criterion is:

Objectivity^D = Stability across Admissible Frame Transports. (A.53)

Appendix B — Master Physics–Finance–LLM Mapping Table

B.1 Core role mapping

General function	Physics	Finance	LLM
Possibility field	State or field configuration space	Possible returns, valuations, orders, and regimes	Candidate tokens, claims, plans, tools, and actions
State representation	Ψ, state vector, density operator	Asset or market state	Latent runtime and candidate-answer state
Complex form	$\Psi = A\exp(i\theta)$	$Z^f = R^f + iQ^f$	$Z^L = R^L + i\mathbf{Q}^L$
Magnitude	State norm or field strength	Total declared value–pressure magnitude	Total semantic commitment and residual magnitude
Relative phase	Physical phase relation	Asset orientation relative to market or factor frame	Candidate orientation relative to intent, evidence, and policy
Projection	Measurement in a selected basis	CAPM, factor, accounting, or valuation projection	Prompt, decoder, retrieval, verifier, and policy projection
Real component	Selected observable channel	Admitted price, return, or value	Emitted answer, action, or memory item
Residual	Unobserved state structure	Unpriced or hidden risk pressure	Unresolved factual, tool, retrieval, and instruction pressure
Gate	Outcome or transition condition	Execution, settlement, default, close, or margin threshold	Answer, tool, clarification, refusal, or memory gate
Trace	Measurement record	Trade, settlement, balance-sheet, or regulatory record	Answer, citation, tool result, audit log, or memory
Ledger	Ordered physical or environmental record	Transaction, accounting, and institutional history	Conversation, action, verifier, and memory history
Coherence	Stable phase relation	Broad synchronization of asset states	Agreement among samples, evidence, tools, and verifiers

General function	Physics	Finance	LLM
Decoherence	Loss of phase relation	Fragmentation and correlation breakdown	Context drift, source conflict, and candidate divergence
Backreaction	Measurement changes future state	Price alters orders, collateral, and expectations	Output alters prompt, memory, tools, and future answers
Internal time	Ordered physical events or records	Consequential market events	Verified semantic commitments
Imaginary-time-like depth	Euclidean time in physics; unresolved phase depth in SMFT	Unresolved market development	Unresolved semantic development
Invariance	Frame-stable physical relation	No-arbitrage or exposure stability	Prompt, source-order, and tool-substitution robustness

B.2 Characteristic mapping with evidence status

Characteristic	Physics meaning	Finance realization	LLM realization	Status outside physics
Complex state	Fundamental state formalism	Engineered R + iQ geometry	Engineered answer + residual geometry	Direct mathematical representation
Superposition	Coherent linear state combination	Alternative valuations or regime ensemble	Candidate-answer ensemble	Usually classical mixture or candidate set
Interference	Phase-dependent probability	Reinforcing or cancelling flows	Reinforcing or conflicting evidence	Direct if phase encoding is explicit
Projection basis	Observable selection	Benchmark, factor, currency, horizon	Prompt, retrieval, policy, schema	Strong operational recurrence
Collapse	Measurement update	Trade or regime commitment	Answer, tool, or memory commitment	Functional analogy
Decoherence	Environmental phase loss	Market fragmentation	Context and evidence fragmentation	Classical/ informational
Entanglement	Quantum nonseparability	Coupled financial dependence	Joint prompt–tool–memory dependence	Analogy only
Noncommutativity	Operator-order dependence	Hedge–liquidation order	Retrieve–draft–verify order	Genuine order sensitivity

Characteristic	Physics meaning	Finance realization	LLM realization	Status outside physics
		effects	effects	
Symmetry breaking	State selection from symmetry	Directional regime selection	Interpretation or plan selection	Strong nonlinear recurrence
Criticality	Divergent susceptibility	Leverage and liquidity instability	Prompt and routing instability	General complex-system property
Attractor	Dynamical basin	Trend or liquidity basin	Semantic or procedural basin	Strong dynamical recurrence
Imaginary time	Euclideanized physical time	Residual market-depth coordinate	Residual semantic-depth coordinate	Engineered analogue
Proper time	Observer-path-dependent duration	Asset-specific internal clock	Task-specific semantic clock	Analogy only
Horizon	Causal or geometric boundary	Liquidity or admissibility boundary	Context, policy, or tool boundary	Analogy only
Gauge invariance	Physical redundancy and symmetry	Equivalent factor or numéraire description	Equivalent prompt or tool frame	Strong engineering principle

B.3 Time-generation mapping

Time-generating stage	Physics	Finance	LLM
Fluctuation	Field or state variation	Asset return and order variation	Token and candidate variation
Interaction	Physical coupling	Correlation, funding, hedging, and flow coupling	Attention, retrieval, tool, and instruction coupling
Synchronization	Collective phase order	Market regime coherence	Candidate and evidence convergence
Clock hand	$\exp(i\Phi^{P})$	$\exp(i\Phi_{m})$	$\exp(i\Phi^{L})$
Cycle count	Unwrapped physical phase	Unwrapped market phase	Unwrapped semantic phase
Gate	Physical transition or measurement condition	Trade, settlement, default, or confirmation	Verifier, policy, tool, or memory threshold
Tick	Recorded event	Consequential market transition	Verified semantic commitment

Time-generating stage	Physics	Finance	LLM
Residual depth	Unresolved phase or Euclideanized direction	Latent pressure and unresolved phase	Unresolved evidence and candidate pressure
Ledger	Physical record	Price, settlement, accounting, and regulation	Conversation, tool, citation, and memory
Feedback	State conditioned by record	Price and history alter behaviour	Output and memory alter future context
Endogenous time	State-dependent internal duration	Market-created cadence	Episode-created semantic cadence

B.4 Failure mapping

General failure	Finance	LLM	Common diagnosis
False closure	Price treated as complete value	Fluent answer treated as complete truth	Residual erased
Premature gate	Weak breakout accepted as regime change	Unsupported claim emitted	Commitment before sufficient evidence
Over-lock	Crowded one-way market	Repetitive semantic attractor	Coherence without diversity
Decoherence	Market fragmentation	Context and source disagreement	Collective orientation lost
Trace pollution	Bad price or model enters institutional systems	Hallucination enters memory or knowledge base	Invalid record bends future decisions
Wrong frame	Inappropriate benchmark or horizon	Misread prompt or wrong evaluation frame	Projection basis misdeclared
Residual coupling	Liquidity amplifies credit stress	Tool failure amplifies factual uncertainty	Hidden off-diagonal risk
Frozen time	Pressure accumulates without repricing	Tokens accumulate without verified progress	Internal movement without ledger advancement
Release cascade	Crash, breakout, or liquidation	Abrupt revision or tool escalation	Residual depth converted into realized event

Appendix C — Derivation of CAPM as Projection Geometry

C.1 Classical excess-return form

The classical CAPM relation is:

$E[r_i] = r^f + \beta_i(E[r_m] - r^f)$. (C.1)

Define excess returns:

$x_i = r_i - r^f$. (C.2)

$x_m = r_m - r^f$. (C.3)

Then:

$E[x_i] = \beta_i E[x_m]$. (C.4)

The statistical beta is:

$\beta_i = Cov(x_i,x_m) / Var(x_m)$. (C.5)

C.2 Vector representation

Suppose n historical observations are collected.

Define centered vectors:

$\mathbf{x}_i = [x_{i,1} - \bar{x_i},...,x_{i,n} - \bar{x_i}]^T$. (C.6)

$\mathbf{x}_m = [x_{m,1} - \bar{x_m},...,x_{m,n} - \bar{x_m}]^T$. (C.7)

Sample covariance is proportional to:

$Cov(x_i,x_m) \propto \mathbf{x}_i^T\mathbf{x}_m$. (C.8)

Market variance is proportional to:

$Var(x_m) \propto \mathbf{x}_m^T\mathbf{x}_m$. (C.9)

Therefore:

$\beta_i = \mathbf{x}_i^T\mathbf{x}_m / \mathbf{x}_m^T\mathbf{x}_m$. (C.10)

This is the scalar coefficient of the orthogonal projection of $\mathbf{x}_i$ onto $\mathbf{x}_m$.

C.3 Projection operator

Normalize the market direction:

$\mathbf{e}_m = \mathbf{x}_m / \|\mathbf{x}_m\|$. (C.11)

The projection operator is:

$\hat{P}_m = \mathbf{e}_m\mathbf{e}_m^T$. (C.12)

The market-aligned component is:

$\mathbf{x}_i,\| = \hat{P}_m \mathbf{x}_i$. (C.13)

The residual projector is:

$\hat{Q}_m = I - \hat{P}_m$. (C.14)

The residual component is:

$\mathbf{x}_i,\perp = \hat{Q}_m \mathbf{x}_i$. (C.15)

The complete decomposition is:

$\mathbf{x}_i = \mathbf{x}_i,\| + \mathbf{x}_i,\perp$. (C.16)

Because the components are orthogonal:

$\mathbf{x}_i,\|^T \mathbf{x}_i,\perp = 0$. (C.17)

Therefore:

$\|\mathbf{x}_i\|^2 = \|\mathbf{x}_i,\|\|^2 + \|\mathbf{x}_i,\perp\|^2$. (C.18)

This is the direct geometric precursor of:

$A_i^2 = R_i^2 + Q_i^2$. (C.19)

C.4 Angle interpretation

Define θ_i as the angle between $\mathbf{x}_i$ and $\mathbf{x}_m$:

$\cos\theta_i = \mathbf{x}_i^T \mathbf{x}_m / [\|\mathbf{x}_i\|\|\mathbf{x}_m\|]$. (C.20)

Let standard deviations be proportional to vector norms:

$\sigma_i \propto \|\mathbf{x}_i\|$. (C.21)

$\sigma_m \propto \|\mathbf{x}_m\|$. (C.22)

Then:

$\beta_i = (\sigma_i/\sigma_m)\cos\theta_i$. (C.23)

The beta coefficient therefore combines:

relative amplitude σ_i/σ_m;

directional alignment $\cos\theta_i$.

The normalized market-aligned component is:

$R_i/A_i = \cos\theta_i$. (C.24)

The normalized residual component is:

$Q_i/A_i = \sin\theta_i$. (C.25)

Thus:

$Z_i = A_i[\cos\theta_i + i\sin\theta_i]$. (C.26)

Or:

$Z_i = A_i\exp(i\theta_i)$. (C.27)

C.5 Passive basis rotation

Let the observer basis rotate by angle α.

The transformed coordinates are:

$R_i' = R_i\cos\alpha + Q_i\sin\alpha$. (C.28)

$Q_i' = -R_i\sin\alpha + Q_i\cos\alpha$. (C.29)

The magnitude is invariant:

$R_i'^2 + Q_i'^2 = R_i^2 + Q_i^2$. (C.30)

The economic interpretation is:

the asset state may remain unchanged;

the benchmark or valuation frame changes;

previously residual structure can become visible.

This is why benchmark choice matters.

C.6 Multi-factor extension

Let the factor matrix be:

$F = [\mathbf{f}_1,\mathbf{f}_2,\ldots,\mathbf{f}_k]$. (C.31)

The projection onto the factor subspace is:

$\hat{P}F = F(F^TF)^{-1}F^T$. (C.32)

The admitted factor component is:

$\mathbf{x}_i,\|F = \hat{P}F\mathbf{x}_i$. (C.33)

The residual is:

$\mathbf{x}_i,\perp F = (I - \hat{P}F)\mathbf{x}_i$. (C.34)

The residual norm is:

$Q_i,F = \|\mathbf{x}_i,\perp F\|$. (C.35)

The admitted norm is:

$R_i,F = \|\mathbf{x}_i,\|F\|$. (C.36)

The total magnitude satisfies:

$A_i^2 = R_i,F^2 + Q_i,F^2$. (C.37)

This demonstrates that the complex residual principle is not limited to one-factor CAPM.

CAPM is the simplest pedagogical bridge.

C.7 Leave-one-out market construction

If asset i is part of the market portfolio, its relationship with the market contains self-inclusion.

Define the leave-one-out market:

$\mathbf{x}_m,-_i = [\Sigma_j\neq_i w_j\mathbf{x}_j] / \Sigma_j\neq_i w_j$. (C.38)

Then:

$\beta_i,-_i = Cov(x_i,x_m,-_i) / Var(x_m,-_i)$. (C.39)

The corresponding angle is:

$\cos\theta_i,-_i = \langle\mathbf{x}_i,\mathbf{x}_m,-_i\rangle / [\|\mathbf{x}_i\|\|\mathbf{x}_m,-_i\|]$. (C.40)

This avoids the simplest circularity when the asset–market relation is interpreted as an internal measurement.

C.8 Discount-depth construction

Let:

$Dbase(h) = (1 + rbase)^{-h}$. (C.41)

Let:

$DCAPM(h) = (1 + rCAPM)^{-h}$. (C.42)

The ratio is:

$DCAPM(h)/Dbase(h) = [(1 + rbase)/(1 + rCAPM)]^h$. (C.43)

Define:

$\kappa = \ln[(1 + rCAPM)/(1 + rbase)]$. (C.44)

Then:

$DCAPM(h)/Dbase(h) = \exp(-\kappa h)$. (C.45)

Define:

$\sigma_{CAPM} = \kappa h$. (C.46)

Then:

$D_{CAPM}/D_{base} = \exp(-\sigma_{CAPM})$. (C.47)

If:

$\cos\theta_{CAPM} = \exp(-\sigma_{CAPM})$. (C.48)

Then:

$\sin\theta_{CAPM} = \sqrt{[1 - \exp(-2\sigma_{CAPM})]}$. (C.49)

The normalized complex state is:

$Z_{CAPM}/A = \exp(-\sigma_{CAPM}) + i\sqrt{[1 - \exp(-2\sigma_{CAPM})]}$. (C.50)

The additive quantity is σ_{CAPM}:

$\sigma_{CAPM}(h_1 + h_2) = \sigma_{CAPM}(h_1) + \sigma_{CAPM}(h_2)$. (C.51)

The angle is bounded:

$0 \le \theta_{CAPM} < \pi/2$. (C.52)

Therefore:

σ_{CAPM} = selection-depth coordinate. (C.53)

θ_{CAPM} = bounded geometric display of that depth. (C.54)

Appendix D — Market Phase-Coherence Estimator

D.1 Objective

The market phase estimator seeks to construct:

$C_m(t) = \rho_m(t)\exp[i\Phi_m(t)]$. (D.1)

From a declared asset universe and feature map.

The estimator must specify:

asset universe;

observation frequency;

rolling window;

benchmark;

normalization;

phase orientation;

weighting;

missing-data rule.

Without these declarations, Φ_m has no stable meaning.

D.2 Data matrix

Let N assets be observed over rolling window W.

For asset i at time t, define feature vector:

$\mathbf{x}_i(t) = [x_{i,1}(t), x_{i,2}(t), \ldots, x_{i,p}(t)]^T$. (D.2)

Possible features include:

excess return;

realized volatility;

volume surprise;

spread;

momentum;

credit change;

option skew;

funding sensitivity.

Standardize each feature:

$\tilde{x}_{i,j}(t) = [x_{i,j}(t) - \mu_j(t)] / \sigma_j(t)$. (D.3)

The normalized state is:

$\mathbf{v}_i(t) = \tilde{\mathbf{x}}_i(t) / \|\tilde{\mathbf{x}}_i(t)\|$. (D.4)

D.3 Reference direction

Define the market reference vector:

$\mathbf{m}(t) = [\Sigma_i w_i(t)\mathbf{v}_i(t)] / \Sigma_i w_i(t)$. (D.5)

Normalize:

$\mathbf{e}_1(t) = \mathbf{m}(t) / \|\mathbf{m}(t)\|$. (D.6)

A second direction is required to assign signed phase.

One option is the leading residual principal direction:

$\mathbf{e}_2(t)$ = FirstPrincipalDirection of residuals orthogonal to $\mathbf{e}_1(t)$. (D.7)

Require:

$\mathbf{e}_1(t)^T\mathbf{e}_2(t) = 0$. (D.8)

The two-dimensional coordinates are:

$a_i(t) = \mathbf{v}_i(t)^T\mathbf{e}_1(t)$. (D.9)

$b_i(t) = \mathbf{v}_i(t)^T\mathbf{e}_2(t)$. (D.10)

The phase is:

$\theta_i(t) = \text{atan2}[b_i(t),a_i(t)]$. (D.11)

The projected magnitude is:

$A_i,_{2D}(t) = \sqrt{[a_i(t)^2 + b_i(t)^2]}$. (D.12)

The out-of-plane residual is:

$Q_i,_{out}^2(t) = 1 - A_i,_{2D}(t)^2$. (D.13)

This residual should be retained rather than ignored.

D.4 Phase sign stability

Principal directions can change sign without changing their geometric meaning.

To preserve phase continuity, orient $\mathbf{e}_2(t)$ such that:

$\mathbf{e}_2(t)^T\mathbf{e}_2(t-1) \geq 0$. (D.14)

If:

$\mathbf{e}_2(t)^T\mathbf{e}_2(t-1) < 0$. (D.15)

Then set:

$\mathbf{e}_2(t) \leftarrow -\mathbf{e}_2(t)$. (D.16)

A similar rule may be applied to $\mathbf{e}_1(t)$.

Without sign alignment, the estimator can create false π-radian jumps.

D.5 Collective coherence

Choose nonnegative weights $w_i(t)$.

Then:

$C_m(t) = [\Sigma_i w_i(t) \exp(i\theta_i(t))] / \Sigma_i w_i(t)$. (D.17)

Coherence is:

$\rho_m(t) = |C_m(t)|$. (D.18)

Collective phase is:

$\Phi_m(t) = \arg[C_m(t)]$. (D.19)

The unwrapped phase is:

$\tilde{\Phi}_m(t) = \text{unwrap}[\Phi_m(t)]$. (D.20)

The phase velocity is:

$\Omega_m(t) = [\tilde{\Phi}_m(t) - \tilde{\Phi}_m(t-1)] / \Delta t$. (D.21)

The phase acceleration is:

$A\Phi,_m(t) = [\Omega_m(t) - \Omega_m(t-1)] / \Delta t$. (D.22)

D.6 Phase entropy

Divide the circle into K bins.

Let $p_k(t)$ be the weighted proportion in bin k.

Then:

$H\text{phase}(t) = -\Sigma_{k=1}^{K} p_k(t) \ln p_k(t)$. (D.23)

Normalize:

$\bar{H}\text{phase}(t) = H\text{phase}(t) / \ln K$. (D.24)

Then:

$0 \leq \bar{H}\text{phase} \leq 1$. (D.25)

Interpretation:

$\bar{H}$phase near 1 = dispersed phases;

$\bar{H}$phase near 0 = concentrated phases.

A diversity-adjusted coherence may be:

$\rho\text{healthy}(t) = \rho_m(t) \bar{H}\text{phase}(t)$. (D.26)

An over-lock indicator may be:

$O\text{lock}(t) = \rho_m(t)[1 - \bar{H}\text{phase}(t)]$. (D.27)

High ρhealthy indicates coherent diversity.

High Olock indicates concentrated synchronization.

D.7 Multi-frame estimator

Compute phase under several admissible weighting frames f:

$C_m^f(t) = \rho_m^f(t)\exp[i\Phi_m^f(t)]$. (D.28)

Examples:

$f \in \{capitalization,equal,liquidity,risk,sector\text{-}neutral\}$. (D.29)

Define phase-frame dispersion:

$Dframe(t) = CircularVariance\{\Phi_m^f(t)\}$. (D.30)

Define coherence-frame dispersion:

$D\rho(t) = Var\{\rho_m^f(t)\}$. (D.31)

A robust market regime should have:

Dframe low and Dρ low. (D.32)

A frame-dependent artefact may have:

Dframe high or Dρ high. (D.33)

D.8 Gate estimator

A transparent gate may be:

$G_m(t) = \sigma[S_m(t)]$. (D.34)

Where:

$S_m(t) = a_0 + a_1\rho_m(t) + a_2H\text{phase}(t) + a_3Breadth(t) + a_4Volume(t) - a_5Fragility(t)$. (D.35)

The logistic function is:

$\sigma(x) = 1 / [1 + \exp(-x)]$. (D.36)

To model saturation, include:

$a_6\rho_m(t)^2$ with $a_6 < 0$. (D.37)

Then gate probability can rise with coherence and later fall under extreme lock.

D.9 Complex market time estimator

The realized increment is:

$\Delta\tau_m(t) = G_m(t)|\Delta\tilde{\Phi}_m(t)| / \Omega ref.$ (D.38)

The unresolved increment is:

$\Delta T_m(t) = [1 - G_m(t)]|\Delta\tilde{\Phi}_m(t)| / \Omega ref.$ (D.39)

The cumulative values are:

$\tau_m(t) = \Sigma_{s \le t} \Delta\tau_m(s).$ (D.40)

$T_m(t) = \Sigma_{s \le t} \Delta T_m(s).$ (D.41)

A decay-adjusted unresolved depth is:

$T_m(t) = \delta T_m(t-1) + \Delta T_m(t) - Release(t).$ (D.42)

Where:

$0 \le \delta \le 1.$ (D.43)

And:

$Release(t) = \eta(t) T_m(t-1).$ (D.44)

D.10 Estimator validity checks

The estimator should be rejected if:

phase changes drastically under minor preprocessing changes;

basis sign flips create artificial jumps;

results depend on a few dominant assets;

the phase merely reproduces market return;

coherence adds no value beyond average correlation;

T_m has no stable relation to later events;

results vanish under alternative admissible frames.

The phase estimator passes a minimum robustness test when:

$Robustness_{phase} \ge R^*.$ (D.45)

Where:

$Robustness_{phase} = 1 - AverageDistance$ across admissible phase constructions. (D.46)

D.11 Minimal estimator output

A practical phase dashboard should report:

Variable	Interpretation
ρ_m	collective coherence
Φ_m	current market phase
$\tilde{\Phi}_m$	accumulated phase chronology
Ω_m	phase velocity
$H_{\bar{p}}$phase	phase diversity
Olock	crowding or saturation
Dframe	cross-frame disagreement
G_m	event-admission strength
τ_m	realized market time
T_m	unresolved market depth
$\|\mathbf{Q}_m\|$	residual pressure magnitude

The dashboard should never report only one scalar “quantum market score.”

That would recreate the false closure the framework was designed to avoid.

Appendix E — Gate-Partitioned Complex Time

E.1 Purpose

The central temporal construction of this article separates system development into two channels:

development that becomes consequential trace;

development that remains unresolved.

The complex time-like coordinate is:

$\zeta^D = \tau^D + iT^D$. (E.1)

Where:

τ^D = realized, trace-producing internal time;

T^D = unresolved, non-trace-producing depth;

D = domain index.

The differential form is:

$d\zeta^D = d\tau^D + i\, dT^D$. (E.2)

The gate partitions accumulated phase development:

$d\tau^D = G^D|d\tilde{\Phi}^D| / \Omega ref^D$. (E.3)

$dT^D = (1 - G^D)|d\tilde{\Phi}^D| / \Omega ref^D$. (E.4)

Therefore:

$d\zeta^D = [G^D + i(1 - G^D)]|d\tilde{\Phi}^D| / \Omega ref^D$. (E.5)

This appendix clarifies what this coordinate means, what it does not mean, and how it can be implemented.

E.2 External process parameter

Let s denote an externally available ordering parameter.

Examples include:

laboratory time;

calendar time;

simulation step;

transaction sequence;

token index;

agent-runtime cycle.

The collective phase evolves as:

$\tilde{\Phi}^D = \tilde{\Phi}^D(s)$. (E.6)

Its phase velocity is:

$\omega^D(s) = d\tilde{\Phi}^D/ds$. (E.7)

The total phase-development magnitude is:

$d\chi^D = |d\tilde{\Phi}^D| / \Omega ref^D$. (E.8)

Where Ωref^D normalizes the scale.

Then:

$d\tau^D = G^D d\chi^D$. (E.9)

$dT^D = (1 - G^D)d\chi^D$. (E.10)

And:

$d\chi^D = d\tau^D + dT^D$. (E.11)

Equation (E.11) is a partition identity, not a conservation law of physical energy.

It states that each declared unit of phase development is classified as either:

admitted;

unresolved;

or partly distributed across both channels under a continuous gate.

E.3 Binary gate

The simplest gate is binary:

$G^D \in \{0,1\}$. (E.12)

If:

$G^D = 1$. (E.13)

Then:

$d\tau^D = d\chi^D$. (E.14)

$dT^D = 0$. (E.15)

The phase movement becomes fully realized.

If:

$G^D = 0$. (E.16)

Then:

$d\tau^D = 0$. (E.17)

$dT^D = d\chi^D$. (E.18)

The phase movement remains unresolved.

The binary model is appropriate when the domain contains a sharp commitment event.

Examples include:

settlement completed or not completed;

tool action executed or not executed;

memory item accepted or rejected;

legal declaration issued or not issued.

E.4 Continuous gate

Many real systems exhibit partial admission.

Define:

$0 \le G^D \le 1$. (E.19)

A continuous gate may represent:

incomplete commitment;

probabilistic admissibility;

graded evidence;

partial market breadth;

incomplete verification;

mixed observer agreement.

A logistic gate is:

$G^D = 1 / [1 + \exp(-S^D)]$. (E.20)

Where S^D is the gate score.

A generic score is:

$S^D = b_0 + \Sigma_j b_j g_j - \lambda \|\mathbf{Q}^D\|$. (E.21)

Where:

g_j = supporting gate factors;

b_j = weights;

$\|\mathbf{Q}^D\|$ = residual magnitude;

λ = residual penalty.

The gate increases when evidence for commitment grows.

It decreases when unresolved pressure grows.

E.5 Gate components

A general gate may be factored as:

$G^D = G_{align}^D G_{diff}^D G_{pressure}^D G_{commit}^D G_{trace}^D$. (E.22)

Where:

Alignment

G_{align}^D measures whether relevant components have sufficient common orientation.

Distinguishability

G_{diff}^{D} measures whether a meaningful gradient remains.

A completely uniform state may be unable to generate a new distinction.

Pressure

$G_{pressure}^{D}$ measures whether accumulated tension exceeds a transition threshold.

Commitment

G_{commit}^{D} measures whether sufficient resources, authority, or control are present.

Trace

G_{trace}^{D} measures whether the event can leave a durable record.

The multiplicative form means that failure of one indispensable condition can close the gate.

Alternative architectures may use:

$G^{D} = \sigma[a_0 + a_1 G_{align}^{D} + a_2 G_{diff}^{D} + a_3 G_{pressure}^{D} + a_4 G_{commit}^{D} + a_5 G_{trace}^{D}]$. (E.23)

The proper form is domain-specific.

E.6 Non-monotonic coherence

The relation between coherence and gate opening may be non-monotonic.

Low coherence can prevent collective organization.

Extreme coherence can eliminate useful difference.

A simple function is:

$G_{coh}(\rho) = 4\rho(1 - \rho)$. (E.24)

Where:

$0 \leq \rho \leq 1$. (E.25)

This function satisfies:

$G_{coh}(0) = 0$. (E.26)

$G_{coh}(0.5) = 1$. (E.27)

$G_{coh}(1) = 0$. (E.28)

The equation is illustrative rather than universal.

Its role is to model:

No Order → Productive Coherence → Saturated Lock. (E.29)

A more flexible form is:

$G_{coh}(\rho) = \rho^{a}(1-\rho)^{b} / B(a+1, b+1)$. (E.30)

Where a and b control the location and width of the productive region.

E.7 Accumulated coordinates

The realized coordinate is:

$\tau^{D}(s) = \int_{s0}^{s} G^{D}(u)|\dot{\Phi}^{D}(u)| / \Omega_{ref}^{D}\, du$. (E.31)

The unresolved coordinate is:

$T^{D}(s) = \int_{s0}^{s} [1 - G^{D}(u)]|\dot{\Phi}^{D}(u)| / \Omega_{ref}^{D}\, du$. (E.32)

The total phase depth is:

$\chi^{D}(s) = \tau^{D}(s) + T^{D}(s)$. (E.33)

The complex coordinate is:

$\zeta^{D}(s) = \tau^{D}(s) + iT^{D}(s)$. (E.34)

The magnitude is:

$|\zeta^{D}| = \sqrt{[\tau^{D2} + T^{D2}]}$. (E.35)

The complex-time angle is:

$\psi^{D} = \operatorname{atan2}(T^{D}, \tau^{D})$. (E.36)

Interpretation:

ψ^{D} near 0 means realized development dominates;

ψ^{D} near $\pi/2$ means unresolved development dominates.

E.8 Realization ratio

Define the realization ratio:

$r^{D} = \tau^{D} / [\tau^{D} + T^{D} + \varepsilon]$. (E.37)

Where ε prevents division by zero.

Then:

$0 \le r^{D} \le 1$. (E.38)

High r^{D} means most development has become trace.

Low r^{D} means most development remains unresolved.

The unresolved ratio is:

$u^D = T^D / [\tau^D + T^D + \varepsilon]$. (E.39)

Therefore:

$r^D + u^D \approx 1$. (E.40)

This ratio may be more interpretable than raw T^D when comparing systems of different scale.

E.9 Residual release

Unresolved depth may later convert into realized time.

Let $\eta^D(s)$ be the release rate:

$0 \le \eta^D(s) \le 1$. (E.41)

The realized coordinate updates:

$d\tau^D = G^D d\chi^D + \eta^D T^D ds$. (E.42)

The unresolved coordinate updates:

$dT^D = (1 - G^D)d\chi^D - \eta^D T^D ds$. (E.43)

A sudden release at s^* may be represented:

$\Delta\tau^D(s^*) = \eta T^D(s-)$. (E.44)

$T^D(s^*+) = [1 - \eta^*]T^D(s^*-)$. (E.45)

Where:

s^*- = state immediately before the release;

s^*+ = state immediately after the release.

This models:

market breakout;

crisis repricing;

decisive tool result;

clarification;

verifier resolution;

institutional declaration.

E.10 Residual decay

Some unresolved structure may dissipate without becoming trace.

Let δ^D be the decay rate:

$$dT^D/ds = (1 - G^D)|\dot{\Phi}^D| / \Omega ref^D - \eta^D T^D - \delta^D T^D. \quad (E.46)$$

The η^D term converts residual into realization.

The δ^D term removes residual from the declared system.

Examples include:

forgotten market tension;

expired option exposure;

abandoned LLM branch;

deleted context;

irrelevant ambiguity.

Decay must not be confused with resolution.

Resolved residual becomes trace.

Decayed residual disappears from the declared state without becoming accepted history.

E.11 Leakage

A declared boundary may fail to capture all residual transfer.

Let Λ^D be leakage:

$$dT^D/ds = Input^D - Release^D - Decay^D - \Lambda^D. \quad (E.47)$$

Leakage may include:

capital leaving the observed market;

hidden over-the-counter activity;

unlogged tool action;

lost conversation state;

inaccessible environmental interaction.

A complete accounting relation is:

$$InputDepth^D = Realized^D + StoredResidual^D + Dissipated^D + Leaked^D. \quad (E.48)$$

The model should not claim closure unless every material channel is measured or bounded.

E.12 Frozen-time-like regime

A frozen-time-like regime is defined by:

$d\tau^D/ds \approx 0$. (E.49)

While:

$dT^D/ds > 0$. (E.50)

The system continues internal development but produces little consequential trace.

Examples include:

Finance

stale price;

trading halt;

volatility suppression;

liquidity disappearance;

fixed-price defence.

LLM

repeated reasoning loop;

tool retry loop;

semantic repetition;

unresolved verifier conflict;

plan without action.

The freeze score is:

$F^D = [dT^D/ds] / [d\tau^D/ds + \varepsilon]$. (E.51)

A large F^D indicates high unresolved development relative to realized progress.

E.13 Accelerated internal time

Define the internal-time rate:

$\gamma^D = d\tau^D/ds$. (E.52)

A high-rate regime satisfies:

$\gamma^D \gg \gamma baseline^D$. (E.53)

Examples include:

crisis market;

rapid regime transition;

high-productivity reasoning episode;

decisive multi-tool execution.

A low-rate regime satisfies:

$\gamma^{D} \ll \gamma\text{baseline}^{D}$. (E.54)

This is an effective internal-time rate.

It is not relativistic time dilation.

E.14 Multiple clocks

Suppose the system contains subsystems a and b.

Their realized times are:

$d\tau_a = G_a |d\tilde{\Phi}_a| / \Omega_a$. (E.55)

$d\tau_\beta = G_\beta |d\tilde{\Phi}_\beta| / \Omega_\beta$. (E.56)

The clock ratio is:

$\chi_{a\beta} = d\tau_a / d\tau_\beta$. (E.57)

If:

$\chi_{a\beta}$ = constant. (E.58)

The clocks are proportionally synchronized.

If:

$d\chi_{a\beta}/ds \neq 0$. (E.59)

Their internal development rates diverge.

This may indicate:

cross-market dislocation;

asynchronous agent modules;

stale memory;

delayed settlement;

tool result arriving after context changed.

E.15 Shared-world synchronization

A system containing multiple trace-bearing observers requires sufficient temporal agreement.

Let observer phases be $\Phi_1,\ldots,\Phi_n$.

Define:

$$C_{time} = [\Sigma_j w_j \exp(i\Phi_j)] / \Sigma_j w_j. \quad (E.60)$$

Then:

$$\rho_{time} = |C_{time}|. \quad (E.61)$$

A shared world requires:

$$\rho_{time} \geq \rho_{time}^*. \quad (E.62)$$

Below the threshold, observers may disagree on:

event order;

current state;

which trace is authoritative;

whether an action is still admissible.

This is a practical concern in distributed AI and financial settlement systems.

E.16 Temporal objectivity

Two observers a and b may use different clocks.

A valid temporal transport is:

$$T_{a\to\beta}: \tau_a \to \tau_\beta. \quad (E.63)$$

A cross-observer invariant may be:

$$\mathrm{Order}(\mathrm{Trace}_1,\mathrm{Trace}_2)_a = \mathrm{Order}(\mathrm{Trace}_1,\mathrm{Trace}_2)_\beta. \quad (E.64)$$

Or:

$$\mathrm{Sign}(\Delta\tau_a) = \mathrm{Sign}(\Delta\tau_\beta). \quad (E.65)$$

Temporal objectivity does not require identical clock values.

It requires stable relations across admissible clock transformations.

E.17 Complex-time state transition

The full state may be:

$S^D = \{Z^D, \tilde{\Phi}^D, G^D, \tau^D, T^D, L^D\}$. (E.66)

The transition is:

$S^D_{,k+1} = F^D(S^D_{,k}, \text{Environment}^D_{,k}, \text{Intervention}^D_{,k})$. (E.67)

Expanded:

$\tilde{\Phi}^D_{,k+1} = \tilde{\Phi}^D_{,k} + \Delta\tilde{\Phi}^D_{,k}$. (E.68)

$G^D_{,k} = \text{Gate}^D(S^D_{,k})$. (E.69)

$\Delta\tau^D_{,k} = G^D_{,k}|\Delta\tilde{\Phi}^D_{,k}| \,/\, \Omega\text{ref}^D$. (E.70)

$\Delta T^D_{,k} = [1 - G^D_{,k}]|\Delta\tilde{\Phi}^D_{,k}| \,/\, \Omega\text{ref}^D$. (E.71)

$L^D_{,k+1} = \text{Update}(L^D_{,k}, \text{Trace}^D_{,k}, \text{Residual}^D_{,k})$. (E.72)

$Z^D_{,k+1} = \text{UpdateState}(Z^D_{,k}, L^D_{,k+1}, T^D_{,k+1})$. (E.73)

This forms a minimal computational kernel.

E.18 Validity conditions

The complex-time model is meaningful only if:

$\tilde{\Phi}^D$ has an operational definition;

G^D is observable or estimable;

trace events can be identified;

residual depth predicts later behaviour;

τ^D adds explanatory value beyond s;

results survive admissible frame changes.

The validity condition is:

$\text{ValidComplexTime}^D \Leftrightarrow \text{PhaseRobust} \wedge \text{GateAuditable} \wedge \text{TraceObservable} \wedge \text{ResidualPredictive} \wedge \text{IncrementalUtility}$. (E.74)

If these conditions fail, ζ^D should be treated as visualization rather than explanatory coordinate.

Appendix F — LLM Residual Vector Schema

F.1 Purpose

The LLM residual vector is designed to replace a single undifferentiated confidence score with typed, actionable uncertainty.

The general state is:

$Z^{L} = R^{L} + i\mathbf{Q}^{L}$. (F.1)

Where:

R^{L} = admitted runtime commitment. (F.2)

$\mathbf{Q}^{L}$ = typed unresolved state. (F.3)

A recommended schema is:

$\mathbf{Q}^{L}$ =
[Qfact,Qsource,Qretrieval,Qinstruction,Qpolicy,Qtool,Qambiguity,Qalternative,Qmemory,Qtempor
(F.4)

Each coordinate should have:

an operational definition;

measurable proxies;

a gate consequence;

a trace field;

a resolution pathway.

F.2 Factual residual

Qfact measures the probability or degree that a claim lacks adequate factual support.

Possible components include:

$Qfact = w_1U + w_2C + w_3D + w_4V$. (F.5)

Where:

U = unsupported-claim rate;

C = contradiction score;

D = cross-sample disagreement;

V = verifier failure.

Normalize:

$0 \leq Qfact \leq 1$. (F.6)

Gate consequence:

High Qfact → verify, qualify, or abstain. (F.7)

Resolution pathways:

retrieve authoritative source;

calculate;

use tool;

remove unsupported claim;

explicitly qualify.

F.3 Source residual

Qsource measures weakness in source provenance.

Possible components are:

$$Q_{source} = a_1 AuthorityGap + a_2 ProvenanceGap + a_3 CitationMismatch + a_4 CoverageGap. \quad (F.8)$$

A source may exist but still be inadequate because:

it is not authoritative;

it does not support the claim;

it covers only part of the answer;

its provenance is unclear.

Gate consequence:

$$\text{High } Q_{source} \rightarrow \text{weaken claim or search for better evidence.} \quad (F.9)$$

F.4 Retrieval residual

Qretrieval measures uncertainty caused by the retrieval process.

A practical form is:

$$Q_{retrieval} = b_1(1 - Relevance) + b_2 Staleness + b_3 Conflict + b_4 MissingCoverage. \quad (F.10)$$

Retrieval residual differs from source residual.

Qsource concerns the quality of obtained evidence.

Qretrieval concerns whether the retrieval process found the right evidence.

Gate consequence:

$$\text{High } Q_{retrieval} \rightarrow \text{reformulate query, broaden search, or disclose retrieval limits.} \quad (F.11)$$

F.5 Instruction residual

$Q_{instruction}$ measures unresolved instruction conflict.

Possible components are:

$$Q_{instruction} = c_1 PriorityConflict + c_2 ScopeAmbiguity + c_3 FormatConflict + c_4 IdentityAmbiguity. \quad (F.12)$$

Examples include:

user instruction conflicts with system constraint;

requested scope is unclear;

two output formats conflict;

the intended entity is ambiguous.

Gate consequence:

High $Q_{instruction}$ → clarify, resolve hierarchy, or select the safest valid interpretation. (F.13)

F.6 Policy residual

Q_{policy} measures uncertainty in admissibility.

Possible components are:

$$Q_{policy} = d_1 RiskClassAmbiguity + d_2 PermissionUncertainty + d_3 JurisdictionGap + d_4 SafetyConflict. \quad (F.14)$$

Gate consequence:

High Q_{policy} → constrain, refuse, or escalate. (F.15)

Policy residual should not be hidden inside factual confidence.

An answer can be factually correct but inadmissible to provide or act upon.

F.7 Tool residual

Q_{tool} measures uncertainty associated with external tools.

Possible components include:

$$Q_{tool} = e_1 Failure + e_2 PartialResult + e_3 ArgumentUncertainty + e_4 CrossToolConflict + e_5 StateStaleness. \quad (F.16)$$

Gate consequence:

High Q_{tool} → retry, substitute, disclose failure, or avoid claiming a tool-derived result. (F.17)

A tool failure should never be converted into fabricated output.

F.8 Ambiguity residual

Qambiguity measures multiplicity of plausible interpretations.

One estimate is:

$$Q_{ambiguity} = \text{Entropy}[\Pr(\text{Interpretation}_j \mid \text{Prompt})]. \quad (F.18)$$

A normalized form is:

$$Q_{ambiguity} = H_{interpretation} / \ln N_{interpretation}. \quad (F.19)$$

Gate consequence:

High Qambiguity → ask one targeted clarification or answer conditionally. (F.20)

F.9 Alternative residual

Qalternative measures the significance of rejected answer branches.

Let candidate scores be $p_1,\ldots,p_n$.

If p_1 is selected, define:

$$Q_{alternative} = 1 - p_1. \quad (F.21)$$

A more informative measure is:

$$Q_{alternative} = \Sigma_{j>1} p_j \text{ValueDifference}(a_j,a_1). \quad (F.22)$$

High alternative residual means that significant alternatives remain.

Gate consequence:

High Qalternative → compare alternatives, disclose assumptions, or request more evidence. (F.23)

F.10 Memory residual

Qmemory measures the risk that stored context is unreliable.

Possible components are:

$$Q_{memory} = f_1\text{ProvenanceGap} + f_2\text{IdentityMismatch} + f_3\text{Staleness} + f_4\text{CurrentConflict} + f_5\text{ConsentGap}. \quad (F.24)$$

Gate consequence:

High Qmemory → do not rely on memory without revalidation. (F.25)

A memory item should ideally include:

$$\text{MemoryItem} = \{\text{content},\text{source},\text{scope},\text{time},\text{confidence},\text{expiry},\text{consent}\}. \quad (F.26)$$

F.11 Temporal residual

Qtemporal measures the probability that the answer depends on information that may have changed.

A simple form is:

$$Q_{temporal} = g_1 ChangeRate + g_2 AgeOfEvidence + g_3 CurrentRoleDependence + g_4 ScheduleDependence. \quad (F.27)$$

Examples include:

current officeholder;

current price;

current law;

current software version;

current schedule.

Gate consequence:

High Qtemporal → obtain current data before answering. (F.28)

F.12 Causal residual

Qcausal measures uncertainty in causal interpretation.

Possible components include:

$$Q_{causal} = h_1 ConfoundingRisk + h_2 ReverseCausality + h_3 SelectionBias + h_4 MechanismGap. \quad (F.29)$$

Gate consequence:

High Qcausal → use correlational language and avoid claiming mechanism. (F.30)

F.13 Consequence residual

Qconsequence measures uncertainty in downstream impact.

A practical form is:

$$Q_{consequence} = j_1 Irreversibility + j_2 AffectedScope + j_3 FinancialCost + j_4 SafetyRisk + j_5 PropagationRisk. \quad (F.31)$$

Gate consequence:

High Qconsequence → require stronger verification, approval, or reversibility. (F.32)

The same factual uncertainty should produce different gates depending on consequence.

A casual recommendation and an irreversible financial transaction should not use the same threshold.

F.14 Residual normalization

Residual components may be normalized:

$q_j = \text{clip}[(x_j - l_j)/(u_j - l_j),0,1]$. (F.33)

Where:

l_j = lower operational bound;

u_j = upper operational bound.

The residual vector is:

$\mathbf{q} = [q_1,\ldots,q_n]^T$. (F.34)

A weighted magnitude is:

$Q_{total} = \sqrt{[\mathbf{q}^T G^L \mathbf{q}]}$. (F.35)

The metric G^L should be estimated on training data and fixed before final evaluation.

F.15 Residual coupling

Off-diagonal terms represent interactions.

For two residuals q_j and q_k:

$\text{Interaction}_{jk} = 2G_{jk}q_jq_k$. (F.36)

Examples include:

$Q_{temporal} \times Q_{fact} \rightarrow$ outdated factual claim. (F.37)

$Q_{tool} \times Q_{consequence} \rightarrow$ dangerous action under tool uncertainty. (F.38)

$Q_{memory} \times Q_{identity} \rightarrow$ wrong-person memory contamination. (F.39)

$Q_{retrieval} \times Q_{causal} \rightarrow$ unsupported causal synthesis. (F.40)

A runtime should identify dominant coupled residuals rather than only total magnitude.

F.16 Gate threshold by consequence

Let risk class be r.

The admission threshold is:

$G^* = G^*(r)$. (F.41)

For low-risk tasks:

$G_{low} < G_{high}$. (F.42)

For high-risk tasks:

$G^*_{high} \to 1$. (F.43)

A residual-adjusted gate is:

$G^L = \sigma[S^L - \lambda(r)Q_{total}]$. (F.44)

Where $\lambda(r)$ increases with consequence.

Thus:

Higher Consequence ⇒ Stronger Residual Penalty. (F.45)

F.17 Output-channel policy

A channel policy may be:

Answer if $Q_{total} \leq q_{answer}$ and $G_{evidence} \geq e^*$. (F.46)

Qualify if $q_{answer} < Q_{total} \leq q_{qualify}$. (F.47)

Retrieve if $Q_{retrieval} \geq q_{retrieve}$. (F.48)

Clarify if $Q_{ambiguity} \geq q_{clarify}$. (F.49)

RetryTool if $Q_{tool} \geq q_{tool}$ and retry is admissible. (F.50)

Escalate if $Q_{consequence} \geq q_{high}$ and Q_{total} remains high. (F.51)

Refuse if policy prohibits the requested action. (F.52)

The thresholds must be calibrated to task costs.

F.18 Residual disclosure schema

A user-facing residual summary can be compact.

Suggested fields:

ResidualSummary = {Known,Uncertain,Missing,Conflicting,ToolStatus,NextResolution}. (F.53)

A structured trace may contain:

$Trace^L = \{protocol,claim,evidence,\mathbf{Q}^L,G^L,channel,tool,memory,version\}$. (F.54)

The user-facing summary need not include private hidden reasoning.

It should include only decision-relevant limits.

F.19 Residual update

At runtime step k:

$\mathbf{Q}^{L}_{,k+1} = U(\mathbf{Q}^{L}_{,k}, Evidence_k, Tool_k, Verifier_k, User_k)$. (F.55)

A linear approximation is:

$\mathbf{Q}^{L}_{,k+1} = A\mathbf{Q}^{L}_{,k} + B\mathbf{e}_k + C\mathbf{t}_k + D\mathbf{v}_k$. (F.56)

Where:

$\mathbf{e}_k$ = new evidence;

$\mathbf{t}_k$ = tool result;

$\mathbf{v}_k$ = verifier result.

A nonlinear update may be:

$\mathbf{Q}^{L}_{,k+1} = NeuralResidualUpdater(\mathbf{Q}^{L}_{,k}, Context_k)$. (F.57)

The updater must still remain auditable at the output level.

F.20 Residual-resolution effectiveness

For residual component q_j:

$ResolutionEffect_j = q_{j,before} - q_{j,after}$. (F.58)

The cost-adjusted effectiveness is:

$Efficiency_j = ResolutionEffect_j / Cost_j$. (F.59)

The runtime should learn which intervention works best for each residual type.

For example:

retrieval may reduce Qfact;

clarification may reduce Qambiguity;

tool substitution may reduce Qtool;

memory deletion may reduce Qmemory.

F.21 Residual ledger

The residual ledger is:

$Lres^{L} = \{\mathbf{Q}^{L},0,\mathbf{Q}^{L},1,\ldots,\mathbf{Q}^{L},_{n}\}$. (F.60)

A useful audit computes:

$Persistence_{j}$ = AverageDuration($q_{j} \geq q^{*}$). (F.61)

$Recurrence_{j}$ = Number of Episodes with $q_{j} \geq q^{*}$. (F.62)

$Escalation_{j}$ = Probability(q_{j} increases after intervention). (F.63)

These measures help identify systemic failure.

F.22 Minimal schema

A low-cost prototype can begin with:

$\mathbf{Q}^{L},min = [Qfact,Qretrieval,Qambiguity,Qtool]^{T}$. (F.64)

The minimal gate is:

$G^{L},min = GevidenceGclarityGtool$. (F.65)

The minimal output is:

Outputmin = {answer,residual_summary,citations,tool_status}. (F.66)

The prototype should first test whether this simple schema reduces unsupported commitments.

Appendix G — Unified Experimental Protocols

G.1 Protocol-first rule

Every experiment must declare:

$P = (B,\Delta,h,u,\varphi,M,T,E)$. (G.1)

Where:

B = system boundary;

Δ = observation rule;

h = horizon;

u = admissible intervention;

φ = feature map;

M = model specification;

T = target variable;

E = evaluation metric.

No variable should be interpreted outside this declaration.

G.2 Claim declaration

Each experiment begins with a falsifiable claim.

A claim record is:

Claim = {Hypothesis,Mechanism,Target,Direction,Horizon,FailureCondition}. (G.2)

Example finance claim:

Higher T_m predicts higher realized volatility over the next 20 trading days. (G.3)

Example LLM claim:

Typed residual gating reduces unsupported committed claims by at least 20% at equal refusal cost. (G.4)

The failure condition must be explicit.

G.3 Data declaration

The data record is:

Data = {Source,Period,Sampling,Universe,Exclusions,MissingRule,Version}. (G.5)

For finance, this includes:

survivorship-bias handling;

delisting;

corporate actions;

transaction-cost assumptions;

market closures.

For LLMs, this includes:

benchmark version;

source documents;

prompt set;

model version;

tool configuration;

random seed.

G.4 Training, validation, and test

The basic split is:

Dataset = Train ⊔ Validation ⊔ Test. (G.6)

No test information may influence:

feature selection;

threshold selection;

residual weights;

gate design;

stopping rule.

The final model is:

M* = Select(M | Train,Validation). (G.7)

The final result is:

Result = Evaluate(M* | Test). (G.8)

G.5 Baseline declaration

Every experiment requires at least one simple baseline.

Let:

M_0 = simplest credible baseline. (G.9)

Let:

M_1 = strongest conventional baseline. (G.10)

Let:

MCRP = Complex Residual Principle model. (G.11)

The incremental gain is:

Δ_0 = Score(MCRP) – Score(M_0). (G.12)

Δ_1 = Score(MCRP) – Score(M_1). (G.13)

The framework has practical value only if:

Δ_1 > ComplexityPenalty. (G.14)

G.6 Null model

A null model destroys the proposed mechanism while preserving simpler statistics.

Possible null operations include:

phase shuffle;

label shuffle;

source-order randomization;

gate randomization;

residual permutation;

trace deletion.

Let:

Mnull = NullTransform(MCRP). (G.15)

The mechanism is supported only if:

Score(MCRP) > Score(Mnull). (G.16)

G.7 Ablation

The full model is:

Mfull = {Phase,Residual,Gate,Trace,Ledger,ObserverUpdate}. (G.17)

Ablations remove one component:

M−Phase. (G.18)

M−Residual. (G.19)

M−Gate. (G.20)

M−Trace. (G.21)

M−Ledger. (G.22)

M−ObserverUpdate. (G.23)

The component contribution is:

$\text{Contribution}_j = \text{Score}(\text{Mfull}) - \text{Score}(M_{-j})$. (G.24)

A component with near-zero contribution should not be presented as essential.

G.8 Frame robustness

Let admissible frames be:

$P_1, P_2, \ldots, P_n$. (G.25)

Compute the target relation under each frame:

$I_j = I(M \mid P_j)$. (G.26)

Frame robustness is:

$R_{frame} = 1 - \text{AverageDistance}(I_j, I_k)$. (G.27)

A robust result requires:

$R_{frame} \geq R^*$. (G.28)

Equivalent frames should preserve the core relation.

Materially different frames may legitimately produce different results.

G.9 Replication

Replication levels are:

Internal replication

Repeat with new seeds or resamples.

Temporal replication

Repeat in a later period.

Cross-domain replication

Repeat in another market, model, or task.

Independent replication

A separate team reproduces the result.

Define replication score:

$R_{rep} = \Sigma_j w_j \text{Success}_j$. (G.29)

A classic article requires more than one successful demonstration.

It requires a reproducible program.

G.10 Intervention test

A mechanism is more credible if controlled intervention produces the predicted effect.

Let u be an intervention.

The causal estimate is:

$\Delta Y(u) = E[Y \mid do(u)] - E[Y \mid do(u_0)]$. (G.30)

Examples include:

Finance simulation

Reduce leverage and test whether residual-release severity declines.

LLM runtime

Increase evidence threshold and test whether premature ticks decline.

Memory system

Require provenance and test whether trace pollution declines.

A theory that predicts but cannot guide intervention remains incomplete.

G.11 Cost adjustment

A more complex system may improve accuracy by using far more resources.

Define total cost:

$Cost = c_{token} + c_{tool} + c_{latency} + c_{human} + c_{compute} + c_{risk}$. (G.31)

Cost-adjusted utility is:

$Utility = Benefit - \lambda Cost$. (G.32)

A normalized efficiency is:

$Efficiency = Benefit / [Cost + \varepsilon]$. (G.33)

The framework should report both raw performance and efficiency.

G.12 Error taxonomy

Errors should be classified.

A generic taxonomy is:

$Error = \{ProjectionError, ResidualError, GateError, TraceError, FrameError, UpdateError\}$. (G.34)

Projection error

The wrong visible structure is extracted.

Residual error

Material unresolved structure is missed or misclassified.

Gate error

The wrong commitment channel is selected.

Trace error

The record is incomplete or corrupted.

Frame error

The protocol or basis is invalid.

Update error

The observer fails to learn from history.

This taxonomy permits mechanism-level diagnosis.

G.13 Statistical significance and practical significance

Statistical significance is not enough.

Let:

pvalue < α. (G.35)

Practical significance requires:

EffectSize ≥ δ*. (G.36)

And:

UtilityGain > 0. (G.37)

A tiny but statistically detectable improvement may not justify the additional complexity.

G.14 Pre-registration template

A pre-registration should include:

hypothesis;

protocol;

data source;

feature map;

phase definition;

residual definition;

gate formula;

target;

baseline;

evaluation metric;

exclusion rule;

stopping rule;

failure criterion.

The declaration hash may be:

Hdecl = Hash(ProtocolDocument). (G.38)

This protects against post-hoc reinterpretation.

G.15 Finance experiment template

A finance experiment record is:

Experiment^{f} = {Universe,Window,Features,PhaseMap,Weights,Gate,Residuals,Target,Baseline,Metric}. (G.39)

A minimal example is:

Universe = 100 liquid equities. (G.40)

Window = rolling 252 trading days. (G.41)

Features = return,volatility,volume,breadth,spread. (G.42)

Target = next-20-day volatility jump. (G.43)

Baseline = GARCH + momentum + breadth. (G.44)

Primary test:

Does T_m add out-of-sample predictive value? (G.45)

G.16 LLM experiment template

An LLM experiment record is:

$Experiment^{L}$ = {Model,PromptSet,Retrieval,Tools,ResidualSchema,Gate,TracePolicy,Target,Baseline,Metric}. (G.46)

A minimal example is:

Task = source-grounded factual question answering. (G.47)

Residuals = Qfact,Qretrieval,Qambiguity,Qtool. (G.48)

Channels = answer,qualify,retrieve,clarify. (G.49)

Target = unsupported committed claim rate. (G.50)

Baseline = confidence-calibrated RAG. (G.51)

Primary test:

Does typed residual gating reduce hallucination at equal refusal rate? (G.52)

G.17 Cross-domain kernel test

The strongest cross-domain experiment tests whether the same abstract kernel predicts both domains.

Define:

Kernel = {Phase,Residual,Gate,Trace}. (G.53)

For each domain D:

$Score^{D}$ = Evaluate(Kernel | $Domain^{D}$). (G.54)

A general kernel should satisfy:

$Score^{f} > Baseline^{f}$. (G.55)

$Score^{L} > Baseline^{L}$. (G.56)

The variables need not be identical.

The functional relations should remain comparable.

G.18 Rejection rule

The framework should be reduced or rejected if:

NoStablePhase ∨ NoResidualCalibration ∨ NoGateUtility ∨ NoTraceBenefit ∨ NoOutOfSampleGain. (G.57)

A modular reduction is permitted.

For example:

retain residual governance;

remove phase terminology;

retain gate and trace;

abandon complex-time claim.

The framework should survive by empirical selection, not by conceptual attachment.

Appendix H — Failure Modes and Invalid Analogies

H.1 Purpose

Cross-domain theories are vulnerable to overextension.

This appendix defines the main ways in which the Complex Residual Principle can fail scientifically, mathematically, or rhetorically.

The governing restriction is:

Functional Homology ≠ Material Identity. (H.1)

A mapping is legitimate only when it improves:

measurement;

prediction;

intervention;

diagnosis;

falsifiability.

H.2 Naming without mechanism

A weak explanation simply renames ordinary variables.

Examples include:

Volatility = quantum uncertainty. (H.2)

Price change = collapse. (H.3)

Prompt sensitivity = superposition. (H.4)

These statements do not specify:

operator;

state space;

measurement;

prediction;

failure condition.

The mechanism requirement is:

ImportedTerm is valid only if OperationalGain > 0. (H.5)

H.3 Complex number decoration

A model may attach i to an arbitrary quantity:

Z = Price + iRisk. (H.6)

This is insufficient.

A valid complex construction should specify:

why the coordinates are paired;

what transformation relates them;

whether a phase angle is meaningful;

what invariant is preserved;

what prediction improves.

Without these:

ComplexNotation = Decoration. (H.7)

H.4 Arbitrary residual definition

If Q is defined only after failure, it can explain everything and predict nothing.

The invalid procedure is:

ObserveFailure → DefineQ to fit Failure. (H.8)

The valid procedure is:

DefineQ before Test → MeasureQ → PredictFailure. (H.9)

Residual must be declared in advance.

H.5 Arbitrary phase

A phase derived from an embedding may depend on:

model choice;

normalization;

axis orientation;

preprocessing;

dimensionality reduction.

If small changes create large phase changes:

PhaseRobustness ≈ 0. (H.10)

Then phase should not be treated as a stable observable.

A valid phase requires:

StableOrientation + ReproducibleReference + PredictiveValue. (H.11)

H.6 Confusing correlation with interference

Two assets moving together may be correlated.

Two sources supporting the same answer may be redundant.

Neither automatically proves phase interference.

A phase-interference claim requires a defined relation:

$$\text{Outcome} = A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\theta. \quad (H.12)$$

And evidence that $\Delta\theta$ contributes beyond ordinary correlation or additive scoring.

Otherwise:

Correlation ≠ Interference. (H.13)

H.7 Confusing dependence with entanglement

Classical dependence is:

$$P(X,Y) \neq P(X)P(Y). \quad (H.14)$$

Quantum entanglement requires stronger state-space and experimental conditions.

Therefore:

Financial Coupling ≠ Quantum Entanglement. (H.15)

Prompt–Memory Dependence ≠ Quantum Entanglement. (H.16)

The term entanglement-like should remain explicitly qualified.

H.8 Confusing order effects with quantum noncommutativity

If:

$AB(X) \neq BA(X)$. (H.17)

Then operation order matters.

This does not automatically establish:

$[\hat{A},\hat{B}] = i\hbar\hat{C}$. (H.18)

Order effects are common in classical, legal, financial, and computational systems.

The valid conclusion is:

Operational Noncommutativity exists. (H.19)

The invalid conclusion is:

Therefore the system obeys quantum observable algebra. (H.20)

H.9 Confusing selection with physical collapse

A transaction selects one executed price.

A decoder selects one token.

A policy selects one action.

These are collapse-like only at the role level.

They do not establish:

physical state-vector reduction;

Born probabilities;

quantum measurement disturbance;

observer-induced physical ontology.

Therefore:

Selection Gate ≠ Quantum Collapse. (H.21)

H.10 Confusing uncertainty with Heisenberg uncertainty

A system may have a trade-off:

More precision in A reduces precision in B. (H.22)

But the Heisenberg relation requires noncommuting operators.

Without a justified commutator:

$\Delta A \Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$. (H.23)

the analogy remains heuristic.

H.11 Confusing Euclidean decay with physical imaginary time

An exponential factor:

$\exp(-\kappa h)$. (H.24)

can resemble Euclidean-time suppression.

But resemblance does not establish physical Wick rotation.

A valid physical claim would require a derivation connecting:

$t \rightarrow -i\tau$. (H.25)

and the governing action or Hamiltonian.

The finance and LLM models instead use:

T = unresolved development depth. (H.26)

This is an engineered analogue.

H.12 Confusing internal time with relativity

A state-dependent internal rate:

$d\tau/ds = f(X)$. (H.27)

does not establish relativistic proper time.

Relativity requires:

spacetime metric;

invariant interval;

Lorentz transformation;

causal structure;

empirical correspondence.

Therefore:

Variable Internal Tick Rate ≠ Relativistic Time Dilation. (H.28)

H.13 Confusing geometry with physical spacetime

Embedding distance is:

$d^2 = (x - y)^T G(x - y)$. (H.29)

This creates geometry.

But not every geometry is physical spacetime.

A financial covariance manifold or semantic embedding space lacks physical status unless a validated mapping is derived.

Therefore:

Relational Geometry ≠ Physical Spacetime. (H.30)

H.14 Confusing strong attractors with black holes

A market or semantic attractor may trap trajectories.

A black hole has specific physical properties involving spacetime, horizons, gravity, and causal structure.

Thus:

Strong Attractor ≠ Black Hole. (H.31)

The phrase black-hole-like should be reserved for a narrow structural analogy:

high attraction;

low escape probability;

information bottleneck;

boundary effect.

H.15 Confusing self-reference with consciousness

A system can update itself:

$\text{Observer}_{k+1} = \text{Update}(\text{Observer}_k, \text{Trace}_k)$. (H.32)

This shows self-reference or adaptation.

It does not prove subjective experience.

Therefore:

H.16 Confusing predictive success with ontological truth

A complex residual model may predict markets well.

That does not prove that markets literally contain an imaginary dimension.

A residual-aware LLM may reduce hallucination.

That does not prove that semantic phase is physically real.

Predictive success supports:

Operational Validity. (H.34)

It does not automatically establish:

Ontological Identity. (H.35)

H.17 Overfitting the gate

A flexible gate can be fitted to past transitions:

$G_m(t) = 1$ near every known crisis. (H.36)

This creates retrospective perfection.

The gate is valid only if:

fixed before test;

applied out of sample;

compared with simpler thresholds;

stable across regimes.

Otherwise:

Gate = Event Labeler rather than Event Predictor. (H.37)

H.18 Residual double counting

Residual components may overlap.

For example:

Qfact and Qretrieval may both include missing evidence.

Qliquidity and Qfunding may both capture the same stress.

If overlap is ignored:

Qtotal is inflated. (H.38)

The remedy is a residual metric:

$Qtotal^2 = \mathbf{Q}^T G \mathbf{Q}$. (H.39)

The metric must account for covariance and conceptual duplication.

H.19 Residual moralization

Residual is not automatically bad.

Residual may include:

option value;

healthy diversity;

alternative interpretation;

protective caution;

unused flexibility.

Therefore:

Residual ≠ Defect. (H.40)

The correct objective is:

Govern Residual, not Eliminate All Residual. (H.41)

A system with zero residual may be over-locked, rigid, or falsely closed.

H.20 Coherence worship

High coherence may improve coordination.

But excessive coherence may produce:

herding;

brittle consensus;

correlated failure;

semantic lock-in;

loss of price discovery.

Therefore:

Maximum Coherence ≠ Maximum Health. (H.42)

A healthy regime may require:

Coherence + Diversity + Adaptive Gate. (H.43)

H.21 Trace fetishism

More logging does not automatically improve governance.

A trace may be:

incomplete;

misleading;

unaudited;

too large to use;

privacy-invasive;

detached from future routing.

A useful trace must satisfy:

TraceQuality = Integrity × Relevance × Reproducibility × GovernedUse. (H.44)

If any factor is near zero, the trace has little value.

H.22 False invariance

A system may appear robust because it always gives the same output.

But invariance requires:

stability under equivalent changes;

sensitivity to meaningful changes.

The correct condition is:

EquivalentFrameChange → StableConclusion. (H.45)

MaterialFrameChange → AppropriateConclusionChange. (H.46)

A constant answer is rigidity, not invariance.

H.23 Invalid universality claim

The framework should not claim:

All systems instantiate the same grammar. (H.47)

The responsible claim is:

Some bounded, interacting, trace-bearing systems may instantiate comparable roles. (H.48)

The domain of validity must be tested.

H.24 Failure classification table

Failure	Symptom	Required correction
Decorative complex notation	No measurable advantage	Remove i or define true geometry
Arbitrary phase	Results change under minor encoding	Redefine phase or abandon it
Residual post-hoc fitting	Q appears only after failure	Pre-register residual
Gate overfitting	Perfect historical event labeling	Use held-out tests
Trace pollution	Wrong records drive future actions	Repair or invalidate ledger
Coherence over-lock	Strong consensus with fragility	Preserve diversity
False invariance	Same answer under all frames	Test material sensitivity
Quantum overclaim	Classical mechanism described as quantum	Downgrade to role analogy
Ontological inflation	Useful model treated as literal reality	Separate operational and foundational claims
Baseline neglect	Complex model beats only weak baseline	Compare with strongest conventional methods

---

H.25 Final discipline rule

Every imported physics term should answer four questions:

What is the domain variable?

What is the mathematical operation?

What prediction changes?

What result would falsify the mapping?

If these cannot be answered:

Use ordinary domain language instead. (H.49)

The Complex Residual Principle becomes stronger, not weaker, when unnecessary quantum

vocabulary is removed.

Appendix I — Blogger-Ready Unicode Formula Conventions

I.1 General rule

Every equation in this article is designed to remain readable without MathJax.

The preferred format is:

$Z = R + iQ.$ (I.1)

Each equation should:

occupy one line;

use Unicode symbols;

end with a numbered tag;

avoid LaTeX delimiters;

remain meaningful as plain text.

I.2 Superscripts

Use Unicode superscripts where available:

$x^2, x^3, x^n.$ (I.2)

For domain labels:

$Z^P, Z^f, Z^L.$ (I.3)

For longer superscripts, plain text may be used:

Ghigh, Glow, Ωref. (I.4)

Readability is more important than typographic purity.

I.3 Subscripts

Use Unicode subscripts where available:

$x_1, x_2, \beta_i, \Phi_m, \tau^L.$ (I.5)

For long labels:

Qfact, Qretrieval, Qinstruction. (I.6)

This is preferable to difficult mixed Unicode strings.

I.4 Vectors

Use bold Unicode when practical:

Q, **x**, **m**, **v**. (I.7)

If bold characters are unavailable, use an arrow:

$\vec{Q}$, $\vec{x}$. (I.8)

The article uses:

$\mathbf{Q} = [Q_1, Q_2, \ldots, Q_n]^T$. (I.9)

I.5 Operators

Preferred symbols include:

Σ = summation. (I.10)

$\int$ = integral. (I.11)

∂ = partial derivative. (I.12)

$\wedge$ = logical AND. (I.13)

$\vee$ = logical OR. (I.14)

$\Rightarrow$ = implication. (I.15)

$\Leftrightarrow$ = equivalence. (I.16)

$\approx$ = approximate relation. (I.17)

$\neq$ = inequality. (I.18)

$\geq$ and $\leq$ = bounds. (I.19)

I.6 Functions

Use plain-text function names:

exp(x). (I.20)

ln(x). (I.21)

sinθ. (I.22)

cosθ. (I.23)

atan2(y,x). (I.24)

arg(Z). (I.25)

Normalize(x). (I.26)

Update(State,Trace). (I.27)

I.7 Fractions

Use slash form when simple:

$\beta_i = \mathrm{Cov}(x_i,x_m) / \mathrm{Var}(x_m)$. (I.28)

Use brackets when needed:

$G = 1 / [1 + \exp(-S)]$. (I.29)

Avoid stacked fraction notation that may break in Blogger.

I.8 Matrices

For short vectors:

$\mathbf{Q} = [Q_{fact},Q_{retrieval},Q_{tool}]^T$. (I.30)

For large matrices, explain the structure in prose rather than relying on multiline layout.

A quadratic form is:

$Q_{total}^2 = \mathbf{Q}^T G \mathbf{Q}$. (I.31)

This is more Blogger-safe than displaying the full matrix.

I.9 Piecewise definitions

Avoid multiline braces when possible.

Use prose followed by separate equations:

If $G = 1$, then $d\tau = d\chi$. (I.32)

If $G = 0$, then $dT = d\chi$. (I.33)

This is more reliable across Blogger themes.

I.10 Equation numbering

Use section-based numbering:

(3.1), (3.2), … (I.34)

Appendices use letter-based numbering:

(A.1), (B.1), … (I.35)

Do not rely on automatic numbering.

The tag should be typed as part of the line.

Appendix J — Minimal Simulation Pseudocode

J.1 Purpose

The following pseudocode illustrates the minimal computational structure shared by finance and LLM implementations.

It is not production code.

It defines the runtime kernel.

J.2 General complex residual kernel

```
INITIALIZE protocol P
INITIALIZE state X
INITIALIZE ledger L
INITIALIZE unresolved depth T = 0
INITIALIZE realized time τ = 0
INITIALIZE previous phase Φ_prev

FOR each process step s:

    features = OBSERVE(X, P)

    phase_states = MAP_TO_PHASE(features, P)

    coherence ρ, collective phase Φ = AGGREGATE_PHASE(phase_states)

    unwrapped increment ΔΦ = UNWRAP(Φ, Φ_prev)

    visible candidate R = PROJECT(X, P)

    residual vector Q = ESTIMATE_RESIDUAL(X, R, P)

    gate G = COMPUTE_GATE(ρ, Q, L, P)

    Δτ = G * ABS(ΔΦ) / Ω_ref
```

```
    ΔT = (1 - G) * ABS(ΔΦ) / Ω_ref

    τ = τ + Δτ

    T = T + ΔT

    channel = SELECT_CHANNEL(G, Q, P)

    trace = COMMIT_OR_DEFER(channel, R, Q, P)

    L = UPDATE_LEDGER(L, trace, Q)

    X = UPDATE_STATE(X, L, Q, T, P)

    Φ_prev = Φ
```

J.3 Finance prototype

```
INPUT:
    asset returns
    volume
    volatility
    breadth
    liquidity proxy

FOR each rolling date t:

    build standardized asset feature vectors

    define market reference direction

    define orthogonal residual direction

    compute asset phases θ_i

    compute market coherence ρ_m

    compute collective phase Φ_m

    unwrap Φ_m to Φ_tilde_m
```

```
    estimate residual vector Q_m:
        liquidity
        leverage
        breadth
        volatility
        credit

    compute gate G_m from:
        coherence
        diversity
        volume
        breadth
        fragility

    Δτ_m = G_m * ABS(ΔΦ_tilde_m) / Ω_ref

    ΔT_m = (1 - G_m) * ABS(ΔΦ_tilde_m) / Ω_ref

    predict future:
        volatility
        drawdown
        regime transition

COMPARE against:
    momentum
    GARCH
    hidden Markov model
    breadth indicators
```

J.4 LLM prototype

```
INPUT:
    user task
    protocol
    retrieval sources
    allowed tools
    output schema

GENERATE candidate answer
```

```
ESTIMATE residuals:
    Q_fact
    Q_retrieval
    Q_ambiguity
    Q_tool

COMPUTE gates:
    G_evidence
    G_clarity
    G_tool

G = G_evidence * G_clarity * G_tool

IF Q_ambiguity is high:
    channel = CLARIFY

ELSE IF Q_retrieval is high:
    channel = RETRIEVE

ELSE IF Q_tool is high:
    channel = RETRY_OR_DISCLOSE

ELSE IF G is above threshold:
    channel = ANSWER

ELSE:
    channel = QUALIFY_OR_ESCALATE

WRITE trace:
    protocol
    answer
    evidence
    residuals
    gate
    tool status
    selected channel
```

J.5 Semantic-time update

```
phase_change = semantic_change_between_steps

Δτ_L = G_L * ABS(phase_change) / Ω_L

ΔT_L = (1 - G_L) * ABS(phase_change) / Ω_L

semantic_time = semantic_time + Δτ_L

unresolved_depth = unresolved_depth + ΔT_L
```

A freeze detector is:

```
freeze_score = unresolved_depth_growth / (semantic_time_growth + epsilon)

IF freeze_score exceeds threshold:
    change strategy
    ask clarification
    reformulate retrieval
    switch tool
    decompose task
    escalate
```

J.6 Residual release

```
IF decisive evidence arrives:

    release_fraction = estimate_release(residual_vector)

    realized_jump = release_fraction * unresolved_depth

    semantic_time = semantic_time + realized_jump

    unresolved_depth = unresolved_depth * (1 - release_fraction)

    update residual vector

    write resolution trace
```

J.7 Minimum audit output

Every run should produce:

```
Protocol
State representation
Phase definition
Coherence
Residual vector
Gate score
Selected channel
Realized-time increment
Unresolved-depth increment
Trace
Ledger update
Prediction or action
```

This output allows the researcher to distinguish:

model success;

projection failure;

residual failure;

gate failure;

trace failure;

observer-update failure.

Final Appendix Summary

The appendices convert the article's central idea into a formal research program.

The domain-independent kernel is:

$Z^D = R^D + i\mathbf{Q}^D$. (J.1)

$C^D = \rho^D \exp(i\Phi^D)$. (J.2)

$d\tau^D = G^D |d\tilde{\Phi}^D| / \Omega^D$. (J.3)

$dT^D = (1 - G^D) |d\tilde{\Phi}^D| / \Omega^D$. (J.4)

$L^D{}_{,k+1} = \mathrm{Update}(L^D{}_{,k}, \mathrm{Trace}^D{}_{,k}, \mathrm{Residual}^D{}_{,k})$. (J.5)

The empirical obligation is equally compact:

Define → Measure → Compare → Intervene → Replicate → Reject or Retain. (J.6)

The full article is therefore not a declaration that physics, finance, and LLMs are one substance.

It is a proposal that they can be studied through one disciplined operational grammar:

Possibility → Phase → Projection → Gate → Trace + Residual → Recursive Time. (J.7)

Appendix K — Candidate Domains for Complex Residual Reconstruction

Where Phase, Thermodynamic Budgets, Effective Geometry, and Trace-Bearing Time May Reappear

K.1 Purpose and scope

Finance and large language models are not necessarily isolated examples of the Complex Residual Principle.

Many mature scientific and engineering domains already contain some combination of:

complex-valued state variables;

phase and synchronization;

active and reactive components;

stored pressure and realized work;

dissipation and entropy production;

transition thresholds;

irreversible records;

state-dependent propagation;

effective geometry;

subsystem-specific clocks.

These systems may therefore support frameworks resembling parts of:

quantum-style possibility, phase, interference, and projection;

thermodynamic work, storage, dissipation, entropy, and critical transitions;

general-relativity-like effective geometry, causal reachability, horizons, and state-dependent internal time.

The word **resembling** is essential.

The proposed domains are not claimed to be physically quantum, thermodynamically equivalent in every respect, or governed by Einstein's field equations.

The appendix asks a narrower question:

> In which mature systems can the complex residual grammar be operationally reconstructed without inventing arbitrary phases, residuals, budgets, or geometries?

The general candidate state is:

$Z^D = R^D + iQ^D$. (K.1)

Or, when the residual is multidimensional:

$Z^D = R^D + i\mathbf{Q}^D$. (K.2)

Where:

R^D = realized, admitted, delivered, expressed, or committed structure;

Q^D = stored, circulating, latent, unresolved, or orthogonal structure;

D = candidate domain.

A candidate domain becomes especially strong when it also supports:

$C^D = \rho^D \exp(i\Phi^D)$. (K.3)

$d\zeta^D = d\tau^D + i\, dT^D$. (K.4)

$\Delta U^D = \text{Input}^D - \text{Work}^D - \text{Dissipation}^D - \text{Leakage}^D$. (K.5)

$ds^{D2} = d\mathbf{x}^T G^D(\mathbf{x}, L) d\mathbf{x}$. (K.6)

These equations respectively represent:

collective phase;

realized and unresolved internal development;

thermodynamic or resource-budget accounting;

effective state-space geometry.

The Gauge Grammar already proposes that systems as different as chemistry, biology, ecosystems, markets, institutions, and AI runtimes repeatedly solve comparable problems of identity, mediation, binding, transition gating, trace, invariance, and observer update. It explicitly treats this as functional role translation rather than physical identity.

The General Life Form framework adds another requirement: candidate systems should possess measurable budgets, couplings, constraints, dissipation, and synchronized ticks rather than being connected only through metaphor.

The Dual Ledger framework further supplies a reusable structure–drive–work geometry in which maintained structure, conjugate drive, inertia, useful work, environmental pressure, and dissipation are jointly measurable.

This appendix therefore functions as a **research map**, not a catalogue of established equivalences.

K.2 Domain admission criteria

A domain should not enter the research program merely because complex numbers can be written somewhere in its mathematics.

A strong candidate should satisfy several conditions.

K.2.1 Natural state multiplicity

The system must contain multiple active states, trajectories, modes, or configurations before commitment.

$\text{Possibility}^{D} = \{x_1, x_2, \ldots, x_n\}$. (K.7)

Examples include:

competing electrical modes;

chemical pathways;

neural interpretations;

distributed-system branches;

legal arguments;

transport routes.

K.2.2 Defensible complex pair

The system should provide a non-arbitrary split:

$Z^{D} = R^{D} + iQ^{D}$. (K.8)

A strong complex pair has one of the following foundations:

native engineering formalism;

analytic signal;

conjugate variables;

orthogonal state projection;

complex eigenmode;

two-dimensional oscillatory plane;

active versus reactive flow.

The pairing is weak when R and Q are chosen only because they sound philosophically compatible.

K.2.3 Meaningful phase

A candidate phase should arise from:

an established oscillator;

Fourier or analytic-signal representation;

coupled-cycle dynamics;

complex eigenvalues;

a declared two-dimensional state plane;

repeated state-transition orientation.

The phase must satisfy:

$\theta^D = \text{PhaseMap}(X^D \mid \varphi^D, P^D)$. (K.9)

The phase is not acceptable if small arbitrary changes in representation produce unrelated results.

K.2.4 Budgeted openness

The system should exchange resources with an environment.

A generic budget is:

$\Delta U^D = I^D - W^D - \Gamma^D - \Lambda^D$. (K.10)

Where:

I^D = input;

W^D = useful work;

Γ^D = dissipation;

Λ^D = leakage or unmeasured transfer.

A candidate becomes thermodynamically mature when the terms have declared units and can be measured within error bounds.

K.2.5 Transition gates

The system must contain identifiable conditions converting possibility into consequence.

Transition^D occurs if $G^D(X,L,P) \geq G^{D*}$. (K.11)

Examples include:

circuit protection;

chemical activation threshold;

membrane channel opening;

immune activation;

consensus commit;

legal admissibility;

traffic-capacity breakdown.

K.2.6 Trace-bearing history

A record must affect future dynamics.

$L^{D}{}_{,k+1} = L^{D}{}_{,k} \sqcup \mathrm{Trace}^{D}{}_{,k}$. (K.12)

And:

$X^{D}{}_{,k+1} = F^{D}(X^{D}{}_{,k}, L^{D}{}_{,k+1})$. (K.13)

A passive log is insufficient.

The trace must alter:

routing;

control thresholds;

memory;

structure;

future admissibility;

future state probability.

K.2.7 Effective geometry

The domain should support a meaningful distance or cost geometry.

A generic metric is:

$ds^{D2} = d\mathbf{x}^{T} G^{D} d\mathbf{x}$. (K.14)

Possible sources of G^{D} include:

covariance;

Fisher information;

control energy;

network resistance;

transport cost;

chemical free energy;

causal reachability;

communication latency.

K.2.8 Falsifiable gain

The imported framework must outperform a simpler domain model in at least one dimension:

prediction;

control;

diagnosis;

compression;

robustness;

failure classification.

$$\text{OperationalGain}^{D} = \text{ScoreComplexResidual}^{D} - \text{ScoreBaseline}^{D}. \quad \text{(K.15)}$$

The domain mapping is useful only if:

$$\text{OperationalGain}^{D} > \text{ComplexityPenalty}^{D}. \quad \text{(K.16)}$$

K.3 Three levels of domain suitability

Candidate domains should be labelled by the maturity of their complex structure.

K.3.1 Level A — Native complex formalism

The domain already uses complex numbers as standard operational variables.

Examples include:

AC power engineering;

signal processing;

communications;

impedance spectroscopy;

frequency-domain control;

wave and plasma dynamics.

The complex residual program extends an existing formalism rather than inventing one.

K.3.2 Level B — Natural phase, engineered residual

The domain possesses established oscillations, synchronization, order parameters, or conjugate state variables, but the exact R + iQ interpretation must be constructed.

Examples include:

chemical reaction networks;

metabolism;

neuroscience;

immune dynamics;

climate oscillations;

ecology.

K.3.3 Level C — Structural residual reconstruction

The domain has gates, traces, budgets, and state-dependent geometry, but complex phase is primarily an engineered representation.

Examples include:

distributed computing;

supply chains;

law;

governance;

cultural diffusion.

These domains may still support powerful models.

Their claims must remain more conservative.

K.3.4 Domain label

Each proposed application should carry one of three labels:

NativeComplex^{D}. (K.17)

EngineeredComplex^{D}. (K.18)

StructuralAnalogy^{D}. (K.19)

The label should appear before any quantum, thermodynamic, or gravitational interpretation.

K.4 The three-layer reconstruction

Each candidate domain should be examined through three distinct layers.

K.4.1 Quantum-style layer

The quantum-style layer asks whether the domain contains:

active alternatives;

relative phase;

constructive or destructive combination;

observer-dependent projection;

transition gates;

unresolved alternatives;

backreaction.

The generic sequence is:

$\text{Field}^{D} \rightarrow \text{Phase}^{D} \rightarrow \text{Projection}^{D} \rightarrow \text{Gate}^{D} \rightarrow \text{Trace}^{D} + \text{Residual}^{D}$. (K.20)

This layer is strongest when complex phase is already native.

K.4.2 Thermodynamic layer

The thermodynamic layer asks whether the domain contains:

energy or resource input;

stored potential;

useful work;

dissipation;

entropy production;

non-equilibrium maintenance;

critical transitions;

hysteresis.

A generic ledger is:

$\text{Input}^{D} = \text{UsefulWork}^{D} + \text{StoredPotential}^{D} + \text{Dissipation}^{D} + \text{Leakage}^{D}$. (K.21)

An effective entropy-production rate may be written:

$\Sigma^{D} = \Sigma_j J_j X_j \geq 0.$ (K.22)

Where:

J_j = flow;

X_j = conjugate driving force.

The exact variables must come from the domain.

K.4.3 General-relativity-like layer

The GR-like layer asks whether the domain possesses:

state-dependent distances;

path-dependent cost;

variable subsystem clock rates;

effective geodesics;

reachability boundaries;

propagation cones;

horizon-like loss of control;

history-induced curvature.

A state-space metric may be:

$ds^{D2} = g_{a\beta}^{D}(\mathbf{x},L)dx^{a}dx^{b}.$ (K.23)

An internal clock may be:

$d\tau^{D} = G^{D}|d\tilde{\Phi}^{D}| / \Omega^{D}.$ (K.24)

A horizon-like boundary may be defined:

$H^{D} = \{x : \text{Reachability}(x \rightarrow \text{SafeSet} \mid u \in U^{D}) = 0\}.$ (K.25)

This is a control or information horizon.

It is not a physical event horizon.

K.5 Tier A — Domains with native complex structure

K.5.1 AC power grids

AC power engineering provides the strongest independent candidate.

Complex power is already written:

$S = P + iQ$. (K.26)

Where:

P = active power;

Q = reactive power;

S = apparent power.

Voltage and current are represented by phasors:

$V = |V|\exp(i\theta V)$. (K.27)

$I = |I|\exp(i\theta I)$. (K.28)

Complex power is:

$S = VI^*$. (K.29)

This is not merely analogous to the Complex Residual Principle.

It is an established complex decomposition separating delivered active work from reactive power that circulates between fields and system components.

Quantum-style route

The grid contains:

phase-bearing oscillators;

constructive and destructive power flows;

synchronization;

mode coupling;

measurement-frame dependence;

threshold-triggered protection;

collapse-like loss of synchrony.

A collective grid order parameter may be:

$Cgrid = [\Sigma_j w_j \exp(i\theta_j)] / \Sigma_j w_j = \rho grid \exp(i\Phi grid)$. (K.30)

Low ρgrid indicates phase fragmentation.

High ρgrid indicates synchronization.

Thermodynamic route

The grid contains:

generation input;

active delivered work;

resistive loss;

reactive storage;

thermal limits;

conversion inefficiency.

A grid budget is:

Pgeneration = Pload + Ploss + Pstorage-change. (K.31)

Reactive power Q can be interpreted as stored or circulating field pressure, but it should not be treated as wasted energy in every context.

GR-like route

A grid has state-dependent electrical distance.

Two buses may be geographically close but electrically far because of:

impedance;

congestion;

phase-angle constraints;

network topology.

An effective metric might be derived from:

admittance matrix;

effective resistance;

control energy;

power-transfer distribution factors.

A grid horizon appears when no admissible control action can prevent loss of synchrony or voltage collapse:

Reachabilitygrid(SafeSet) → 0. (K.32)

First research program

Define P as active realization and Q as reactive residual.

Estimate coherence ρgrid.

Define protection and stability gates.

Construct τgrid from accepted power transfers and switching events.

Construct Tgrid from phase movement unable to become stable delivery.

Test whether Tgrid predicts instability earlier than conventional indicators.

Main caution

Reactive power is a precise electrical quantity.

It should not be broadened into a metaphor for every hidden grid risk.

K.5.2 Signal processing and communications

Complex analytic signals are standard:

$z(t) = x(t) + i\mathcal{H}[x(t)]$. (K.33)

Where $\mathcal{H}[x(t)]$ is the Hilbert transform.

The polar form is:

$z(t) = A(t)\exp[i\theta(t)]$. (K.34)

Where:

$A(t)$ = instantaneous amplitude;

$\theta(t)$ = instantaneous phase.

Quantum-style route

The domain naturally contains:

superposed modes;

interference;

phase coherence;

projection onto bases;

filtering;

measurement noise;

decoding gates.

A receiver projects a signal through:

frequency basis;

modulation basis;

matched filter;

coding scheme;

decision threshold.

The recovered bit or symbol is the admitted result.

The unresolved quadrature, noise, and ambiguity form residual.

Thermodynamic route

Communication has measurable:

transmitted energy;

channel loss;

noise;

decoding work;

information entropy;

error-correction cost.

An information–energy budget may be:

Einput = Ereceived + Edissipated + Eerror-correction + Eleakage. (K.35)

GR-like route

The communication network has:

propagation delay;

bandwidth-dependent distance;

congestion curvature;

reachability cones;

causal ordering.

An information horizon may arise when:

Signal-to-Noise Ratio < Decoding Threshold. (K.36)

Beyond that boundary, the message remains physically present but operationally unrecoverable.

First research program

Use the analytic signal as the native complex state.

Treat successfully decoded structure as R.

Treat unresolved quadrature and decoding ambiguity as Q.

Define gate opening through error-corrected decoding.

Define trace through accepted symbol and receiver-state update.

Test whether complex residual depth predicts burst errors and loss of synchronization.

Main caution

Noise, phase uncertainty, and semantic uncertainty are different residual classes.

They should not be collapsed into one Q.

K.5.3 Control systems and robotics

Control engineering already uses complex poles, frequency responses, and phase margins.

A transfer function is evaluated as:

$H(i\omega) = \mathrm{Re}[H(i\omega)] + i\mathrm{Im}[H(i\omega)]$. (K.37)

System modes may have eigenvalues:

$\lambda_j = \sigma_j + i\omega_j$. (K.38)

Where:

σ_j = growth or decay rate;

ω_j = oscillation frequency.

Quantum-style route

The control system contains:

multiple modes;

phase relations;

interference;

state projection;

command-selection gates;

sensor backreaction;

observer-state updates.

The admitted component may be:

Rcontrol = achieved controlled motion. (K.39)

The residual may include:

Qcontrol = tracking error + stored oscillatory pressure + unobserved mode. (K.40)

Thermodynamic route

A robot or controller pays:

actuation energy;

computation;

friction;

thermal loss;

correction cost.

A control budget is:

Einput = Euseful-motion + Ecorrection + Edissipation + Ereserve-change. (K.41)

GR-like route

Control cost defines effective geometry.

For dynamics:

$\dot{x} = f(x) + B(x)u$. (K.42)

A minimum-energy path may define:

$ds_{control}^2 = dx^T G_{control}(x) dx$. (K.43)

A control horizon occurs when the target state is no longer reachable under admissible inputs.

First research program

Identify oscillatory modes and their phases.

Define R as task-relevant controlled output.

Define Q as uncorrected modal energy or residual error.

Define gates from stability margin, sensor certainty, and actuator authority.

Define internal control time through successful state commitments.

Compare the model with conventional robustness and model-predictive-control metrics.

Main caution

The framework must add value beyond established state estimation, Lyapunov analysis, and robust control.

K.5.4 Electrochemistry and impedance systems

Electrochemical impedance is naturally complex:

$$Z(\omega) = Z'(\omega) + iZ''(\omega). \quad (K.44)$$

Where:

Z' = resistive component;

Z'' = reactive or phase-lag component.

Quantum-style route

The system contains:

frequency-dependent projection;

phase lag;

multiple transport pathways;

interface barriers;

reaction gates;

unresolved charge-storage dynamics.

Thermodynamic route

Electrochemical systems directly contain:

chemical potential;

electrical work;

heat;

diffusion;

activation loss;

entropy production;

stored chemical energy.

A generic free-energy balance is:

$\Delta G = W_{electrical} + W_{chemical} - T d S_{production}$. (K.45)

GR-like route

The effective state geometry may be derived from:

impedance spectrum;

diffusion distance;

chemical-potential metric;

reaction-network resistance.

A horizon-like boundary may correspond to a passivation layer, transport blockade, or reaction state that cannot be reversed under admissible voltage and temperature limits.

First research program

Use measured complex impedance directly.

Separate dissipative and stored components.

identify reaction gates and trace-producing degradation.

construct an internal ageing clock from irreversible electrochemical events.

test whether unresolved reactive depth predicts later capacity fade or failure.

Main caution

Electrical reactance, electrochemical storage, and degradation residual are related but not identical.

K.5.5 Fluid dynamics, plasma, and wave systems

Complex amplitudes are widely used for waves:

$\psi(\mathbf{x},t) = A(\mathbf{x},t)\exp[i\theta(\mathbf{x},t)]$. (K.46)

Quantum-style route

These systems naturally display:

superposition;

interference;

phase-locking;

mode conversion;

coherence loss;

tunnelling-like evanescent transport;

analogue horizons.

Thermodynamic route

They contain:

kinetic and potential energy;

viscosity;

turbulence;

heat;

entropy production;

driven non-equilibrium states.

A simplified energy balance is:

dE/dt = PowerInput – Dissipation – FluxOut. (K.47)

GR-like route

Certain fluid and wave systems can support genuine analogue-gravity mathematics in which perturbations propagate according to an effective metric.

Even here, the effective metric governs waves in a medium rather than physical spacetime itself.

First research program

Select a mature wave or plasma system.

identify coherent modes and phase order.

define R as propagating or measured mode.

define Q as trapped, evanescent, or unresolved modal structure.

identify dissipation and transition gates.

test whether the complex residual improves prediction of turbulence, mode conversion, or instability.

Main caution

Because these systems already possess rich established mathematics, the framework must be formulated with domain specialists to avoid merely renaming known wave theory.

K.6 Tier B — Natural phase and thermodynamic systems

K.6.1 Chemical reaction networks

Chemical networks contain multiple pathways, oscillatory reactions, activation barriers, chemical potentials, and non-equilibrium steady states.

A reaction-state complex variable may be constructed:

$$Z_{chem} = R_{flux} + iQ_{potential}. \quad (K.48)$$

Where:

R_{flux} = realized reaction flux;

$Q_{potential}$ = unresolved chemical-potential imbalance.

Quantum-style route

alternative pathways act as active possibilities;

oscillatory reactions supply phase;

catalysts and activation barriers supply gates;

products and concentration changes create trace;

feedback alters later reaction rates.

Thermodynamic route

For reaction r:

$$Affinity_r = -\Delta_r G. \quad (K.49)$$

Entropy production is:

$$\Sigma_{chem} = \Sigma_r J_r Affinity_r / T \geq 0. \quad (K.50)$$

Where J_r is reaction flux.

This is a strong thermodynamic foundation.

GR-like route

Reaction networks can be given effective geometry through:

thermodynamic length;

Fisher information;

chemical-potential distance;

minimum-dissipation paths.

A reaction horizon may occur when no admissible pathway returns the system to a prior basin.

Plausible development approach

Choose an oscillatory or bistable reaction network.

derive phase from concentration cycles.

define R as realized product flux.

define Q as chemical disequilibrium or blocked pathway pressure.

define gates from activation thresholds.

define chemical time from reaction events rather than laboratory time.

compare with classical non-equilibrium thermodynamics.

K.6.2 Cellular regulation and metabolism

A cell maintains structure by continuously spending free energy.

A candidate state is:

Zcell = Rexpressed + i**Q**regulatory. (K.51)

Where:

Rexpressed = realized phenotype or metabolic output;

Qregulatory = latent transcriptional, signalling, energetic, and stress pressure.

Quantum-style route

gene-expression possibilities form a state field;

regulatory cycles possess phase;

receptors and membranes project the environment;

transcription and translation gates commit expression;

epigenetic and metabolic states leave trace.

Thermodynamic route

Cells are open systems with:

nutrient input;

ATP expenditure;

heat;

chemical work;

transport loss;

entropy production.

The General Life Form framework is directly relevant because it treats living systems as open, budgeted processes with measurable flows, constraints, geometry, dissipation, and synchronized ticks.

GR-like route

State-space geometry may be derived from:

Fisher information;

gene-expression covariance;

metabolic control coefficients;

energy landscape curvature.

A viability horizon appears where no admissible regulatory intervention can return the cell to a viable state.

Plausible development approach

Declare a feature map for maintained structure.

identify cell-cycle and metabolic phases.

define R as realized expression or function.

define Q as regulatory and energetic residual.

define gates from transcription, membrane, and checkpoint mechanisms.

construct a dual energy–information ledger.

test whether internal cell time predicts transition better than external clock time.

K.6.3 Immune systems

The immune system is a distributed observer network.

A candidate complex state is:

$Z_{immune} = R_{response} + i\mathbf{Q}_{latent}$. (K.52)

Where:

Rresponse = expressed immune action;

Qlatent = antigen uncertainty, tolerance conflict, latent activation, and exhaustion pressure.

Quantum-style route

many antigen interpretations remain possible;

receptor binding projects molecular patterns;

activation thresholds act as gates;

cytokine networks synchronize;

memory cells preserve trace;

prior trace alters future response.

Thermodynamic route

Immune response consumes:

metabolic energy;

cellular resources;

tissue integrity;

signalling capacity.

Excessive response produces dissipation and collateral damage.

GR-like route

Immune-network geometry may be based on:

receptor similarity;

cytokine connectivity;

tissue transport;

activation energy;

clonal distance.

A tolerance horizon or exhaustion horizon may mark a region from which ordinary regulatory control cannot restore balanced response.

Plausible development approach

define antigen and immune-state feature spaces;

derive phase from cytokine or cell-population oscillations;

define R as committed response;

define Q as unresolved recognition and tolerance pressure;

model immune memory as trace;

test whether residual depth predicts flare, exhaustion, or delayed response.

K.6.4 Neuroscience and brain dynamics

Neuroscience already measures complex analytic signals:

$z_k(t) = A_k(t)\exp[i\theta_k(t)]$. (K.53)

Phase synchronization is studied across neural regions.

A decision-level state may be:

$Z_{brain} = R_{percept} + i\mathbf{Q}_{competing}$. (K.54)

Where:

$R_{percept}$ = committed perception, decision, or action;

$\mathbf{Q}_{competing}$ = latent neural alternatives and unresolved evidence.

Quantum-style route

multiple candidate interpretations coexist;

neural phases interfere and synchronize;

attention provides projection;

decision thresholds provide gates;

memory creates trace;

perception changes future sampling.

Thermodynamic route

The brain has measurable:

metabolic energy;

signalling cost;

heat;

information-processing load;

synaptic maintenance;

dissipation.

GR-like route

Neural manifolds and information geometry provide effective curvature.

Internal processing time can vary with:

attention;

arousal;

evidence quality;

prediction error;

network synchrony.

A cognitive horizon may occur when a representation cannot be reached or recovered under current attention, memory, and metabolic constraints.

Plausible development approach

combine EEG or MEG phase with decision data;

define neural coherence and phase diversity;

define R as decision commitment;

define Q as unresolved competing neural state;

identify evidence-accumulation gates;

compare complex semantic time with reaction time;

test whether residual depth predicts revision or perceptual switching.

Main caution

The framework does not establish quantum consciousness.

Neural phase dynamics are measurable classical biological processes unless stronger evidence is supplied.

K.6.5 Climate and ocean–atmosphere systems

Climate systems contain oscillatory modes, delayed coupling, stored heat, tipping points, and path dependence.

A candidate state is:

Zclimate = Robserved + i**Q**stored. (K.55)

Where:

Robserved = admitted surface or atmospheric regime;

Qstored = subsurface heat, moisture, chemical imbalance, and unrealized tipping pressure.

Quantum-style route

multiple climatic modes coexist;

oscillations possess phase;

modes reinforce or oppose;

measurement depends on spatial and temporal frame;

tipping gates select regimes;

historical state alters future response.

Thermodynamic route

Climate is fundamentally an energy-transport system.

A simplified planetary budget is:

EnergyIn = EnergyOut + HeatStorageChange. (K.56)

Entropy production arises through:

radiation;

convection;

diffusion;

phase change;

friction.

GR-like route

Effective climate geometry may derive from:

transport distance;

causal influence;

covariance;

information flow;

energy required to move between regimes.

A tipping horizon occurs where feasible intervention can no longer prevent transition.

Plausible development approach

choose a specific coupled mode such as ocean–atmosphere oscillation;

derive phase from established indices;

define R as observed regime;

define Q as stored heat and unexpressed imbalance;

define tipping gates;

construct climate internal time from consequential regime events;

compare with conventional early-warning indicators.

K.6.6 Ecology and evolution

Ecological systems contain cycles, resource budgets, competition, mutualism, succession, extinction thresholds, and evolutionary memory.

A candidate state is:

$$Z_{eco} = R_{population} + i\mathbf{Q}_{niche}. \quad (K.57)$$

Where:

$R_{population}$ = realized population and niche configuration;

$\mathbf{Q}_{niche}$ = latent resource pressure, selection pressure, and unrealized ecological potential.

Quantum-style route

multiple niches and strategies coexist;

predator–prey cycles possess phase;

species interactions reinforce or suppress;

environmental observation is local and bounded;

reproductive and survival thresholds create gates;

succession and genetic inheritance create trace.

Thermodynamic route

Ecosystems process:

solar input;

biomass;

nutrient flows;

work;

heat;

waste;

entropy.

GR-like route

Ecological geometry may derive from:

niche distance;

trophic-network resistance;

migration cost;

adaptive landscape curvature;

evolutionary accessibility.

An extinction horizon appears when no admissible trajectory returns the population above a viability threshold.

Plausible development approach

select a well-measured ecosystem;

derive phases from population cycles;

define R as realized ecological structure;

define Q as resource and selection residual;

define reproduction, migration, and extinction gates;

model succession as trace;

test whether ecological internal time predicts regime change better than calendar time.

K.7 Tier C — Networked engineered systems

K.7.1 Distributed computing and consensus

Distributed systems offer one of the strongest non-physical applications because they contain explicit candidate states, local observers, multiple clocks, commitment gates, forks, and ledgers.

A candidate state is:

Zdist = Rcommitted + i**Q**uncommitted. (K.58)

Where:

Rcommitted = shared committed state;

Quncommitted = pending messages, divergent replicas, unresolved conflicts, and fork pressure.

Quantum-style route

several candidate histories may coexist;

message ordering changes outcomes;

local observers possess different frames;

consensus selects one committed history;

losing branches remain residual;

commitment changes future admissibility.

Thermodynamic route

Distributed systems consume:

computation;

communication;

storage;

cooling;

retry effort;

consensus overhead.

A computational budget is:

Ecompute = Euseful + Econsensus + Erecovery + Edissipation. (K.59)

GR-like route

The system possesses:

vector clocks;

causal order;

latency-dependent distance;

reachability cones;

partition horizons;

subsystem-specific time.

A network horizon occurs when communication delay or partition makes agreement unreachable within the required time.

Plausible development approach

model each replica as a bounded observer;

encode branch orientation or logical-clock phase;

define R as committed state;

define Q as divergence and uncommitted state;

define consensus as gate;

treat ledger commit as trace;

define internal time by commits rather than processor ticks;

test whether Tdist predicts failure, rollback, or fork persistence.

Main caution

Distributed-system theory already possesses rigorous concepts of consensus, causality, and logical clocks.

The complex residual model must add diagnostic or control value beyond them.

K.7.2 Cybersecurity systems

A security system continually distinguishes trusted from untrusted states under incomplete observation.

A candidate state is:

Zsecurity = Rtrusted + i**Q**compromise. (K.60)

Where:

Rtrusted = currently admitted system state;

Qcompromise = unresolved anomaly, hidden persistence, and attack-path pressure.

Quantum-style route

multiple attack hypotheses remain active;

sensors project partial evidence;

authentication gates select admissible identity;

security controls create measurement backreaction;

incident records change future policy.

Thermodynamic route

Security consumes:

computation;

monitoring;

analyst attention;

redundancy;

recovery capacity.

Attackers impose dissipation through:

resource exhaustion;

entropy increase;

corruption;

forced reconfiguration.

GR-like route

Attack graphs create effective geometry.

A privilege boundary acts as a causal horizon.

A compromised credential may curve future shortest paths through the network by lowering attack cost.

Plausible development approach

define trusted-state projection;

define Q as typed unresolved compromise risk;

derive phase from periodic attack and defence cycles or state-space modes;

define authentication and response gates;

write forensic trace;

construct a risk geometry from attack cost;

test whether residual depth predicts breach escalation.

K.7.3 Supply chains and logistics

Supply chains contain flow, delay, storage, bottlenecks, backlogs, cascading failure, and path-dependent recovery.

A candidate state is:

$$Z_{supply} = R_{fulfilled} + i\mathbf{Q}_{backlog}. \quad (K.61)$$

Where:

$R_{fulfilled}$ = delivered throughput;

$\mathbf{Q}_{backlog}$ = unfulfilled demand, inventory imbalance, and disruption pressure.

Quantum-style route

alternative routes and suppliers coexist;

shipment cycles possess phase;

flows can reinforce or cancel;

fulfilment gates convert orders into completed delivery;

residual backlog persists after local closure.

Thermodynamic route

The system consumes:

fuel;

labour;

storage;

refrigeration;

computation;

capital.

Dissipation includes:

spoilage;

delay;

idle capacity;

rework;

congestion.

GR-like route

Logistics already uses path geometry.

The effective metric can include:

time;

cost;

risk;

capacity;

carbon;

reliability.

A supply horizon appears when no feasible route can meet the required deadline or quantity.

Plausible development approach

encode flow cycles and bottleneck phases;

define R as fulfilled throughput;

define Q as backlog and inventory pressure;

define shipment completion as gate and trace;

construct network curvature from congestion;

define operational time through completed fulfilment events;

test whether Tsupply predicts cascade and recovery delay.

K.7.4 Transport and traffic networks

Traffic systems naturally produce waves, phase transitions, queues, and capacity collapse.

A candidate state is:

Ztraffic = Rflow + iQqueue. (K.62)

Where:

Rflow = realized movement;

Qqueue = stored mobility pressure.

Quantum-style route

traffic waves interfere;

routes form competing alternatives;

signals provide phase control;

intersections act as gates;

congestion changes subsequent route choice.

Thermodynamic route

Traffic consumes fuel and electrical energy.

Dissipation appears as:

braking;

idling;

heat;

delay;

stop–start waves.

GR-like route

Travel time defines an effective metric.

Congestion curves the network because the cost of movement depends on the global traffic state.

A transport horizon appears when no route reaches the destination within the admissible time or energy budget.

Plausible development approach

use traffic-wave phase;

define R as completed movement;

define Q as queue pressure;

define signal and capacity gates;

construct traffic internal time from completed trips;

test whether unresolved depth predicts jam formation and recovery.

K.7.5 Cloud and data-centre systems

Cloud systems contain workload waves, queues, thermal constraints, scheduling gates, and failure

cascades.

A candidate state is:

Zcloud = Rcompleted + i**Q**queued. (K.63)

Where:

Rcompleted = completed computation;

Qqueued = pending work, thermal pressure, memory contention, and failure risk.

Quantum-style route

multiple jobs compete for execution;

scheduling projects workload onto resources;

admission control provides gates;

cache and state history alter later routing.

Thermodynamic route

Data centres directly consume electrical energy and produce heat.

A budget is:

Pinput = Pcompute + Pcooling + Ploss + Pstorage-change. (K.64)

GR-like route

Latency defines effective distance.

Congestion and resource contention curve computational geometry.

A service horizon occurs when deadlines become unreachable under available resources.

Plausible development approach

derive phase from workload cycles;

define R as completed useful work;

define Q as queued and thermal residual;

define scheduler and admission gates;

model trace through job history and autoscaling;

construct internal computational time from verified task completion;

test whether Tcloud predicts overload and tail latency.

K.8 Tier D — Institutional and social systems

These domains may support the observer–gate–trace grammar strongly, but their complex phases require greater care.

K.8.1 Law and judicial systems

Law converts disputed possibilities into institutionally declared outcomes.

A candidate state is:

Zlaw = Rdeclared + i**Q**unresolved. (K.65)

Where:

Rdeclared = accepted fact, judgment, or operative rule;

Qunresolved = evidentiary conflict, interpretation, appeal, enforcement, and policy residual.

Quantum-style route

competing legal interpretations coexist;

evidentiary frames project different facts;

admissibility and burden thresholds act as gates;

judgment commits one institutional result;

dissent and appeal preserve residual;

precedent changes future projection.

Thermodynamic route

Institutional resources are spent through:

investigation;

hearings;

legal labour;

delay;

enforcement;

administrative friction.

The analogy is an institutional resource budget rather than literal physical thermodynamics.

GR-like route

Precedent and procedure create effective path geometry.

Some legal routes become easier, harder, or inaccessible because prior cases change the institutional landscape.

A jurisdictional horizon marks a boundary beyond which a court cannot produce enforceable action.

Plausible development approach

declare the legal protocol and jurisdiction;

define R as operative declaration;

define Q as typed unresolved legal residual;

identify admissibility and authority gates;

record precedent as trace;

define institutional cost geometry;

test whether residual-aware models predict appeal, reversal, delay, or enforcement failure.

Main caution

Law contains normative judgment and authority.

It cannot be reduced to an energy-minimization system.

K.8.2 Organizations and governance

Organizations transform proposals, information, and conflicts into decisions and coordinated work.

A candidate state is:

Zorg = Renacted + i**Q**institutional. (K.66)

Where:

Renacted = implemented policy or completed work;

Qinstitutional = unresolved disagreement, procedural debt, coordination pressure, and hidden risk.

Quantum-style route

proposals coexist;

departments project different frames;

approval chains act as gates;

decisions create trace;

prior decisions alter future authority and procedure.

Thermodynamic route

Organizations consume:

labour;

capital;

attention;

information;

coordination capacity.

Dissipation appears as:

rework;

bureaucracy;

conflict;

delay;

duplicated effort;

staff burnout.

The Dual Ledger framework is relevant because it treats maintained structure, conjugate drive, useful work, inertia, environmental pressure, and dissipation as portable audit objects rather than as substrate-specific metaphors.

GR-like route

Organizational geometry may be defined by:

communication delay;

authority distance;

approval cost;

trust;

dependency.

A bureaucracy horizon occurs when a useful action cannot reach implementation within the available authority, time, or budget.

Plausible development approach

define the organization boundary and objective;

identify decision cycles;

define R as implemented work;

define Q as unresolved coordination residual;

define approval gates;

treat policy and postmortem changes as trace;

construct internal time from completed decisions;

test whether residual depth predicts failure, turnover, or delayed execution.

K.8.3 Public administration

Public administration resembles organizational governance but operates under legal authority, public accountability, and multiple stakeholder frames.

A candidate state is:

$Z_{public} = R_{implemented} + i\mathbf{Q}_{pending}$. (K.67)

Potential residuals include:

pending cases;

legal challenge;

implementation gap;

public opposition;

budget shortfall;

cross-agency conflict.

The strongest framework would combine:

Protocol → Evidence → Authority Gate → Implementation → Public Trace → Revision. (K.68)

A public-administration horizon appears when legal, fiscal, or institutional constraints make implementation unreachable.

The complex representation remains engineered rather than native.

K.8.4 Epidemiology and social diffusion

Epidemiological systems contain latent states, transmission networks, delays, intervention gates, and irreversible health records.

A candidate state is:

Zepi = Robserved + iQlatent. (K.69)

Where:

Robserved = detected cases or visible adoption;

Qlatent = unobserved infection, susceptibility, or transmission pressure.

Quantum-style route

alternative transmission pathways coexist;

periodic behaviour creates phase;

contact networks mediate interaction;

testing projects latent state into observed state;

intervention thresholds act as gates.

Thermodynamic route

Literal thermodynamic interpretation is limited.

A resource-budget layer can still track:

healthcare capacity;

testing;

treatment;

immunity;

intervention cost.

GR-like route

Contact-network distance and transmission delay create effective causal geometry.

An epidemiological horizon appears where intervention cannot prevent widespread propagation before detection.

Plausible development approach

define latent and observed state carefully;

derive phase from seasonal or behavioural cycles;

define R as observed transmission;

define Q as latent burden;

define testing and intervention gates;

model public-health memory as trace;

test whether Tepi predicts outbreak acceleration.

K.8.5 Cultural and media systems

Cultural systems contain competing narratives, attention cycles, publication gates, memory, and attractors.

A candidate state is:

Zculture = Rdominant + i**Q**counter. (K.70)

Where:

Rdominant = publicly admitted or dominant narrative;

Qcounter = latent counter-narratives, unresolved disagreement, and suppressed semantic pressure.

Quantum-style route

narratives compete and reinforce;

attention cycles possess phase;

media channels project events;

publication and platform rules act as gates;

historical narratives alter future interpretation.

Thermodynamic route

An attention budget may track:

cognitive effort;

media capacity;

production cost;

signal amplification;

noise;

information degradation.

This is informational thermodynamics only when variables and units are carefully specified.

GR-like route

Semantic attractors create effective curvature in meaning space.

A narrative horizon appears when alternative interpretations become practically unreachable under the prevailing attention and platform structure.

Plausible development approach

define semantic feature space;

derive narrative phase from time-series embeddings or topic cycles;

define R as dominant public trace;

define Q as residual narrative pressure;

identify publication and attention gates;

measure trace through archives and institutional memory;

test whether Q predicts narrative reversal or polarization.

Main caution

This domain is highly vulnerable to post-hoc interpretation and political bias.

It should enter the research program only after the more measurable domains are established.

K.9 Comparative candidate table

Domain	Complex status	Natural phase	Thermodynamic maturity	Effective geometry	Gate and trace maturity	Initia priori
AC power grids	Native	Very high	High	High	High	1
Signal and communications	Native	Very high	High	High	High	2
Control and robotics	Native	High	High	Very high	High	3
Electrochemistry	Native	High	Very high	High	High	4
Fluid and plasma systems	Native	Very high	Very high	Very high	Medium	5

Domain	Complex status	Natural phase	Thermodynamic maturity	Effective geometry	Gate and trace maturity	Initia priori
Chemical reaction networks	Engineered	High	Very high	High	High	6
Cellular metabolism	Engineered	High	Very high	High	Very high	7
Neuroscience	Engineered	Very high	High	Very high	High	8
Distributed consensus	Engineered	Medium	Medium	Very high	Very high	9
Climate systems	Engineered	High	Very high	High	High	10
Ecology	Engineered	High	Very high	High	High	11
Immune systems	Engineered	High	High	High	Very high	12
Supply chains	Engineered	Medium	Medium	Very high	Very high	13
Traffic networks	Engineered	High	High	Very high	Very high	14
Cloud systems	Engineered	High	Very high	Very high	Very high	15
Law	Structural	Low to medium	Low	High	Very high	Later
Organizations	Structural	Medium	Medium	High	Very high	Later
Epidemiology	Structural/ engineered	High	Medium	Very high	High	Later
Cultural systems	Structural	Medium	Low	High	High	Explorat

The ranking is not a judgment of the domain's scientific maturity.

It measures suitability for the **specific complex residual research program**.

K.10 Reusable domain-development protocol

A common procedure should be used across all candidate domains.

K.10.1 Step 1 — Declare the protocol

$P^D = (B^D, \Delta^D, h^D, u^D, \varphi^D)$. (K.71)

Where:

B^D = boundary;

Δ^D = observation rule;

h^D = horizon;

u^D = admissible interventions;

φ^D = feature map.

A domain claim without these declarations is unstable.

K.10.2 Step 2 — Identify the admitted coordinate

Define what counts as realized or committed:

$R^D = \mathrm{Admit}^D(X^D \mid P^D)$. (K.72)

Examples include:

active power delivered;

decoded symbol;

reaction flux;

expressed phenotype;

committed database state;

implemented policy.

K.10.3 Step 3 — Identify the residual

Define:

$\mathbf{Q}^D = \mathrm{Residual}^D(X^D, R^D \mid P^D)$. (K.73)

The residual should be typed.

For example:

$\mathbf{Q}\mathrm{grid} = [Q\mathrm{reactive}, Q\mathrm{thermal}, Q\mathrm{stability}, Q\mathrm{congestion}]^T$. (K.74)

$\mathbf{Q}\mathrm{cell} = [Q\mathrm{metabolic}, Q\mathrm{regulatory}, Q\mathrm{stress}, Q\mathrm{repair}]^T$. (K.75)

$\mathbf{Q}\mathrm{dist} = [Q\mathrm{fork}, Q\mathrm{latency}, Q\mathrm{conflict}, Q\mathrm{pending}]^T$. (K.76)

The residual must not be defined only after failure.

K.10.4 Step 4 — Test whether complex representation is justified

A complex representation is justified when:

R and Q transform together;

phase has operational meaning;

magnitude or invariant is useful;

cross terms predict interaction.

A minimum test is:

$A^{D2} = R^{D2} + \mathbf{Q}^{DT} G^{D} \mathbf{Q}^{D}$. (K.77)

The representation should be rejected if no stable metric G^{D} can be defined.

K.10.5 Step 5 — Define phase independently

Phase should be derived from:

native phasor;

analytic signal;

complex mode;

oscillatory cycle;

oriented state plane.

Then:

$\theta_j^{D} = \operatorname{atan2}[b_j^{D}, a_j^{D}]$. (K.78)

Phase should not be inferred from the target outcome it is later used to predict.

K.10.6 Step 6 — Construct collective coherence

Define:

$C^{D} = [\Sigma_j w_j \exp(i\theta_j^{D})] / \Sigma_j w_j$. (K.79)

Then:

$\rho^{D} = |C^{D}|$. (K.80)

$\Phi^{D} = \arg(C^{D})$. (K.81)

The phase diversity should also be measured:

$H_{phase}^{D} = -\Sigma_k p_k \ln p_k$. (K.82)

The pair $(\rho^{D}, H_{phase}^{D})$ distinguishes productive coherence from over-lock.

K.10.7 Step 7 — Construct the thermodynamic or resource ledger

Define:

$\Delta U^{D} = I^{D} - W^{D} - \Gamma^{D} - \Lambda^{D}$. (K.83)

Every term should have:

units;

measurement method;

error bounds;

sampling interval.

When literal thermodynamics is not justified, the ledger should be called:

ResourceLedger^{D} rather than ThermodynamicLedger^{D}. (K.84)

K.10.8 Step 8 — Identify gate and trace

The gate is:

$G^{D} = \mathrm{Gate}^{D}(X^{D},\mathbf{Q}^{D},L^{D},P^{D})$. (K.85)

The trace is:

$\mathrm{Trace}^{D},_{k} = \{\mathrm{event},\mathrm{protocol},\mathrm{commitment},\mathrm{residual},\mathrm{metadata}\}$. (K.86)

The ledger update is:

$L^{D},_{k+1} = L^{D},_{k} \sqcup \mathrm{Trace}^{D},_{k}$. (K.87)

A mature model must distinguish gate from trace.

The gate selects.

The trace preserves and bends the future.

K.10.9 Step 9 — Define internal time

The realized increment is:

$d\tau^{D} = G^{D}|d\tilde{\Phi}^{D}| \,/\, \Omega^{D}$. (K.88)

The unresolved increment is:

$dT^{D} = (1 - G^{D})|d\tilde{\Phi}^{D}| \,/\, \Omega^{D}$. (K.89)

The complex time-like coordinate is:

$d\zeta^{D} = d\tau^{D} + i\, dT^{D}$. (K.90)

The model should test whether τ^{D} organizes system events better than external time.

K.10.10 Step 10 — Construct effective geometry

Select a metric from domain science:

$G^D = \text{Covariance}^{-1}$. (K.91)

Or:

$G^D = \text{FisherInformation}$. (K.92)

Or:

$G^D = \text{ControlCostMetric}$. (K.93)

Or:

$G^D = \text{NetworkResistanceMetric}$. (K.94)

Then:

$ds^{D2} = d\mathbf{x}^T G^D d\mathbf{x}$. (K.95)

The metric should predict:

transition cost;

reachability;

robustness;

path dependence;

recovery difficulty.

K.10.11 Step 11 — Define horizon-like boundaries

A control horizon is:

$H_{control}^D = \{x : \min_u \text{Cost}(x \to \text{SafeSet}) > \text{Budget}^D\}$. (K.96)

An information horizon is:

$H_{info}^D = \{x : \text{RecoverableInformation}(x) < \text{Threshold}^D\}$. (K.97)

A causal horizon is:

$H_{causal}^D = \{x : \text{Influence cannot arrive before Deadline}^D\}$. (K.98)

These should never be called event horizons without qualification.

K.10.12 Step 12 — State the falsifier

A domain mapping fails if:

$\mathrm{NoStablePhase}^{D} \vee \mathrm{NoResidualUtility}^{D} \vee \mathrm{NoBudgetClosure}^{D} \vee \mathrm{NoGateTrace}^{D} \vee \mathrm{NoGeometryGain}^{D}$. (K.99)

The model should then be reduced.

For example:

retain the budget;

remove the phase;

retain the residual;

abandon the GR-like analogy.

K.11 Domain suitability score

A preliminary suitability score can guide research priority.

Define normalized components:

C^{D} = maturity of complex formalism;

P^{D} = phase measurability;

B^{D} = budget closure;

G^{D} = gate observability;

T^{D} = trace maturity;

M^{D} = metric or geometry maturity;

F^{D} = falsifiability.

Then:

$\mathrm{Suitability}^{D} = wCC^{D} + wPP^{D} + wBB^{D} + wGG^{D} + wTT^{D} + wMM^{D} + wFF^{D}$. (K.100)

Where:

$\Sigma w_j = 1$. (K.101)

A possible classification is:

TierA if $\mathrm{Suitability}^{D} \geq 0.80$. (K.102)

TierB if $0.60 \leq \mathrm{Suitability}^{D} < 0.80$. (K.103)

TierC if $0.40 \leq \mathrm{Suitability}^{D} < 0.60$. (K.104)

Exploratory if $\mathrm{Suitability}^{D} < 0.40$. (K.105)

This score is a research-planning tool.

It should never replace the underlying domain measurements.

K.12 Recommended research priority

The recommended development sequence is:

First wave — Native complex systems

AC power grids

communications and signal processing

control systems and robotics

electrochemistry

fluid, plasma, and wave systems

These domains provide the strongest test of whether the Complex Residual Principle adds value beyond already mature complex-number formalisms.

Second wave — Living and chemical systems

chemical reaction networks

cellular metabolism and regulation

neuroscience

immune systems

climate and ecology

These systems provide the strongest test of the integration between:

complex residual;

non-equilibrium thermodynamics;

information geometry;

trace-bearing internal time.

Third wave — Engineered observer networks

distributed consensus

cloud and data-centre systems

cybersecurity

supply chains

traffic systems

These domains provide the strongest test of:

multiple local observers;

asynchronous clocks;

gates;

committed ledgers;

causal horizons;

trace-guided recovery.

Fourth wave — Institutional and social systems

law

organizations

public administration

epidemiology and social diffusion

cultural systems

These should be developed only after the measurement grammar is stabilized in more quantitative domains.

K.13 The anti-universality gate

The breadth of this appendix creates a serious risk.

A framework that can be mapped onto electricity, cells, brains, climate, software, law, and culture may appear to explain everything.

That appearance can indicate either:

a genuinely reusable abstract grammar;

an unconstrained vocabulary that cannot fail.

The framework must therefore impose an anti-universality gate.

Define:

$G_{anti\text{-}universal}^{D} = G_{native\text{-}variable}^{D} G_{measurement}^{D} G_{incremental\text{-}value}^{D} G_{falsifier}^{D}$. (K.106)

A domain enters the serious research program only if:

Ganti-universal^{D} ≥ G*. (K.107)

Where:

Gnative-variable tests whether the variables arise naturally;

Gmeasurement tests whether they can be estimated;

Gincremental-value tests whether the framework improves domain science;

Gfalsifier tests whether the mapping can be rejected.

The exclusion rule is:

If ImportedVocabulary > OperationalContent, reject the mapping. (K.108)

The discipline remains:

Functional Homology ≠ Physical Identity. (K.109)

Complex Form ≠ Quantum Substance. (K.110)

Resource Budget ≠ Literal Thermodynamics. (K.111)

Effective Geometry ≠ Physical Spacetime. (K.112)

Variable Internal Time ≠ Relativistic Proper Time. (K.113)

Control Horizon ≠ Black-Hole Event Horizon. (K.114)

K.14 Cross-domain research matrix

Each future domain paper could use the following compact matrix.

Question	Domain requirement
What is possible?	State or trajectory field
What is admitted?	R^{D}
What remains unresolved?	$\mathbf{Q}^{D}$
Where does phase come from?	Native oscillator or declared phase map
What creates coherence?	Coupling and synchronization
What is spent?	Energy, resource, computation, attention, or free energy
What is stored?	Potential, reactive power, backlog, latent pressure
What is dissipated?	Heat, friction, error, rework, conflict, or loss
What opens the gate?	Threshold, evidence, authority, capacity, or activation
What becomes trace?	State-changing record
What defines distance?	Information, control, network, or transport metric

Question	Domain requirement
What defines a horizon?	Loss of recoverability, reachability, or causal access
What is internal time?	Ordered consequential transitions
What would falsify the model?	No incremental predictive or control gain

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K.15 Final synthesis

Finance and LLMs appear to belong to a wider family of systems that can be described through:

possibility fields;

relative orientations;

complex or conjugate states;

stored and realized components;

transition gates;

dissipative budgets;

trace-bearing history;

state-dependent geometry;

internal clocks.

The candidate-domain generator is:

Possibility Field → Complex Orientation → Coupled Phase → Projection → Gate → Work + Dissipation + Residual → Trace → Effective Geometry → Internal Time. (K.115)

The strongest candidates already contain complex numbers natively.

The next strongest contain mature phase and thermodynamic structures.

The most speculative contain mainly gates, trace, and observer-relative residual.

The framework should expand outward only as quickly as operational definitions and falsifiable tests permit.

The final proposition of this appendix is:

> The Complex Residual Principle may describe a broad family of phase-bearing, budgeted, gated, and trace-forming systems, but the strength of the correspondence must be earned separately in every domain.

Or, in compact form:

Common Grammar must be demonstrated through Domain-Specific Measurement. (K.116)

And the final restriction remains:

A Reusable Grammar Is Not a Universal Physical Identity. (K.117)

Reference

Finance Geometry: Complex Valuation, Risk Pressure, and the Hidden Coordinate Behind Mature Finance Filters
https://osf.io/yucvm/files/osfstorage/6a4abb8fcaf0a0c36ddaa3e3

Imaginary Time as Admissibility Depth: A Ledger Ontology of Wick Rotation, Macro Systems, and Physical Time
https://osf.io/mvq6e/files/osfstorage/6a405c693e12266e39804e08

The True Nature of Technical Analysis - An Operator-First Interpretation of Market Charts, Volume, Waves, Gann Geometry, and Financial Self-Reference
https://osf.io/ne89a/files/osfstorage/6a3689cb33b86e3d1a86e142

The Imaginary Axis of Technical Analysis: How Complex Numbers Turn Chart Folklore into Market Pressure Geometry
https://osf.io/yucvm/files/osfstorage/6a4b942006735c3ce6daa274

A Rigorous Mathematical Grammar And Checklist That Ensure Nature-Inspired Systems Are Stable, Bounded, And Economically Viable
https://osf.io/hj8kd/files/osfstorage/6a500b7bbdb5870c2c7afb69

From Fundamental Physics to Purpose-Matched AI Agents
4π Spinor Closure, Hidden Control Stacks, and Environment-Aware Runtime Design
https://osf.io/hj8kd/files/osfstorage/6a4f89f3eef0d1166c5b9338

From Physics to AI Design: A Rosetta Stone for Runtime Architecture
https://osf.io/hj8kd/files/osfstorage/69d5023f5cdefa314c3eb654

Proto-Eight Dynamics (P8D): a small, testable model of how growth actually works 【先天八卦動力學】
https://osf.io/9rdsc/files/osfstorage/68b71c00b65e7b0e352c22f6

From Interfaces to Isomorphisms: A Protocol-Bound Theory of World Formation
How Bounded Observers Turn Fields into Operational Worlds — and Why Physics, Life, Organizations, Finance, Law, and AI Reuse the Same Grammar
https://osf.io/ae8cy/files/osfstorage/69ffbfc888878a0f3e78fda2

Philosophical Interface Engineering 1 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

Philosophical Interface Engineering 2 - Turning Deep Ideas into Testable Worlds, Thought

Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

Philosophical Interface Engineering 3 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

Life as a Dual Ledger: Signal – Entropy Conjugacy for the Body, the Soul, and Health
https://osf.io/s5kgp/files/osfstorage/690f973b046b063743fdcb12

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.